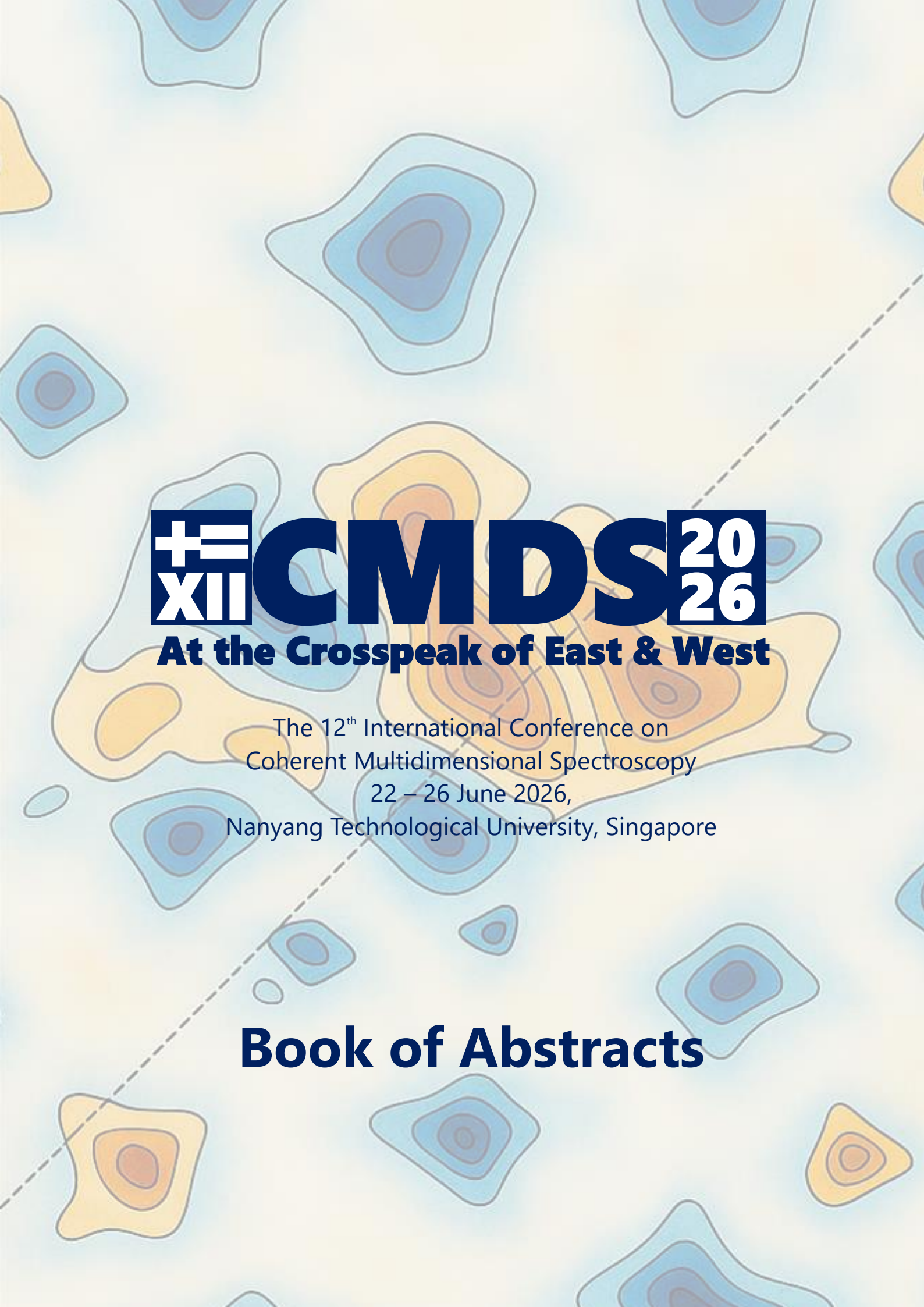




+ xii **CMDS** **20 26**

At the Crosspeak of East & West

The 12th International Conference on
Coherent Multidimensional Spectroscopy
22 – 26 June 2026,
Nanyang Technological University, Singapore

Book of Abstracts

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The 12th International Conference on Coherent Multidimensional Spectroscopy (CMDS2026)

CMDS is a major international forum to discuss the latest results in multidimensional optical spectroscopy. Conference topics will include the developments & applications, in theory and experiment, of multidimensional optical spectroscopy and related nonlinear optical techniques, to chemistry, biology, physics and materials science.

The CMDS series is organized biennially and CMDS 2026 is the twelfth in the series. CMDS was inaugurated in 2002 in Seoul, Korea and has since been held at Madison, USA (2004), Rigi-Kulm, Switzerland (2006), Kyoto, Japan (2008), Minneapolis, USA (2010), Berlin, Germany (2012), Eugene, USA (2014), Groningen, Netherlands (2016), Seoul, Korea (2018), Austin, USA (2022) and York, UK (2024).

The conference begins on 22nd June with an Early Career Symposium (ECS) followed by the CMDS 2026 main program from 23rd-26th June. The inaugural CMDS Innovation Award and the CMDS Achievement Award will also be presented at CMDS 2026. Please check out the nomination procedures and other details [here](#).

CMDS 2026 will serve as a platform to renew friendships, forge new collaborations among researchers, and to welcome young and aspiring scientists venturing into the multidimensional optical spectroscopy arena.

Humid and warm throughout the year, Singapore is a cosmopolitan city-state just north of the equator. Lush with greenery and a confluence of many cultures and traditions, we hope you will have the chance to sample some bits of our multicultural society. So, in the four official languages of Singapore, I bid you...

Selamat Datang ke Singapura!

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欢迎来到新加坡!

Welcome to Singapore!

Howe-Siang Tan,
On behalf of the CMDS2026 organizing committee

Committees

CMDS International Organizing Committee

- Minhaeng Cho (Korea University, Korea)
- Thomas Elsaesser (Max-Born-Institute, Germany)
- Peter Hamm (University of Zurich, Switzerland)
- David Jonas (University of Colorado-Boulder, USA)
- Shaul Mukamel (University of California-Irvine, USA)
- Yoshitaka Tanimura (Kyoto University, Japan)
- Andrei Tokmakoff (University of Chicago, USA)
- Keisuke Tominaga (Kobe University, Japan)
- Thomas L.C. Jansen (University of Groningen, Netherlands)
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- Howe-Siang Tan (Nanyang Technological University, Singapore)
- Tze-Chien Sum (Nanyang Technological University, Singapore)
- Yang Zhao (Nanyang Technological University, Singapore)
- Jeff Davis (Swinburne University of Technology, Australia)
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- Zac Faltz (University of Wisconsin-Madison, USA)
- Hyunmin Jang (Korea University, Korea)
- Sundar Krishnaswamy (University of Groningen, Netherlands)

Organizing Institutions



School of Chemistry, Chemical
Engineering and Biotechnology
College of Engineering • College of Science



School of Physical and
Mathematical Sciences
College of Science

Conference Venue

CMDS 2026 will be held at Nanyang Technological University, Singapore.

The conference venue is the School of Physical and Mathematical Sciences Building (SPMS), Nanyang Technological University, Singapore. The talks will be in Lecture Theatre 1 (LT1), SPMS. The poster sessions, exhibitions, and lunch/dinner/break will be at the atrium in the SPMS building. (Note: There are several other LT1s in the NTU campus. Look specifically for SPMS-LT1)

How to get from Nanyang Executive Centre (NEC) to the SPMS building:

Walk along Nanyang View to Nanyang Avenue (The hotel reception should be able to point you in the correct direction). Turn right at the junction and walk to the "Nanyang Heights" campus bus stop. The walk from NEC to this bus stop is about 400m.

Take the "Campus Loop-Blue" bus and get off at the "Opp. SPMS" stop. Click [here](#) for information about the bus route. You can also follow the google map instructions [here](#). Alternatively, you can take a 25-30min walk from NEC to the SPMS building (See Google map above).

How to get from Genting Hotel Jurong to the SPMS building:

Daily bus transport will be provided for conference attendees between the Genting Hotel Jurong and SPMS, NTU (Both to and fro). You are welcome to take the bus even if you are not staying at Genting Hotel Jurong.

By taxi: Quote this postal code (637371) to the taxi driver, and mention "School of Physical and Mathematical Sciences Building (SPMS), Nanyang Technological University" if need be.

By public transport: Take the subway system (MRT) from to "Pioneer" Station. Take Exit A from Pioneer Station, and you will arrive at a bus stop. Take bus 179. After 10 stops, alight at the "School of Communications & Information" stop. Follow signs to the School of Physical and Mathematical Sciences Building (SPMS), which is about 400 m walk away. (Apologies as the place near the bus stop may be a bit messy now, due to on-going construction).



Old Administrative Building (Currently, Chinese Heritage Centre), Nanyang Technological University.



School of Physical and Mathematical Sciences, Nanyang Technological University.

Code of Conduct at CMDS

The conference organisers at CMDS promote a respectful, diverse and inclusive environment.

This document sets out our expectations of acceptable and unacceptable behaviour for attendees and speakers at the CMDS 2026 conference and associated Early Career Symposium (ECS). All participants agree to be bound by this document when registering for the conference. We expect cooperation and support from all to help ensure a safe, harassment-free and respectful environment for everybody. This applies to the main event venue itself, break-out venues, refreshment areas, and even to places that are not part of the official event programme including but not limited to, visitors' accommodation and social interaction spaces.

Examples of harassment include but are not limited to:

- offensive comments (including those related to gender, age, sexual orientation, disability, physical appearance, body size, race or religion) either in person or *via* the Internet;
- inappropriate language and images in talks;
- deliberate intimidation or following;
- harassing photography or recording;
- sustained disruption of talks or other events;
- inappropriate physical contact;
- unwelcome sexual or other forms of attention.

Participants asked to stop any unacceptable behaviour are expected to comply immediately.

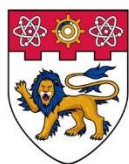
If a participant engages in unacceptable behaviour, the organising committee will take any action they deem appropriate, including warning the offender and if necessary, expelling them from the event (with no refund, for events with a registration fee). In the most severe cases, the organising committee reserves the right to contact the offender's home institution and / or the police.

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We want to once again say "THANK YOU!" to the members of this great community who have come together in Singapore this year. CMDS is always an incredible conference, and we could not be more excited to meet you and hear about the new things you've been up to recently.

We are incredibly grateful to be able to support the exciting and groundbreaking spectroscopy you do. We love seeing you create new methods that wouldn't have been possible five or ten years ago, or seeing you extend established 2D methods to new questions or stubborn old ones. We love it whether or not you use any PhaseTech products in your research. But it's especially fun to see when one of our pulse shapers or detectors or spectrometers was involved.

And while we are honored to be able to support your science, we also know that PhaseTech would not be who we are without you. We keep this in mind as we work to improve existing designs and to create new ones that can serve the evolving needs of the ultrafast and 2D communities.

So once again, THANK YOU! and we are looking forward to seeing and talking with you at CMDS 2026!

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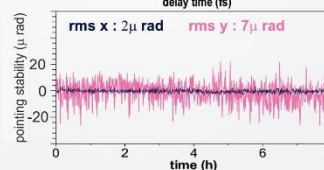
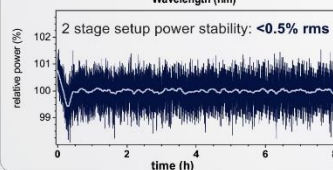
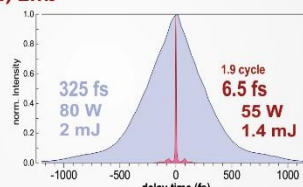
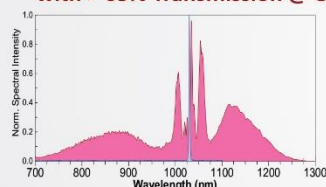
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*compressed pulse duration can be preserved if ONLY the input energy is varied

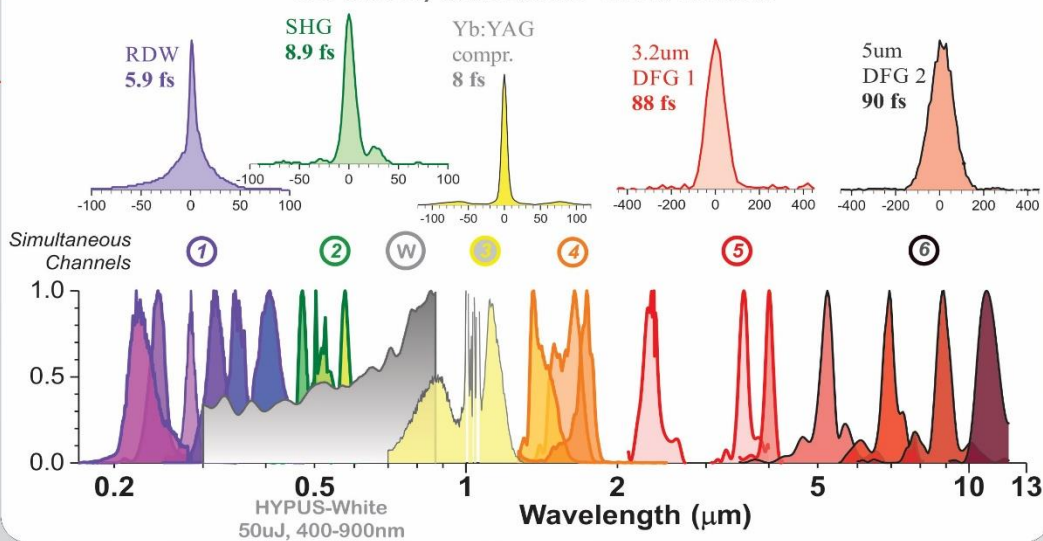
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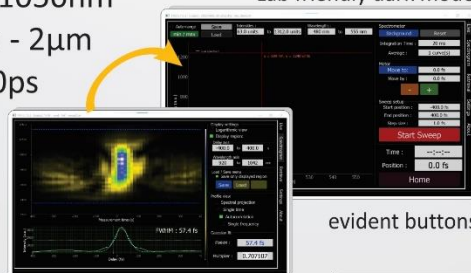
SHG Range - 700nm - 2μm

Long Scan Range - 20ps

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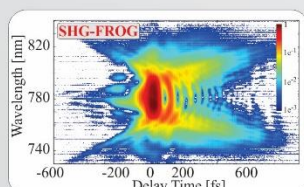
User oriented software:

Lab friendly dark mode

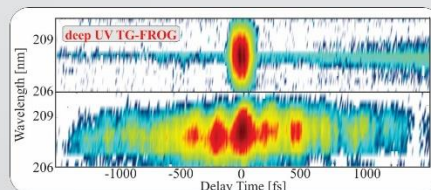
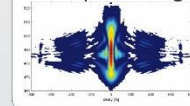
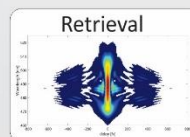
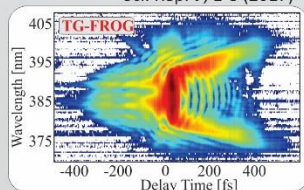


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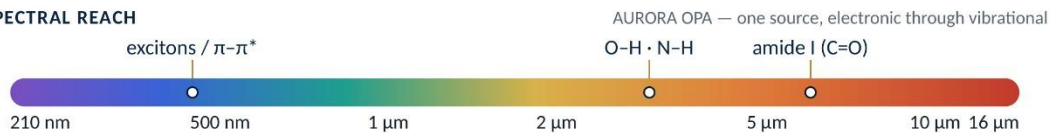
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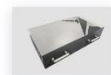
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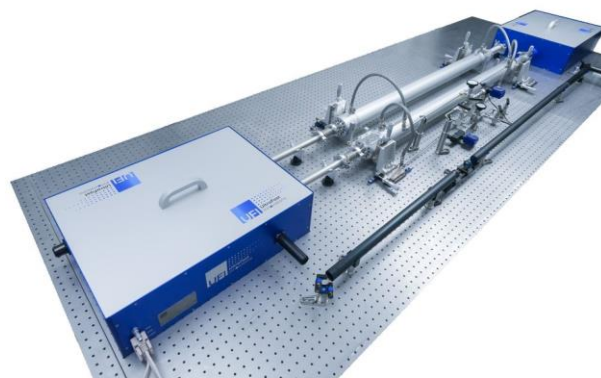


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Program

Early Career Symposium Program

(For ECS Talk abstracts, please refer to the poster abstracts numbers, P## or Q##, in brackets)

Monday, 22nd June 2026 (Morning)

0830-0850 Registration

0850-0900 Opening Remarks

	Chair: Sundar Raj Krishnaswamy
E01 (Q13) 0900-0920	Thanh Nhut Do pH-Dependent Structural Alterations in LHCII Revealed by Two-Dimensional Infrared Spectroscopy
E02 (P09) 0920-0940	Johanna L. Hall Photosystem II Subunit Design Revealed by 2DEV Spectroscopy
E03 (P10) 0940-1000	Ajeet Kumar Ultrafast Excitation-Energy Dynamics in Intact Cells of Gemmatimonas Phototrophica Probed by Two-Dimensional Electronic Spectroscopy
E04 (P12) 1000-1020	Caleb J. C. Jordan Exploring Chromophore-Environment Interactions of Metalloporphyrins in De Novo Proteins
1020-1050	COFFEE
	Chair: Hyunmin Jang
E05 (Q12) 1050-1110	Avinash Chettri A single GLFG repeat which serves as a model system for NuPs forms hydrogel with reversible fibers
E06 (Q01) 1110-1130	Gerrit N. Christenson 2D Electronic-Vibrational Spectroelectrochemistry: Vibronic Coupling in Organic Mixed Conductors
E07 (Q02) 1130-1150	Lara Denninger Same Same but Different: Vibronic Coupling to Selectively Target Subpopulations
E08 (P02) 1150-1210	Atandrita Bhattacharyya Multiexcitonic triplet state maintains significant singlet character throughout singlet fission
1210-1340	LUNCH

Monday, 22nd June 2026 (Afternoon)

		Chair: Hoang Long Nguyen
E09 (Q18) 1340-1400	Kyle Boschen Excitonic Interactions and Dynamics in Mono- and Bi-Layer WSe ₂	
E10 (P28) 1400-1420	Gijsbert A.H. ten Hoven Fourth Order Extension of Multichromophoric FRET	
E11 (Q25) 1420-1440	Zachary Faltz New Polarization Tricks For Background Removal In Action Detection Spectroscopies	
E12 (P33) 1440-1500	De-Ran Zhang Extracting high-order signals in two-dimensional spectroscopy via the mouse-tracking algorithm	
1500-1530	COFFEE	
1530-1700	Panel Sharing Session: Amber Krummel, Pavel Malý, Yin Song, and Michal Maj	
1730-1900	DINNER	

CMDS2026 Program

Tuesday, 23rd June 2026 (Morning)

0830-0840 Opening Remarks

Chair: Amber Krummel

V01 Neil Hunt

0840-0910 Combining Experimental 2D-IR with Machine Learning for Protein Analysis

C01 Jianping Wang

0910-0930 Unveiling Local Structures and Dynamics in Biomolecular and Pharmaceutical Systems by Ultrafast 2D Infrared Spectroscopy

C02 Michal Maj

0930-0950 2D IR Structural Fingerprinting of Amyloid Fibril Polymorphs

C03 Bei Ding

0950-1010 Monitoring the Active-site Response upon Photoactivation of the AppA BLUF Photoreceptor Using Two-Dimensional Infrared Spectroscopy and an N₃F Probe

S01

1010-1020 Light Conversion

1020-1050 COFFEE

Chair: Jasper Knoester

C04 Luca Bolzonello

1050-1110 Action-detected Multidimensional Spectroscopy: from fundamentals to *operando* device characterization

C05 Pavel Malý

1110-1130 Photo-thermal 2DIR and 2DES

C06 Alexei Kananenka

1130-1150 When Simple Kinetic Models are not Enough to Accurately Simulate Action-detected 2D Spectra

V02 Akihito Ishizaki

1150-1220 Two-Dimensional Fluorescence Spectroscopy with Quantum Entangled Photons and Time- and Frequency-Resolved Two-Photon Coincidence Detection

1220-1350 LUNCH

Tuesday, 23rd June 2026 (Afternoon)

		Chair: Tom Oliver
V02 1350-1420	Tobias Brixner Isolating Multi-Particle Correlations in Time and Space	
C07 1420-1440	Jacob Krich Order-Separating Spectroscopies: Revealing Multi-Exciton Spectra and Dynamics	
C08 1440-1500	JunWoo Kim Interaction Pathway-Selective 2DES with Undetected Photons	
C09 1500-1520	Jeff Davis Accessing New Information from the Phase Evolution in CMDS Measurements	
S02 1520-1530	PHASETECH	
1530-1630	EQUATORIAL FRUIT FEAST & COFFEE	
		Chair: Keisuke Tominaga
V04 1630-1700	Vivek Tiwari Disordered Light-harvesting Aggregates Can Host Functional Vibronic Couplings at Room Temperature	
C10 1700-1720	Carlos Baiz Hydrogen-Bond Dynamics at Electrochemical Interfaces: The Role of Cosolvents	
C11 1720-1740	Bogdan Dereka Dynamics of Strong Proper C–H Hydrogen Bonds	
P## 1800-2000	Poster Session 1 & DINNER	

Wednesday, 24th June 2026 (Morning)

		Chair: Donatas Zigmantas
V05 0830-0900	Antonieta De Sio Vibronic coupling-driven ultrafast excited state dynamics in quadrupolar molecules	
C13 0900-0920	Yuxiang Weng Quantum biological effects in photosynthesis revealed by two-dimensional electronic spectroscopy	
C14 0920-0940	Masaaki Tsubouchi Two-dimensional electronic spectroscopy of phycobiliproteins synthesized by recombinant technique	
C15 0940-1000	Hoang Long Nguyen Two-Dimensional Electronic Anisotropy to Reveal Isoenergetic Energy Transfer Dynamics in Photosynthetic Light-Harvesting Complexes	
S03 1000-1010	Few Cycle	
1010-1040	COFFEE	
		Chair: Minhaeng Cho
C16 1040-1100	Kyungwon Kwak Interchain structure around Li ⁺ ion in Solid Polymer Electrolytes	
C17 1100-1120	Nicholas Lewis Following electron transfer and confined water dynamics in Prussian Blue using electrochemical 2D IR	
C18 1120-1140	Daniel Kuroda Using Dual IR Reporters and 2DIR Spectroscopy to Map Interactions and Dynamics in Highly Heterogeneous Environments	
V06 1140-1210	Martin Zanni Are exciton dynamics in thin films representative of actual solar cells? Addressing the question using artifact-free ultrafast 2D photocurrent spectroscopy	
1210-1400	LUNCH	

Wednesday, 24th June 2026 (Afternoon)

	Chair: Vivek Tiwari
V07 1400-1430	David Jonas 2D Spectroscopy, Generalized Einstein Relations, and Spectroscopic Thermodynamics
C19 1430-1450	Jens Bredenbeck How IR Vibrational Excitation Reshapes UV/Vis Electronic Spectra
C20 1450-1510	Benjamin Fingerhut Quantitative Access to Molecular Excited State Properties of Conjugated Polyenes
C21 1510-1530	Thomas L.C. Jansen A Versatile Mapping Program for Modelling Two-Dimensional Infrared and Electronic Spectroscopy
1530-1600	COFFEE
	Chair: Thomas L.C. Jansen
C22 1600-1620	Tomáš Mančal Unified System–Bath and Light–Matter Perturbation Theory for Nonlinear Optical Response
C23 1620-1640	Garth Jones System–Bath Partitioning and Emergent Quantum Resources in 2DES
C24 1640-1700	Deependra Jadoun Using Twin-Beams for an Instantaneous Detection of Quantum Coherences in 2D Electronic Spectroscopy
C25 1700-1720	Andres Montoya-Castillo Short-lived memory in multidimensional spectra encodes full signal evolution
C36 1720-1740	Yin Song Tracking relaxation dynamics of polaritons and reservoir states in organic exciton-polaritons
Q## 1800-2000	Poster Session 2 & DINNER

Thursday, 25th June 2026 (Morning)

	Chair: Maxim Pchenitnikov
V08 0830-0900	Minhaeng Cho Orbital Angular Momentum-Resolved Nonlinear Spectroscopy with Vortex Beams
C26 0900-0920	James Lloyd-Hughes CMDS of Vibrational Correlations in Layered Organic-Inorganic Perovskites
C27 0920-0940	Rohan Singh Frequency-Resolved Quantitative Lineshape Analysis in Two-Dimensional Coherent Spectroscopy
C28 0940-1000	Donghai Li Revealing Ultrafast Excited-State Dynamics of Quantum Dots by 25 kHz Shot-to-Shot Phase-Cycling 2DES
1000-1030	COFFEE
	Chair: Jianping Wang
V09 1030-1100	Peter Hamm 2D-IR spectroscopy with a missing <i>i</i> : Universal Structure in the Relaxation of Photoactive Proteins
C29 1100-1120	Amber Krummel Bridging Molecular and Macroscopic Timescales: 2D IR Imaging of Copper Redox Dynamics in Ionic Liquids
C30 1120-1140	Sean Garrett-Roe CO ₂ Jumps Through Polymerized Ionic Liquids
C31 1140-1200	Yung Sam Kim 2D IR Study of Conformational Thermodynamics (ΔH and ΔS) and Fermi Resonance in the <i>syn</i> - and <i>anti</i> -Isomers of Acetic Acid in Acetonitrile
C32 1200-1220	Paul Donaldson Salt-pellet 2D-IR spectroscopy: a helping hand for vibrational spectroscopy of inorganic solids?
1220-1400	LUNCH

Thursday, 25th June 2026 (Afternoon)

	Chair: Steve Meech
V10 1400-1430	Hong-Guang Duan Correlating Wave-Packet Dynamics with Structural Evolution Using Ultrafast Spectroscopic and Diffractive Methods
C33 1430-1450	Donatas Zigmantas Polarization-controlled 2DES for extracting Q-band relaxation in chlorophyll <i>a</i>
C34 1450-1510	Giovanni Bressan Conformational Switching Modulates Excited-State Pathways in a Cofacial Perylene Dimer
1530-1900	CONFERENCE EXCURSION: Singapore River and Marina Bay Cruise.
1900-2130	CONFERENCE BANQUET: Permata Singapore. Nusantara Cuisine, in a historic mansion in Kampung Glam, Singapore.

Friday, 26th June 2026 (Morning)

	Chair: Antonietta De Sio
C35 0900-0920	Qiang Shi Open quantum system approach to dynamics and spectroscopy for strong light-molecule interaction
C37 0920-0940	Daniel Timmer Ultrafast collective multimode Rabi oscillations in an open cavity probed by 2DES
C38 0940-1000	Kevin Kubarych Correlating Vibrational Dynamics with Vibrational Strong Coupling Reaction Modification

1000-1030 COFFEE

	Chair: Jeff Davis
V11 1030-1100	Maxim Pchenitchnikov Excitons in the Making: Tracking Self-Assembly by 2DES
C39 1100-1120	Tõnu Pullerits Operando CMDS of Carrier Cooling in Perovskite Solar Cells
C40 1120-1140	Navendu Mondal Unravelling the Origin of Ultrafast Vibronic coupling in Hybrid Perovskite Nanosystems

1140-1200 Poster Awards & Closing Remarks

1200-1330 LUNCH

	Post-conference Excursion. Short Hike up Bukit Timah, Singapore's highest point. 1600hrs Bus from NEC to Bukit Timah Nature Reserve 1600hrs Bus from Genting Hotel Jurong to Bukit Timah Nature Reserve Afternoon ~1830hrs Somewhere in the Bukit Timah area for dinner.
--	--

POSTER SESSION I

Tuesday, 23rd June 2026 (Evening)

P01	Sundar Raj Krishnaswamy In Situ 2DES of Hierarchical Supramolecular Self-Assembly
P02	Atandrita Bhattacharyya Multiexcitonic Triplet State Maintains Significant Singlet Character Throughout Singlet Fission
P03	Alexey Kuevda Trimming Excitonic Landscape with Light
P04	Indranil Roy Anisotropic Quantum Beats Report Room Temperature Functional Vibronic Couplings in LHAs
P05	Rishabh Tripathi Excitation-Intensity-Driven Quantum Pathway Control in Two-Dimensional Coherent Spectroscopy
P06	Imran Mustapha A Kinetic Monte Carlo Study of Equilibration, Anisotropy Decay, Exciton-Exciton Annihilation and First Passage Dynamics in Photosynthetic Antenna Complexes
P07	Dominik Bäuerle New Insights on Dynamics in Chlorophyll-f Containing Far-red Photosystem I
P08	Junhua Zhou Non-Monotonic Temperature Dependence of Energy Transfer in a Photosynthetic Antenna Complex
P09	Johanna Hall Photosystem II Subunit Design Revealed by 2DEV Spectroscopy
P10	Ajeet Kumar Ultrafast Energy Transfer in Intact Gemmatimonas Phototrophica Probed by 2D Electronic Spectroscopy
P11	Mengyuan Cui Excitonic Energy Transfer in Red Algal Photosystem I Reveals an Evolutionary Bridge between Cyanobacteria and Plants
P12	Caleb Jordan Exploring Chromophore-Environment Interactions of Metalloporphyrins in <i>De Novo</i> Proteins
P13	Matteo Bruschi Multi-Exciton Processes in Ultrafast Nonlinear Optical Spectroscopy

P14	Jeswin Sunny Multiple Quantum Coherences of Electronic States and Vibrational Modes in Cyanocobalamin
P15	Alina Bach Energy Transfer in a Vibronically Coupled System: Transient Anisotropy of Chlorophyll c
P16	Nelson Kim Soon Ong Revealing Anti-Correlated Vibrational Coherences in Zinc Phthalocyanine via Polarization-Resolved Two-Dimensional Electronic Spectroscopy
P17	Maria Francesca Cherchi Selective Q-Band Excitation Modulates Exciton–Exciton Annihilation in H ₂ TPPS J-Aggregates
P18	Vittoria Burigana Substitution and Environmental Effects on Ultrafast Relaxation Dynamics of Red-absorbing Squaraines
P19	Ramandeep Kaur Two-Dimensional Electronic Spectroscopy of a System with a Double-Well Excited-State Potential
P20	Jinhyuk Ma Probing Quantum Indeterminacy in Ultrafast Excited State Intramolecular Proton Transfer
P21	Stephen Meech Remote Control of Fluorescent Protein Photoswitching
P22	Amitav Sahu Solvent-Driven Spectral Diffusion in IR-125 Probed by Full and Reduced Fluorescence-Detected 2DES
P23	Michal Ptacek Ultrafast Non-Adiabatic Dissipative Energy Dynamics of Porphyrins: from Explicit to Implicit Model
P24	Ting-Chien Ho Using Polarization Dependent 2D Electronic Spectroscopy to Reveal Hidden Vibronic Mixing in Photosynthetic Systems: A Theoretical Perspective
P25	Dario Baldon 2D-ES Analysis of Colloidal Plexcitonic Nanohybrids
P26	Sankaran Ramesh Plasmon-Exciton Hybridization and Helicity-Controlled Ultrafast Dynamics in Chiral Plexcitons
P27	Stephanie C. Gonzalez Migoni Coarse-Graining F-2DES: Back in Action

P28	Gijsbert ten Hoven Fourth Order Extension of Multichromophoric FRET
P29	Kennet Rueda Espinosa Theory and Simulations of Fluorescence-Detected Two-Dimensional Electronic Spectroscopy
P30	Sayan Ghosh Prominent Signatures of Energy Transfer in Action-Detected Spectra of a Photosynthetic Protein
P31	Muyi Zhang Ultrabroadband Fluorescence-Detected 2DES with Dual Rapid-Scanning Interferometers
P32	Jacob Higgins Directly Observing Many-Body Correlations with Two-Dimensional Electronic X-Ray Spectroscopy
P33	De-Ran Zhang Extracting High-Order Signals in Two-Dimensional Spectroscopy via the Mouse-Tracking Algorithm
P34	Valentina Shumakova Generation of Few-Cycle Pulses from Industrial Grade Yb:Laser Through Cascaded Post-Compression
P35	Dasol Im Spatially Resolving Energy Transfer in Carbon Nanotube Films with Ultrafast 2D Confocal Microscopy
P36	Jiading Zou Two-Dimensional Ultraviolet Spectroscopy with Conventional Optics
P37	Bhaskar De CNN-Based Full Lineshape Analysis in Two-Dimensional Coherent Spectroscopy
P38	Sachin Prasad Incorporating population-to-population, coherence-to-population, and coherence-to-coherence conversion processes in the response function expressions for multi-level systems in four-wave mixing experiments
P39	Chenlin Ma Multiple Davydov Ansatz for accurate system-bath dynamics in the presence of multiple baths with differing temperatures

POSTER SESSION II

Wednesday, 24th June 2026 (Evening)

Q01	Gerrit Christenson 2D Electronic-Vibrational Spectroelectrochemistry: Vibronic Coupling in Organic Mixed Conductors
Q02	Lara Denninger Same Same but Different: Vibronic Coupling to Selectively Target Subpopulations
Q03	Joseph Davies Intra-Defect and Defect-Lattice Coupling in Diamond Revealed by Two-Dimensional Infrared Spectroscopy
Q04	Tuhin Samanta Probing Supercritical Water Along the Widom Line via Infrared Spectroscopy
Q05	Helena Poulouse Tracking reaction kinetics with 2D-IR spectroscopy
Q06	Akhil Mon Ultrafast Vibrational Dynamics in Concentrated Electrolyte Solutions
Q07	Keisuke Tominaga Vibrational Population Relaxation and Spectral Diffusion of the CN Stretching Mode of [Ru(bpy)(CN) ₄]
Q08	Giorgio Scattolini Bringing the bulk to the surface: towards the application of surface enhanced 2D-IR spectroscopy for the study of solvation dynamics
Q09	BoHyeon Park Identification of Reactive Mode via Polarization Dependent Coherent Vibrational Spectroscopy
Q10	Samuel Bird Investigating Pre-Catalyst Solvation in Ionic Liquids using Ultrafast 2D-IR Spectroscopy
Q11	Timo Stern Making Proteins Transparent: Selective FEIR Spectroscopy of a Chromophore in Complex Protein Environments
Q12	Avinash Chettri A single GLFG repeat which serves as a model system for NuPs forms hydrogel with reversible fibers
Q13	Thanh Nhut Do pH-Dependent Structural Alterations in LHCII Revealed by Two-Dimensional Infrared Spectroscopy

Q14	Till Stensitzki Tracking enzymatic activity with 2D-IR spectroscopy
Q15	Riyanka Karmakar Carrier-donor Scattering Enhanced Multiple Carrier Generation at Exceptionally Low Energy Threshold
Q16	Manas Sarangi Relevance of Electron and Hole Transfer in Lead Halide Perovskite Nanocrystals on Current Conduction
Q17	EunSeo Jang Investigation of Coherent Phonon Dynamics and ASE Performance in FAPbBr ₃ with MACl Additives
Q18	Kyle Boschen Excitonic Interactions and Dynamics in Mono- and Bi-Layer WSe ₂
Q19	Yao Li Ultrafast charge generation dynamics via interfacial energetic modulation for single-component OSCs
Q20	Gi Rim Han Tracking the impact of ultrafast measurement on strongly confined perovskite nanoplatelets using asynchronous and interferometric transient absorption
Q21	Adam Alfrey Two-Dimensional Coherent Spectroscopy Provides Unambiguous Separation of Linewidths in MoSe ₂
Q22	Justin Wei Xiang Lim Investigating the Ultrafast Carrier Dynamics of 2-Dimensional Semiconductors using Time- and Energy-Resolved Photoemission Electron Microscopy
Q23	Bruno Schmidt Femtosecond Hyperspectral Source – HYPUS
Q24	Ross Milverton From Pulse Shaping to Batteries: New Approaches and Applications in 2D Raman Spectroscopy
Q25	Zac Faltz New Polarization Tricks for Background Removal in Action Detection Spectroscopies
Q26	Garth Jones SpecSuite: A toolbox for modelling ultrafast experiments
Q27	Paul Donaldson The new HiLUX-ULTRA instruments at the Central Laser Facility

Q28	Hyunmin Jang Asynchronous and Interferometric Nonlinear Spectroscopy of Indium Arsenide Nanoparticle
Q29	James Harris Probing Coupling to and Through Metal-Metal Multiple Bonds with DOVE Spectroscopy
Q30	Natalia Gray Disentangling Electronic and Vibrational Signatures via Simulated Femtosecond Coherence Spectroscopy
Q31	Junyan Sun Kramers–Kronig–Constrained Reconstruction of Optical Absorption from Interference-Dominated Transmission Spectra
Q32	Christine Chun Hui Ling Ultrafast Vibrational Dynamics of the Aqueous Adenine Radical Induced by Photodetachment
Q33	Bani Hakim Mohamed Ultrafast Vibrational Wavepacket Dynamics of Aqueous Phe-Ala
Q34	EunHo Hong Excited-State-Selective 2D Electronic Spectroscopy: A Role-Reversed Time-Dependent Perturbation Approach with Quantum Photonic States
Q35	Seyyed Jabbar Mousavi Anharmonic Phonon-Phonon Coupling in LaAlO_3 Revealed by Two-Dimensional Raman-Terahertz Spectroscopy
Q36	Luis Herrera Decoding the THz–IR–Vis Spectrum of Water
Q37	Minhyuk Lee Two-dimensional Raman-infrared Spectroscopy for Reaction Dynamics in the Electronic Ground state

Abstracts

For ECS Talks abstracts, please refer to their corresponding poster abstract numbers (The P## or Q## number in brackets beside the given ECS Talk code on the program table)

Invited Talks

Combining Experimental 2D-IR with Machine Learning for Protein Analysis

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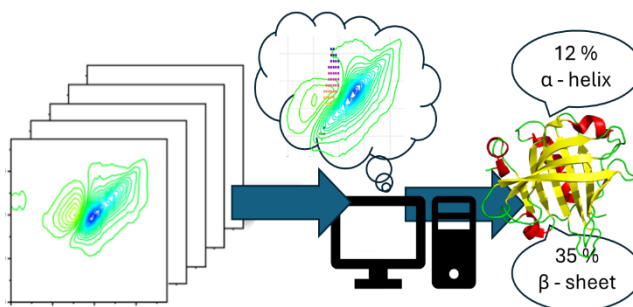
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Multidimensional laser spectroscopy methods such as two-dimensional infrared (2D-IR) produce large, information-rich datasets, while recent developments in laser technology have brought substantial increases in data collection rates. This combination of data depth and quantity creates the opportunity to unite advanced data science approaches, such as Machine Learning (ML), with 2D-IR to reveal insights that extend those from established data interpretation methods. To demonstrate this, we combined ML and 2D-IR spectroscopy in two applications: quantification of protein secondary structure based on a library of label-free amide I 2D-IR spectra [1] and diagnostically relevant classification of blood serum samples collected from patients with melanoma [2].

Ultrafast 2D-IR spectroscopy of the protein amide I band yields a spectral fingerprint that derives directly from the 3D backbone structure. However, transforming 2D-IR fingerprints into quantitative, solution-phase protein structures relies on decoding the fundamental link between the atomistic structure and the 2D spectrum. We show that a 2D-IR data library consisting of 6732 spectra of 35 proteins, spanning range of secondary structures, in H₂O can be used to train Support-Vector Machine (SVM) models to classify unknown protein samples according to structural content and measure quantities of α -helix and β -sheet with an RMS error of $\leq 7\%$ [1]. The potential for hybrid 2D-IR-ML tools to predict the number and length of helices in a protein and identify the presence of parallel and antiparallel β -sheets from the 2D-IR fingerprint is also demonstrated. These results lay the groundwork for rapid, quantitative analysis of dynamic protein structures under physiologically relevant conditions.



A Partial Least Squares-SVM model enabled classification of amide I 2D-IR spectra of blood serum samples taken from 40 patients with melanoma according to the presence, absence or later development of metastatic disease. Area under the receiver operating characteristic curve (AUROC) values of 0.75 and 0.86 were obtained when identifying samples from patients who were radiologically cancer free and with metastatic disease respectively [2]. The model was also able to classify (AUROC = 0.80) samples from a third group of patients who were radiologically cancer-free at the point of testing but would go on to develop metastatic disease within five years. This ability to identify post-treatment patients at higher risk of relapse from a spectroscopic measurement of biofluid protein content shows the potential for hybrid 2D-IR-ML analyses to contribute to risk stratification for melanoma patients.

References

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- [2] K. Brown, A.L. Farmer, s. Gurung, M.J. Baker, R. Board, N.T. Hunt, *Chem. Sci.*, 16, (2025), 8394-8404

V02

Two-dimensional fluorescence spectroscopy with quantum entangled photons and time- and frequency-resolved two-photon coincidence detection

Yuta Fujihashi,¹ Ozora Iso,² Ryosuke Shimizu,² Akihito Ishizaki^{1*}

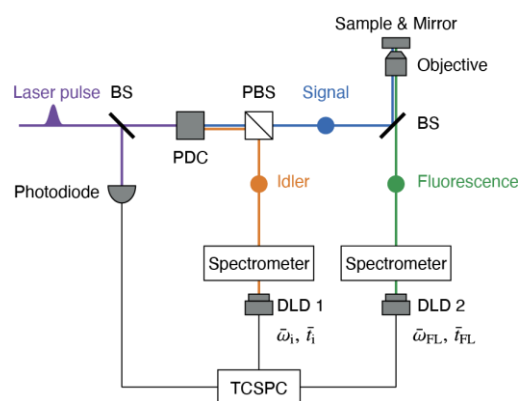
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Recent theoretical developments in quantum spectroscopy have underscored the potential of non-classical correlations in entangled photon pairs to selectively address specific nonlinear optical processes [1]. Despite this promise, experimental implementation of time-resolved spectroscopy with entangled photons remains unrealized, primarily due to the extremely low signal intensities generated under such conditions.

In this talk, we propose a theoretical scheme for quantum spectroscopy based on time-resolved fluorescence detection, which is fully compatible with existing photon detection technologies. This approach offers two key advantages over conventional two-dimensional electronic spectroscopy. First, it enables the acquisition of two-dimensional spectra without requiring precise control over multiple ultrafast laser pulses [2,3]. Second, it simplifies spectral interpretation by focusing on spontaneous emission pathways that selectively reflect nonlinear optical processes [4]. While conceptually related to previous proposals, our method yields signal intensities sufficient for detection with current technologies, thus making experimental realization feasible [5]. These findings provide a promising step toward the real-time observation of molecular dynamics using entangled photon pairs.



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- [3] Y. Fujihashi, K. Miwa, M. Higashi, A. Ishizaki, *J. Chem. Phys.* **159**, 114201 (2023)
- [4] Y. Fujihashi, A. Ishizaki, R. Shimizu, *J. Chem. Phys.* **160**, 104201 (2024).
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Isolating Multi-Particle Correlations in Time and Space

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Time-resolved spectroscopy is used to detect dynamics in quantum systems: An excitation is placed into the system by a pump pulse or, in the case of 2D spectroscopy, by a pair of pump pulses, and its evolution is then detected by a time-delayed probe pulse. Generally, however, it cannot be avoided that the interaction with the pump pulse(s) leads to multiple excitations. This can lead to artifacts and erroneous interpretation of time-resolved spectra and kinetics if not accounted for properly. On the other hand, new opportunities arise because the spectroscopic data contains information about multi-particle correlations that are relevant in many fundamental phenomena and applications.

We introduced the method of “intensity cycling” that allows us to separate the various perturbative orders of response in transient absorption, accessing multi-particle dynamics [1–5]. In this talk, I will show the generalization from transient absorption to 2D spectroscopy [6]. This is accomplished by systematic variation of pump-pulse intensities and specific data evaluation. We obtain one-quantum and two-quantum 2D spectra separated into the nonlinear orders. Experiments were carried out on squaraine polymers and squaraine dimers, and we obtained the third, fifth, seventh, and ninth nonlinear order of response. Since this removes artifacts from unresolved higher orders, it is possible to reach quantitative agreement with simulations of the dimer. We obtain information on its energetic structure, the coupling parameters, and the transition dipole moments, including information on double-excited states.

In action-detected spectroscopy, using fluorescence as an observable, phase cycling is generally used to separate nonlinear signal contributions. The number of phase measurements can be reduced using cogwheel phase cycling [7]. To access multi-particle correlations, we established two-quantum fluorescence-detected pump-probe spectroscopy [8]. Applying a data post-processing strategy eliminates incoherent mixing contributions as well as other “parasitic” signals that result from pulse-overlap ambiguities. Thus, we obtain background-free spectral and dynamical information of doubly excited electronic states. Care must be taken that nonlinearities in pulse shapers used to generate pulse sequences do not influence the measurement results [9]. Finally, we extend the concept of nonlinear order separation from coherent detection [1-6] to action detection. Thus, we obtain fluorescence-detected 2D spectra separated into the various orders of response.

Altogether, both coherently detected and action-detected 2D electronic spectra can now be separated into the different orders of response. Quantitative comparison with simulations provides new access to multi-particle correlations. We currently transfer these techniques to microscopy.

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V04

Disordered Light-harvesting Aggregates Can Host Functional Vibronic Couplings At Room Temperature

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Photosynthetic energy transfer occurs within overlapping vibrational–electronic manifolds, yet whether such vibronic couplings remain functional in large, disordered aggregates at room temperature remains unresolved. Here we employ polarization-controlled two-dimensional electronic spectroscopy to selectively [1,2] probe the coupled Qx–Qy states in self-assembled porphyrin nanotubes [3], a chlorophyll-like artificial template. Early-time cross peaks, their rapid peak broadening across the entire Q band within ~ 100 fs, and crucially, the isolation of anisotropic vibrational quantum beats provide direct evidence for functional vibronic couplings within the Q band at ambient conditions. Complementary vibronic exciton modeling reveals a vital functional role of energetic disorder: it enables Qx–Qy electronic mixing and dramatically enhances the spectral range of intraband vibronic couplings. These results demonstrate that disorder-enhanced vibronic coupling can robustly mediate ultrafast energy transfer in large photosynthetic aggregates and suggest a design principle wherein energetic disorder comparable to dense low-frequency vibrations with weak reorganization energies optimizes transport.

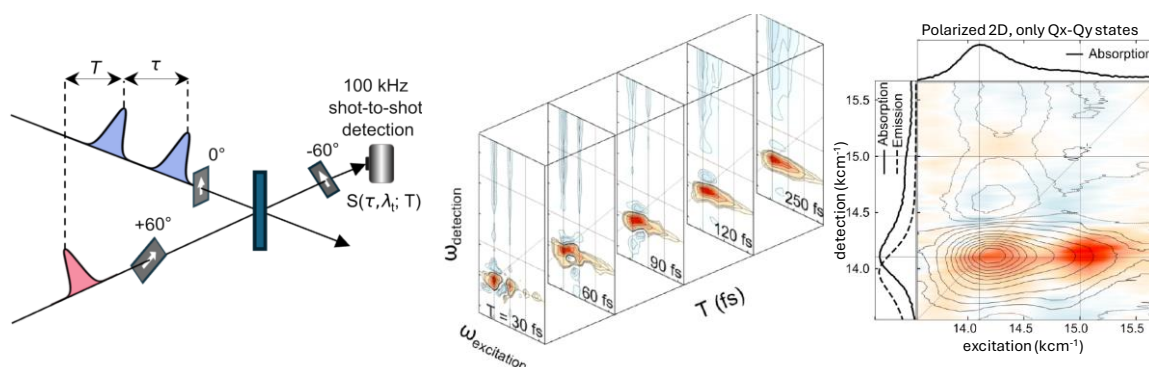


Fig. 1. (left) Polarization-controlled two-dimensional electronic spectroscopy on a model porphyrin nanotube template for photosynthetic nanotubes. (middle) Rapid electronic delocalization is reported by sub-100 fs peak broadening in the lower cross peak region. (right) Selectively probing Qx–Qy coupled states reveals anisotropic quantum beats. Adapted from ref. [3].

References

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Vibronic coupling-driven ultrafast excited state dynamics in quadrupolar molecules

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Ultrafast non-equilibrium dynamics following photoexcitation of functional organic materials arise from a complex interplay of electronic and vibrational motion, often involving their strong interaction. Detailed understanding of how vibronic couplings influence the initial energy redistribution, relaxation and localization of excitations is crucial for controlling nanoscale energy transport and for the rational design of new materials for applications in energy conversion and emerging quantum technologies. Most of the mechanisms underlying these processes occur on ultrafast, few 100s fs timescales, thus demanding methods that combine high time resolution and the ability to resolve couplings. To this aim, two-dimensional electronic spectroscopy (2DES) with sub-vibrational-cycle time resolution is ideally suited.

Quadrupolar acceptor-donor-acceptor (A-D-A) molecules constitute an important class of organic functional materials, promising for applications in photovoltaics, nonlinear optics, and bioimaging. Beyond their technological relevance, they provide a versatile material platform for investigating light-induced nonadiabatic dynamics mediated by complex vibronic couplings.

In this talk, I will give an overview of some of our recent results using ultrafast 2DES to probe vibronically mediated excited state dynamics in A-D-A molecules undergoing excited state symmetry breaking [1-3]. We identify spectral and dynamical signatures of vibronic coupling to an intramolecular high-frequency antisymmetric mode, resulting in a double-minimum shape of the excited state potential energy surface, governing the sub-100-fs relaxation dynamics. In aggregates of these A-D-A molecules, 2DES reveals ultrafast wavepacket motion through a conical intersection, highlighting a complex interplay of vibronic coupling and intermolecular interactions.

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V06

Are exciton dynamics in thin films representative of actual solar cells? Addressing the question using artifact-free ultrafast 2D photocurrent spectroscopy

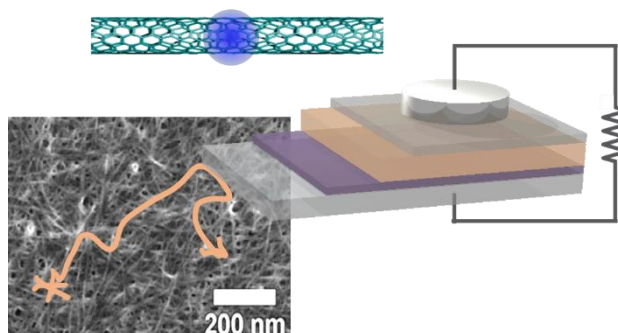
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The performance of many photoactive devices is impacted by the dynamics of excitons within the photoactive layer. Typically, these dynamics are deduced by independently conducted transient absorption measurements on films outside of the device with the assumption that the photophysics in free standing films inform the photophysics of the film inside the device. But there are interfaces in devices that are not present in films and the morphology and atomic packing might be impacted by device construction. Moreover, photoabsorption experiments measure all excitons whereas the best photovoltaic devices are perhaps 20% efficient, meaning that only 1 in 5 excitons actually produce electricity. Thus, the conclusions drawn from free standing films might not be directly relevant to devices. This talk will cover results made possible by new advances in ultrafast 2D photocurrent spectroscopy that enable ultrafast exciton dynamics to be directly compared between films and devices. In photovoltaics made from semiconducting carbon nanotubes, we find that the conclusions drawn from films versus devices are quite different, leading us to reinterpret long standing experiments and reevaluate our goals for device architecture.

Fig. 1. A collage of pictures that encompass the topics covered in this talk, including cartoon of a carbon nanotube with an exciton, an SEM of a thin film of carbon nanotubes with a potential trajectory of an exciton, and a photovoltaic device made using semiconducting carbon nanotubes.



V07

2D Spectroscopy, Generalized Einstein Relations between Absorption and Emission, and Spectroscopic Thermodynamics

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Two-dimensional Fourier transform electronic spectra of quantum dots[1] led us to a single-molecule generalization of Einstein's relationships between absorption and emission that resolves the conflict between the time-energy uncertainty principle and Einstein's relations between line-spectra in vacuum.[2] A transition in one direction (say R to S) between two broadened molecular bands R and S cannot always be classified as absorption or emission. In consequence, for transitions between two broadened bands, the set of 3 Einstein coefficients must be replaced by a set of 4 Einstein coefficient spectra and we use positive vs. negative frequencies to distinguish absorption vs. emission. These equilibrium spectra depend on a single total strength but replace Einstein's degeneracy ratio and transition frequency with a change in standard chemical potential for the transition ($\Delta\mu_{R \rightarrow S}^0$) and a single underlying lineshape that is manifested differently in the four spectra. In the electric-dipole approximation, the relationships for dipole-strength spectra are

$$d_{S \rightarrow R}^2(-\nu) = d_{R \rightarrow S}^2(\nu) \exp[-(h\nu - \Delta\mu_{R \rightarrow S}^0)/kT] \quad \text{and} \quad a_{S \rightarrow R}^{\nu}(-\nu) = (8\pi h \nu^3 / c^3) n(\nu) d_{S \rightarrow R}^2(-\nu) \theta(\nu) / 6\hbar^2,$$

which give the stimulated transition cross sections and the spontaneous radiative rates

$$\sigma_{R \rightarrow S}(\nu) = (h\nu / cn(\nu)) [d_{R \rightarrow S}^2(\nu) - d_{R \rightarrow S}^2(-\nu)] / 6\hbar^2 \quad \text{and} \quad A_{S \rightarrow R} = \int_0^\infty a_{S \rightarrow R}^{\nu}(-\nu) d\nu.$$

Because equilibration within the lowest excited electronic state is usually much faster than electronic state population relaxation for molecules in room-temperature solutions,[3] the equilibrium relationships are expected to apply to non-equilibrium photoluminescence for molecular ensembles with a single molecular form. Inhomogeneously broadened ensemble spectra do not exhibit the absorption-emission relationship (their ensemble Stokes' shift is too small). 2D spectra with a waiting time that is longer than needed for equilibration within the excited electronic state but shorter than the time needed for electronic energy transfer can reveal the single-molecule relationships (Fig. 1). To fully test these relationships, we are using spectroscopic measurements of $\Delta\mu_{0 \rightarrow x}^0$ to construct thermodynamic cycles that predict equilibrium constants for photochemical reactions.

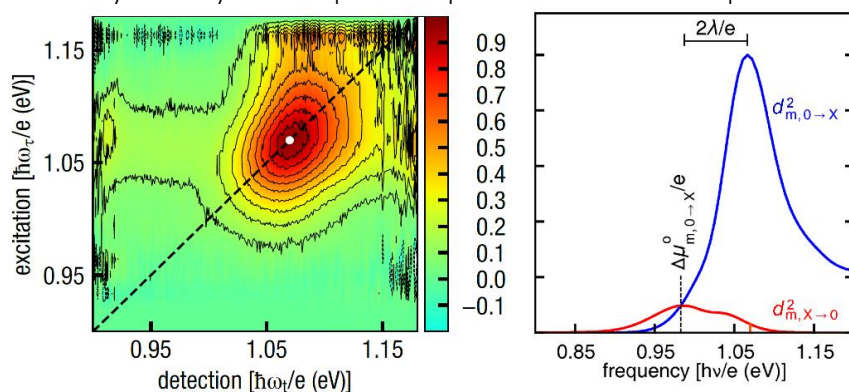


Fig. 1. (left) 2D FT relaxation spectrum at 1 ps waiting time for ~ 4 nm diameter colloidal PbS quantum dots, half-capped with oleate, in tetrachloroethylene. Pulse spectra and 2D propagation functions have been divided out. (right) the absorption (blue) and stimulated emission (red) dipole-strength spectra for quantum dots with the most probable bandgap. Adapted from ref. [1].

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Orbital Angular Momentum-Resolved Nonlinear Spectroscopy with Vortex Beams

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Time-resolved nonlinear microspectroscopy merges high spatial resolution microscopy with ultrafast spectroscopy to probe localized molecular and exciton dynamics on ultrafast timescales. By integrating ultrafast techniques like pump-probe and coherent multidimensional spectroscopy with high numerical aperture lenses and structured beams, it offers label-free chemical contrast and reveals transient phenomena in complex systems. Advances in adaptive optics, tailored beam profiles, spatial light modulators, and metasurfaces have enhanced spatial and temporal control, enabling novel coherent multidimensional microspectroscopy.

We present a theoretical framework for time-resolved nonlinear microspectroscopy using Laguerre-Gaussian (LG) beams with orbital angular momentum. Analytical expressions for pump-probe and photon echo signals reveal how beam parameters affect nonlinear responses, capturing spatial diffusion, ultrafast relaxation, and inhomogeneity-induced dephasing. Unlike pump-probe methods, photon echo is highly sensitive to the spatial structure of LG beams due to coherence evolution within inhomogeneously broadened ensembles. Our results show that customized ultrafast pulses and structured light fields enhance spatial separation of photon echo signals, improve resolution and sensitivity, and boost quantum information density. This work advances both nonlinear spectroscopy and quantum information science through structured light and ultrafast optics.

2D-IR spectroscopy with a missing i: Universal Structure in the Relaxation of Photoactive Proteins

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Over the last decade, we have performed time-resolved transient infrared experiments on a wide set of photoactive proteins, i. e., various PDZ domains and MCL 1/peptide complexes that are artificially photo-switchable protein systems, as well as naturally occurring cyanobacteriochromes. These proteins have in common that they are relatively small, compact and rigid single-domain proteins, but in terms of details of their structure, the embedded chromophore and their biological function they are in part very different. In either case, an embedded chromophore photo-isomerizes after electronic excitation on an ultrafast femtosecond to picosecond timescale, initially perturbing the structure of the protein directly around the chromophore. This local perturbation propagates over the protein in a cascade of events, which spread over a wide range timescales from picoseconds up to seconds.

Analyzing their time response via essentially an inverse Laplace transformation (instead of a Fourier transformation) very structured responses are obtained, see Fig. 1. These 2D maps reveal a series of kinetic steps, that appear to be coherent on a logarithmic time-axis (they are not “coherent” in the common sense). In a very universal manner for all investigated protein systems, the separation in time between kinetic steps is about one per decade [1].

To explain that observation, we first evaluate the inherent resolution power to disentangle exponential relaxation processes. Concluding that this is not limiting, various generic models of protein dynamics are discussed. Diffusion on a rugged free energy landscape along a one- or low-dimensional progress variable may explain the universal response, where the quasi-randomness of the kinetic matrix thins out eigenstates that contribute to transport, with some analogies to Anderson localization known from solid state physics. The separation in time of kinetic steps is a measure of the typical barrier heights, which by comparison to the universal patterns observed experimentally is found to be in the range of $k_B T$. Such barrier heights give a protein the flexibility to quickly structurally rearrange, yet provide some level of structural stability, which is relevant for example in the context of allosteric communication.

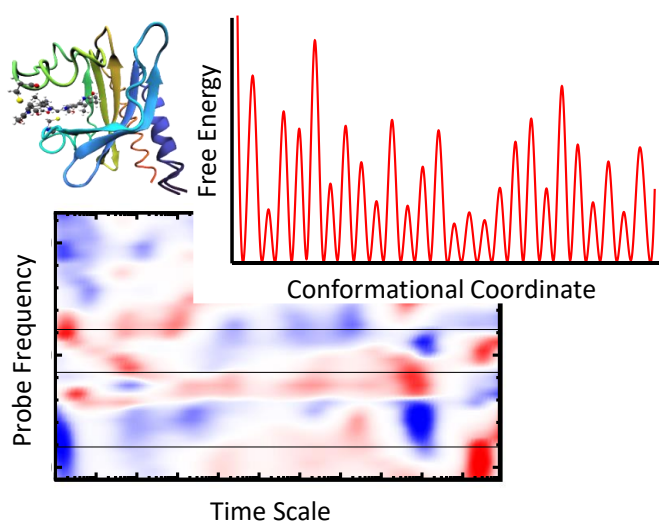


Fig. 1. The peaked structure in the lifetime spectra of the cyanobacterio-chrome TePixJ can be explained by diffusion on a rugged energy landscape [1].

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Correlating Wave-Packet Dynamics with Structural Evolution Using Ultrafast Spectroscopic and Diffractive Methods

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It is a central challenge of elucidating the intrinsic connection between quantum states of matter and their excited-state dynamics in ultrafast science. We have developed a high-temporal-resolution two-dimensional electronic spectroscopy (2DES) based on a NOPA with the full white-light spectral range. By implementing broadband pulse compression and active phase stabilization, we achieved a temporal resolution better than 5 fs, enabling direct access to electronic and vibronic coherent dynamics.

We first employed 2DES to photosynthetic protein complexes and investigated the relationship between quantum coherence and energy transfer. Temperature-dependent measurements of coherence lifetimes and energy transfer rates revealed that natural photosynthetic systems exhibit optimal transfer efficiency within the regime of 20-50 K, providing experimental insights into the functional relevance of quantum coherence in biological energy conversion.

To further expand the capability of 2D spectroscopy, we advanced its theoretical framework and numerical simulation methods and developed approaches applicable under strong light-matter coupling in optical cavities. These developments enable the exploration of cavity-modified quantum states and novel excited-state phenomena under photonic control.

We integrated 2D coherent spectroscopy with ultrafast electron diffraction (UED) to directly correlate electronic coherence with structural dynamics during photoinduced processes. This combined approach establishes a unified picture linking electronic, vibrational and structural degrees of freedom, offering new perspectives for understanding functional responses in complex materials.

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Excitons in the Making: Tracking Self-Assembly by 2DES

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Natural light-harvesting antennae in photosynthetic organisms, including green algae and bacteria, achieve remarkable energy-transfer efficiencies by organizing chromophores into dense, strongly coupled arrays. This organization supports delocalized Frenkel excitons, whose migration through the antenna enables efficient excitation-energy transport. Molecular double-walled nanotubes (DWNTs), formed by the self-assembly of amphiphilic C8S3 dye molecules in aqueous solution, provide a structurally well-defined biomimetic analogue of such antenna systems. Despite their soft molecular nature, DWNTs exhibit exceptionally low structural disorder and can support exciton delocalization over hundreds of chromophores. This combination of molecular softness, high structural order, and extended excitonic coherence raises a fundamental question: how do such ordered supramolecular architectures emerge from initially disordered molecular precursors in aqueous solution?

We address this question by following the optical properties of intermediate supramolecular species as they form during self-assembly. Our central premise is that these emerging optical properties arise from resonance interactions, or excitonic coupling, between the transition dipole moments of individual molecules. Optical spectra therefore do not merely indicate whether aggregates are present; they encode the evolving molecular organization, the strength and geometry of intermolecular coupling, and the degree of structural and energetic disorder within the assembling system.

To establish this connection, we combine molecular exciton theory, two-dimensional electronic spectroscopy, and microfluidics. Molecular exciton theory provides the framework for relating spectral signatures to delocalized electronic excitations in strongly coupled chromophore assemblies. Two-dimensional electronic spectroscopy directly probes couplings between electronic transitions and thereby reveals excitonic structure, disorder, and supramolecular architecture in intermediate species. Microfluidics maps self-assembly time onto a spatial coordinate, allowing transient intermediates to be separated from fully assembled nanotubes. These methods are complemented by cryo-TEM, which identifies double-walled structures, and single-DWNT polarization microscopy, which probes excitonic anisotropy at the level of individual nanotubes. Among these approaches, 2DES offers a particularly favorable balance between experimental complexity and structural insight.

We find that DWNTs form through a stepwise, asymmetric self-assembly pathway, in which the outer NT structures first and the inner NT follows only after a delay. This intermediate stage is marked by the rapid formation of molecularly ordered domains that remain mutually disordered, while the nanotubular assemblies continue to elongate steadily. As the NTs grow longer, structural irregularities, particularly end-cap defects caused by imperfect molecular packing at the termini, increasingly impede further elongation and ultimately give rise to the broad NT length distribution observed after two days. Together, these findings allow hierarchical supramolecular self-assembly to be followed as it occurs and connect the emergence of optical function to the underlying excitonic interactions, providing a framework for understanding how biomimetic light-harvesting architectures acquire their high degree of order and efficient excitation-energy transport.

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Contributed Talks

Unveiling Local Structures and Dynamics in Biomolecular and Pharmaceutical Systems by Ultrafast 2D Infrared Spectroscopy

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Understanding the intricate relationship between structure, dynamics, and function in complex molecular systems, from membrane proteins to drug-drug cocrystals, is a grand challenge. Conventional structural biology methods often lack the time resolution to capture fast dynamics or the site-specificity to resolve local motifs within a large assembly. Here, we demonstrate how ultrafast two-dimensional infrared (2D IR) spectroscopy, combined with strategic vibrational probes, overcomes these limitations to reveal critical local structural details and dynamics in two distinct systems.

First, to resolve the conformation and function of Lysine residues on the cytoplasmic surface of the microbial photoreceptor bacteriorhodopsin (bR), we employed seven site-specific $^{13}\text{C},^{15}\text{N}$ -lysine labelings. The isotope labelings created a set of unique vibrational signatures, red-shifting the amide-I frequency of the labeled residues. Ultrafast 2D IR spectroscopy, supported by 2D solid-state NMR, isolated a high-frequency amide-I mode at ca. 1693 cm^{-1} originating from K159. Its vibrational lifetime, shorter than the bulk helix, indicated a water-accessible, hydrogen-bonded γ -turn structure with E161. This work shows how 2D IR can map residue-specific structures crucial for protein function, such as stabilizing the E-F loop for proton uptake.

Second, we investigated the core structure of S086, a novel angiotensin receptor-neprilysin inhibitor drug cocrystal, which exists in two forms (α and ξ). Using the asymmetric stretching modes of carboxylate groups (COO^-) from its two constituent molecules (EXP3174 and sacubitril) as intrinsic IR probes, we differentiated their binding geometries to Ca^{2+} ions. Linear IR spectroscopy quantified the molar ratio of "bidentate" to "bridging" COO^- groups from EXP3174, finding a 1:3 ratio in the α -crystal versus a 2:2 ratio in the ξ -crystal. 2D IR spectroscopy then revealed a more rigid core structure for the α -crystal through spectral diffusion analysis. Furthermore, by tracking vibrational energy transfer between the COO modes of EXP3174 and sacubitril, we determined faster transfer rates in the α -crystal, suggesting a more tightly packed core with shorter intermolecular distances. This integrated IR approach provides critical dynamical parameters for understanding drug stability and release.

These studies demonstrate that combining linear and nonlinear 2D IR spectroscopies with targeted vibrational probes provides a unique window into site-specific local structures and ultrafast dynamics. This approach offers unparalleled insights into protein functional mechanisms and the structural basis for the properties of complex pharmaceuticals.

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2D IR Structural Fingerprinting of Amyloid Fibril Polymorphs

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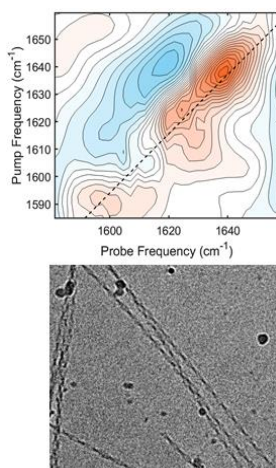


Fig. 1. 2D IR spectrum of a disease-associated helical fibril polymorph of hIAPP along representative cryo-EM micrograph of this monomorph state

Amyloid fibrils found in patients with distinct pathologies exhibit a restricted set of unique folds. In contrast, fibrils grown *in vitro* display extensive structural polymorphism. The molecular determinants that drive this *in vivo* structural selectivity remain largely unknown. Here, we combine femtosecond two-dimensional infrared (2D IR) spectroscopy and cryo-electron microscopy (cryo-EM) to decipher the mechanisms governing polymorph selection in human Islet Amyloid Polypeptide (hIAPP). While cryo-EM provides atomic-level resolution, we demonstrate that 2D IR provides a rapid, structure-sensitive fingerprint that resolves distinct polymorph populations in solution. We find that conditions favoring secondary nucleation, specifically high protein concentrations, drive structural diversification, where newly spawned fibrils do not structurally replicate the parent template. Strikingly, by introducing the C-terminal domain of DNAJB6 chaperone, we selectively inhibit secondary nucleation and steer aggregation toward a monomorphous, disease-relevant state. These findings suggest that molecular chaperones act as biological filters, which restrict the landscape of accessible polymorphs in various amyloid diseases.

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Monitoring the Active-site Response upon Photoactivation of the AppA BLUF Photoreceptor Using Two-Dimensional Infrared Spectroscopy and an N₃F Probe

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The blue light using flavin (BLUF) domain is a crucial photosensor unit in microbial photoreceptors. Its photoactivation mechanism involves an electron-coupled proton relay mediated by conserved tyrosine (Tyr) and glutamine (Gln) residues, ultimately leading to the reorganization of the hydrogen-bond network surrounding the flavin mononucleotide (FMN) cofactor [1]. However, how the active-site dynamics changes upon photoactivation remains unknown so far. In this study, we employed azidophenylalanine (N₃F), an infrared probe with a high molar extinction coefficient, and site-specifically incorporated it near the active center (position W104) and on the protein surface (position F55, as a control) of the AppA BLUF domain. Although the Fourier transform infrared (FTIR) spectrum of the N3 group exhibits complex multi-peak features due to Fermi resonance, we successfully resolved this interference using two-dimensional infrared (2DIR) spectroscopy. Analysis of the W104N3F mutant indicates that its spectral characteristics originate from the coupling between the fundamental N3 stretching mode and combination/overtone bands. By constructing and analyzing a three-level Fermi resonance Hamiltonian model, we successfully extracted the unperturbed fundamental frequency of the N3 stretch. The analysis reveals a 4 cm⁻¹ red shift in the fundamental frequency of the N3 probe at the W104 site from the dark to the light state, which reflects changes in the local electrostatic environment induced by photoactivation. Furthermore, lineshape analysis of the 2DIR diagonal peaks shows that the environment of the active center W104N3F, in both the dark and light states, is close to the homogeneous limit and exhibits the narrowest bandwidth, revealing that this site is embedded within a highly compact protein structure. Molecular dynamics (MD) simulations further confirm that the frequency shift primarily results from changes in the local electric field caused by alterations in the relative orientation and distance between the key residue Q63 (which undergoes tautomerization and rotation) and the N₃F probe upon light activation. This study demonstrates that, even in the presence of Fermi resonance, critical structural and dynamic information during protein functional processes can be effectively extracted from the N₃F probe by combining 2DIR spectroscopy with theoretical analysis, providing new insights into the photoactivation and signal transduction mechanisms of BLUF domains.

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Action-Detected Multidimensional Spectroscopy: From Fundamentals to Operando Device Characterization

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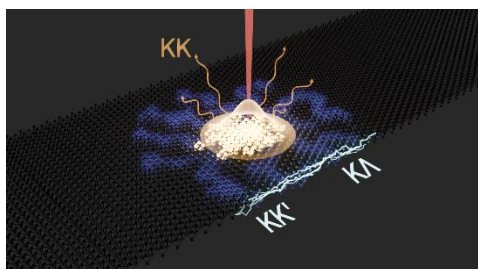
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Action-detected two-dimensional electronic spectroscopy (A-2DES) has emerged as a powerful complement to conventional coherent 2DES, measuring incoherent observables, fluorescence, photocurrent, photoelectrons, or photoions, arising from excited-state populations rather than phase-matched coherent fields [1]. By replacing spatial phase matching with phase cycling in a fully collinear beam geometry, action detection enables integration with high-numerical-aperture microscopy (sub-micron spatial resolution), measurements on highly scattering and heterogeneous samples, and operando characterization of functioning optoelectronic devices under realistic operating conditions.

These advantages come with fundamental trade-offs. Action detection measures fourth-order populations weighted by quantum yields rather than third-order polarizations, introducing additional Liouville pathways and incoherent mixing contributions that alter signal interpretation, particularly in multichromophoric systems, where excited-state signals scale as $1/N$ relative to ground-state bleaching [2]. Understanding when these differences enrich and when they complicate the spectroscopic picture is essential for the correct application of the technique.



Extending beyond the multidimensional framework, we developed dual action spectroscopy, a combined photocurrent and photoluminescence excitation scheme, and applied it to WSe₂ devices from monolayer to bulk [3]. This approach disentangles the contributions of bright and momentum-dark excitons to the material's photoresponse, showing that dark KK' and KA excitons dominate charge

generation and transport over diffusion lengths exceeding 5 μm , while bright excitons govern the luminescence channel. These results, inaccessible by conventional optical techniques, demonstrate the unique capacity of action detection to reveal structure–function relationships in photoactive materials.

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Photo-thermal 2DIR and 2DES

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Two-dimensional (2D) spectroscopy allows full characterization of the third-order nonlinear response, 2DIR of the vibrational modes and 2DES of the electronic ones. Novel variants of 2D spectroscopy, such as those with incoherent observables, widen its scope of applicability and augment its capabilities. Examples are recently popular fluorescence- or photocurrent-detected 2DES [1], or tag-loss detected 2DIR [2]. In all these variants of 2D spectroscopy, the generated heat is seen as a loss channel.

We explore as of yet practically unexplored variant of 2D spectroscopy which detects the photo-generated heat: photo-thermal 2D (PT-2D) spectroscopy. Our theoretical formulation of PT-2D in terms of response functions is general, applicable both to PT-2DIR spectroscopy, which has been recently realized in the form of peak-force 2DIR (PF-2DIR) [3], and PT-2DES spectroscopy, which we propose as a novel spectroscopy technique.

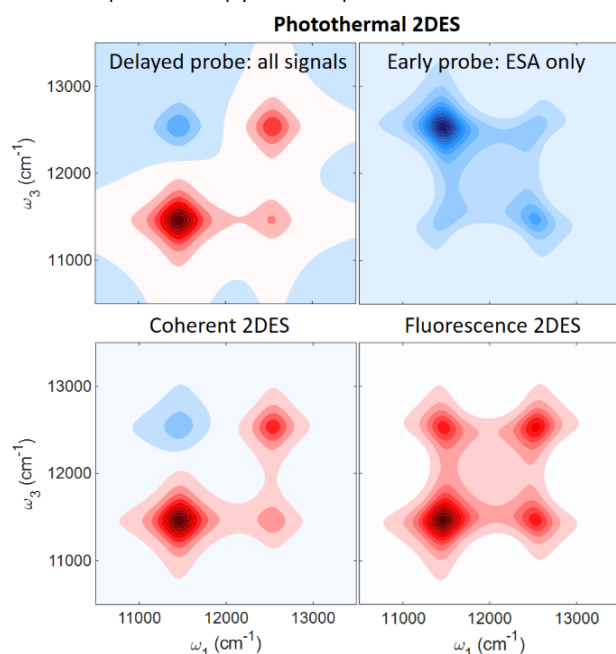


Fig. 1 Absorptive photothermal PT-2DES spectra of a coupled dimer (top), contrasted with coherent 2DES (bottom left) and F-2DES (bottom right). For the PT-2DES, delayed off-resonant probe of all generated heat (top left) retrieves a signal akin to coherent 2DES, while early-time probe (top right) isolates the excited-state absorption contribution.

generated heat, the PT-2DES spectra approach the coherently detected 2DES, including sensitivity to electronic coupling and energy transfer. Furthermore, we demonstrate that variable timing of the off-resonant photothermal probe pulse allows isolation of a single excited-state absorption type pathway, providing access to excited-state response. We conclude by outlining experimental realizations of PT-2DES, including our first steps in this direction.

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The detection of the generated heat gives PT-2D spectroscopy some unique properties. For instance, since generation of multiple excitations typically does produce proportionally more heat, PT-2D does not suffer from so-called incoherent mixing of excitations during signal emission, which plagues other types of action-detected 2D spectroscopy.

For PT-2DIR, we show that the 2D spectra are sensitive to mode anharmonicity and coupling, as in coherent 2DIR. We verify our calculations by direct comparison with the experimental PF-2DIR results of the collaborating Xiaoji Xu group [3], and contrast these to the tag-loss 2DIR spectra [2]. PT-2DIR as PF-2DIR thus enjoys both high spatial resolution of the AFM tip detection and sensitivity of 2DIR.

Next, we introduce the visible, electronic version of PT-2D, coherent PT-2DES. Contrasting PT-2DES with coherently and fluorescence- detected 2DES, see Fig. 1, we highlight its favorable properties. Detecting all

When Simple Kinetic Models are not Enough to Accurately Simulate Action-detected 2D Spectra

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Action-detection two-dimensional electronic spectroscopy (A-2DES) is rapidly becoming an incisive tool for measuring ultrafast dynamics in molecular aggregates, semiconductors, solid-state devices, and other systems. In A-2DES a sequence of four collinear laser pulses prepares the system in an excited-state population. The resulting observable is an action signal emitted during the typically long detection time. Because A-2DES is more complicated and less established than coherently detected 2DES (C-2DES), simulations are necessary to connect the spectral features to underlying physical processes. It is however computationally expensive to perform quantum dynamics simulations for long detection times. A practical approach to simulating A-2DES relies on using kinetic modeling of population dynamics after the fourth light-matter interaction. I will present recent work from my group which derives and analyzes a kinetic model [1] that leads to the commonly used simple formula for calculating A-2DES spectra proposed by A. Marcus [2] in which A-2DES signals are approximately written in terms of ground-state bleach, stimulated emission, and excited-state absorption signals of C-2DES spectra. This approach works well for most typical molecular aggregates where the exciton-exciton annihilation is efficient. I will show when such a simple method breaks down and more quantum-mechanically rigorous approach is necessary [1]. Specifically, I will relate the accuracy of this approach to competing processes in excited-state manifolds. Additionally, I will present our newest unpublished (but summarized in Fig. 1) work in which we used simulations to determine polarizations schemes that remove the coherent signal from A-2DES spectra and expose the spectroscopic signatures of multi-exciton processes such as exciton-exciton annihilation occurring during the detection time.

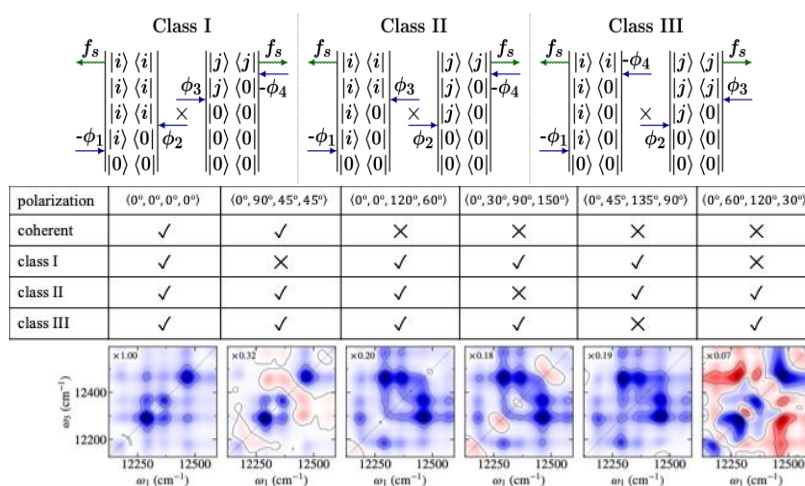


Fig. 1. (Top) Three classes of incoherent signals in action 2D spectroscopy as classified in Ref. [3]. (Middle) Table shows the presence of coherent and various classes of incoherent signals in simulated fluorescence-detected 2D electronic spectra of Fenna-Matthews-Olson light-harvesting complex for various polarizations of the four laser pulses.

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Order-Separating Spectroscopies: Revealing Multi-Exciton Spectra and Dynamics

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Transient absorption (TA) and two-dimensional electronic spectroscopy (2DES) are powerful tools for uncovering spectral and dynamical information about excited states. Most studies use sufficiently low pulse intensities that the measured signals are third order in the pulse amplitudes, in which the system interacts twice with the pump and once with the probe. Increasing pump intensities improves signal-to-noise ratios but also increases contributions to the signal from molecules that are excited multiple times, which can be a challenge to interpret. The measured signals can be treated as a power series in the pump intensity, with the usually desired third-order signal as the lowest order.

I will describe a general method to systematically separate spectroscopic orders of response – that is, the terms in the power series – by acquiring spectra with multiple pump-pulse intensities, with applications in TA [1] and 2D spectroscopy [2]. These experiments pose no additional technical challenges over standard TA and 2DES. I will show how to choose pulse intensities to give the best extractions of response orders, given the noise present, and to quantify the errors in the extracted orders. This method allows acquisitions with increased pump intensities that improve signal-to-noise while systematically removing signal contaminations from higher-order processes. I will show experimental demonstrations from TA and 2DES and will discuss the rich information present in high-order responses, including examples in multiple polymer systems of how they encode exciton-exciton-annihilation kinetics.

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Interaction Pathway-Selective 2DES with Undetected Photons

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Two-dimensional electronic spectroscopy (2DES) is a powerful technique for investigating light-induced dynamics in complex material systems such as photosynthetic light-harvesting complexes. In molecular excitonic systems, 2DES signals typically include overlapping contributions from ground-state bleach (GSB), stimulated emission (SE), and excited-state absorption (ESA). The simultaneous presence of these interaction pathways provides rich information about the energetic structure and relaxation processes of the system. However, when these contributions cannot be clearly separated, especially in densely packed spectra, the resulting spectral congestion reduces both the selectivity and interpretability of the measurement.

To address this challenge, recent theoretical studies have proposed quantum-optical approaches to femtosecond time-resolved spectroscopy that enable interaction pathway selectivity by exploiting the nonclassical properties of quantum light [1,2]. Building on our previous work, we proposed single-photon transient stimulated emission (SP-TSE), a pathway-selective spectroscopic method that uses the particle nature of single photons to isolate the SE contribution [3,4]. SP-TSE offers strong pathway selectivity by eliminating coherent artifacts and suppressing unwanted nonlinear signals. This capability not only isolates SE from GSB and ESA but also enables accurate analysis of ultrafast kinetics, which is often compromised by coherent artifacts in conventional wave-mixing 2DES.

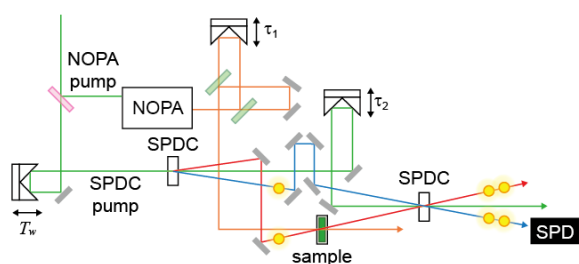


Fig. 2. Schematic illustration of QO-2DES. NOPA, non-collinear parametric amplifier; SPDC, spontaneous parametric down conversion; SPD, single-photon detector.

However, since SP-TSE employs single photons as both pump and probe, it is experimentally challenging to apply to molecular or excitonic systems that require ensemble measurements, due to their extremely small absorption cross sections. To overcome this limitation, we now introduce a new method, referred to as two-dimensional electronic spectroscopy with undetected photons (UP-2DES; Fig. 1).

UP-2DES uses entangled photon pairs to achieve ensemble-compatible measurements while preserving light-matter interaction pathway

selectivity, particularly with respect to GSB and ESA, and eliminating background signals such as pump scattering. This is accomplished by correlating the detection of one photon in an entangled pair with the undetected interaction of its partner with the sample. UP-2DES enables access to material properties that are not observable in conventional semiclassical 2DES, thereby highlighting the unique advantages of quantum-optical approaches. More broadly, this work demonstrates how quantum photonic concepts can expand the scope of multidimensional spectroscopy beyond the limitations of classical frameworks.

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Accessing New Information From the Phase Evolution in CMDS Measurements

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One of the key requirements of Coherent Multidimensional Spectroscopy (CMDS) is measurement of the phase of the third-order signal, and its evolution. This phase information ensures a unique solution to the Fourier transform of the data, and helps separate the real and imaginary parts of the third order susceptibility. In recent work, we have shown how carefully tracking the phase evolution in this type of measurement can yield details of adiabatic changes to the energy of specific states or bands, and mixing between them, which are hidden when the data is viewed in the frequency domain [1]. In that case, the transient changes were induced by an additional, red-detuned pulse driving a valley-selective optical Stark shift in monolayer WS_2 .

In this presentation we will describe new measurements that take this concept further and reveal how monitoring the evolution of the 3rd-order signal phase can yield insight into quantum phase transitions. Two different material systems will be considered. The first is an extension of our previous work on monolayer WS_2 , where an additional blue-detuned pump pulse modifies the band-structure. Considered within a Floquet formalism, this driving is expected to induce a transition to an excitonic insulator phase, or to a topological insulator phase, depending on the magnitude of the detuning [2]. The measured phase evolution under these conditions shows signatures consistent with a transient, reversible, quantum phase transition driven by the Floquet pump pulse.

Secondly, we will show how tracking the phase evolution in CMDS measurements on the high-temperature superconductor $\text{Bi}_2\text{Sr}_2\text{Ca}_1\text{Cu}_2\text{O}_{8+\delta}$ can provide new insight into photo-melting of the superconducting phase. This insight comes from an additional response that is evident in the imaginary part of the signal, but hidden by other contributions in real part (see Fig. 1). This contribution appears to follow the inferred dynamics of the melted superconducting phase and places new constraints on the processes that could be responsible for the quenching the macroscopic coherence of the superconducting condensate.

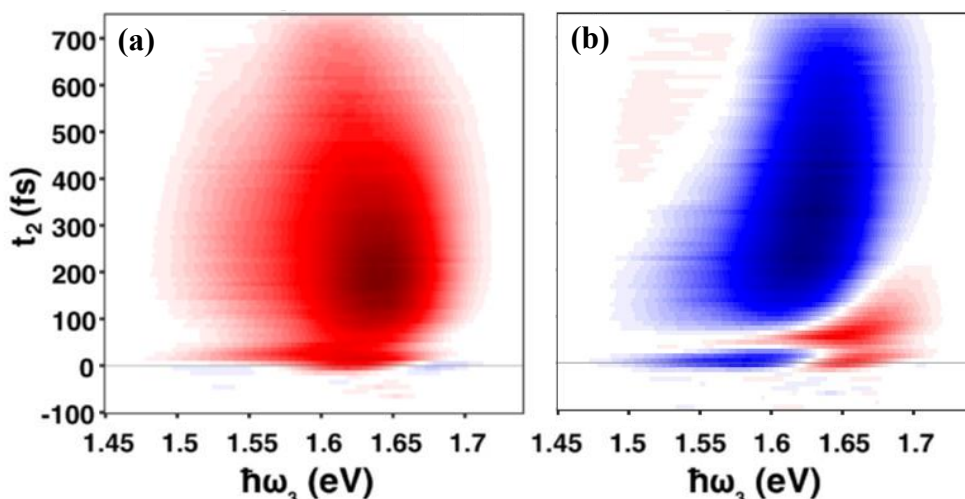


Fig. 1. (a) and (b) show the real and imaginary part of the third order response from $\text{Bi}_2\text{Sr}_2\text{Ca}_1\text{Cu}_2\text{O}_{8+\delta}$, respectively. The imaginary part shows an additional positive (red) contribution at early times that is associated with the melted superconducting phase.

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C10

Hydrogen-Bond Dynamics at Electrochemical Interfaces: The Role of Cosolvents

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Electrochemical reactions are driven by a thin, nanometer-scale interfacial layer whose hydrogen-bond (H-bond) connectivity, dynamics, and local electric fields set the energy landscape for charge transfer processes. While applied potential can be used to control interfacial environments, chemical composition offers an orthogonal and under-explored control knob. This can reshape interfacial H-bond populations and lifetimes. Two-dimensional infrared (2D IR) spectroscopy provides a direct route to measure solvent dynamics: by tracking the spectral diffusion of a local reporter, 2D IR can connect ultrafast line-shape evolution to H-bond lifetimes. Surface-tethered probes provide access to interfacial dynamics under operando-relevant conditions.

We use surface-enhanced 2D IR (SE-2D IR) to quantify how dimethyl sulfoxide (DMSO) modulates *interfacial and bulk* H-bond dynamics using the CN stretching vibration of a 4-mercaptobenzonitrile (4-MBN) functionalized Au electrode. The nitrile stretch serves as a reporter, and center-line-slope (CLS) analysis resolves a fast (~ 1 ps) component associated with local inertial fluctuations and a slower component assigned to activated H-bond making/breaking processes. At open-circuit potential, in pure water, the interfacial CLS decay is biexponential (0.93 ps and 9.90 ps), but, suprisingly, adding DMSO selectively accelerates the slow H-bond exchange component while leaving the ~ 1 ps component essentially unchanged: at only 5 mol% DMSO the slow component shortens to 5.56 ps, and at 20 mol% DMSO it collapses to 1.84 ps, indicating a strongly labile interfacial H-bond network (Figure 1).[1] In sharp contrast, bulk 2D IR measurements using MeSCN show that DMSO slows long-time spectral diffusion and introduces signatures of dynamic heterogeneity consistent with microdomain formation at intermediate-to-high DMSO concentrations. These opposing trends imply that DMSO acts as a H-bond network disruptor whose consequences invert under interfacial confinement, potentially amplified by preferential interfacial enrichment of DMSO and the reduced dimensionality/under-coordination of interfacial water.

The presentation will summarize the results, including applied electrostatic potentials together with a set of cosolvents, such as trifluoroethanol (TFE). Interpretations are based on MD simulations of these systems. Generally, these results establish cosolvent composition as a practical “lever” for tuning electrochemical solvation dynamics and motivate combined “molecular + electrostatic” strategies to engineer interfacial environments relevant to electrocatalytic selectivity and rates.

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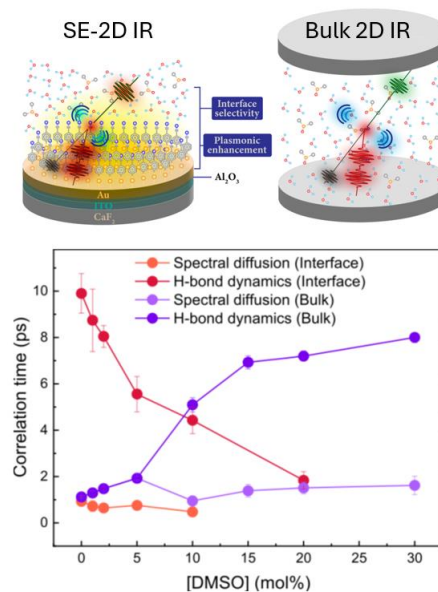


Fig. 1. Comparison between surface and bulk 2D IR measurements. (*upper*) Plasmonic cell for SE-2D IR, versus conventional bulk 2D IR. (*lower*) Nitrile correlation times from CLS analysis of spectra as a function of DMSO concentration. Interfacial H-bond lifetimes become shorter while bulk H-bond lifetimes increase.

Dynamics of Strong Proper C–H Hydrogen Bonds

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Hydrogen bonds (HBs) represent one of the most fundamental molecular interactions. Typically, HBs are associated with highly polarized moieties, whereas C–H bonds are generally not considered as H-bond donors: they either do not form hydrogen bonds at all or form only very weak and improper hydrogen bonds, markedly distinct from conventional (proper) HBs. Characteristically, these improper C–H hydrogen bonds result in a blueshift of the C–H stretch frequency and a shortening of the C–H bond length.

In this presentation, I will demonstrate that C–H groups can form strong, proper hydrogen bonds that are highly reminiscent of their O–H and N–H counterparts. Using a combination of mid-IR, NMR and sub-ps broadband THz spectroscopy and supported by ab initio molecular dynamics simulations and the gold standard of symmetry-adapted perturbation theory we observed characteristic signatures of C–H hydrogen bonding not only to strongly electronegative heteroatoms, but also to aromatic π -systems and, most surprisingly, even single C=C double bonds in non-activated non-polar molecules. Furthermore, we devised a purely experimental decomposition method to disentangle the electric field effect from the short-range non-electrostatic contributions to C–H hydrogen bonding that allows clear quantification of these effects in various environments.

With the help of ultrafast two-dimensional infrared (2D IR) spectroscopy we directly observe hydrogen bond switching events and compare their dynamics between C–H and C–D bonds. We discuss two different coupling mechanisms to other modes and demonstrate how one type of coupling uniquely enables measuring C–H dynamics selectively in conditions that make conventional 2D IR measurements infeasible.

Given a strong interest in expanding the selection of local IR markers available for sensing, our results open new opportunities for site-selective probing of structural dynamics in molecules and materials.

C12

Deciphering Solvent Dynamics and Solvation Structures in "Water-in-DES" Electrolytes for Zn-Ion Batteries

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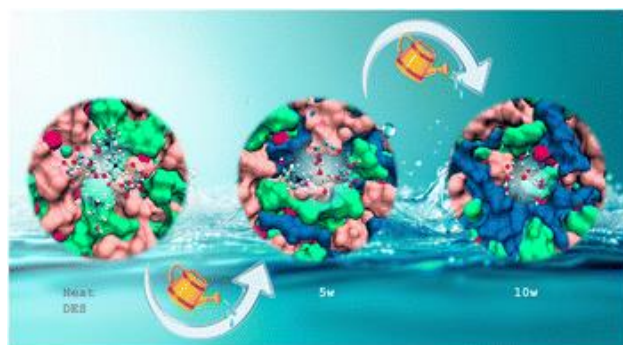
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Aqueous Zn-ion batteries (ZIBs) represent a sustainable frontier in energy storage, yet their commercial viability is hindered by poor anode reversibility and dendrite growth. While Water-in-Deep Eutectic Solvent (DES) electrolytes offer a promising "green" solution, the molecular-level mechanisms governing their performance—specifically the disruption of hydrogen-bonding networks and the



evolution of the Zn²⁺ solvation shell, remain elusive. In this work, we employ a synergistic approach using ultrafast 2D IR and 2D NMR spectroscopy to map the vibrational and chemical dynamics of water-in-DES systems across a wide concentration gradient. By probing the spectral diffusion and cross-peak dynamics, we resolve how incremental water addition modulates the DES network and influences ion-pairing kinetics. Complementary molecular dynamics simulations provide a

structural blueprint for these interactions, correlating microscopic rearrangements with macroscopic transport properties. Our results identify a critical threshold at 10 moles of water per mole of DES, where the system achieves optimal ionic conductivity and electrochemical stability. This study provides a fundamental framework for designing next-generation electrolytes by bridging the gap between ultrafast molecular spectroscopy and battery engineering.

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Quantum biological effects in photosynthesis revealed by two-dimensional electronic spectroscopy

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The concept of quantum coherence is derived from the superposition principle in quantum mechanics. The advent of ultrashort laser pulse having a broad band spectrum can easily create superposition of a number of excited states undergoing quantum coherence. For photosynthetic light-harvesting complexes, coherent energy transfer is expected to have a higher efficiency than the classical pathway, while elucidating the nature of the coherence, i.e., electronic, electronic-vibrational coupling and vibrational, is still challenging. Two-dimensional electronic spectroscopy (2DES) is a powerful tool in elucidating the coherent energy transfer, exciton delocalization and many other coherent processes. In this talk, I will briefly introduce the principle and implementation of 2DES, and then report the recent progresses in exploring biological quantum effects either by using 2DES in our group: (1) quantum phase synchronization via electronic-vibronic energy dissipation sustains long-lived coherence to support coherent energy transfer in allophycocyanin (APC) trimer which is a photosynthetic light-harvesting pigment-protein complex located in the phycobilisomes (PBS) of cyanobacteria and red algae; (2) Classical energy transfer in PC620, a main light harvesting antenna in the PBS rod with its structure being very similar to that of APC; (3) Electronic-vibrational coherence in cryptophyte phycoerythrin 545 (PE545) antenna.

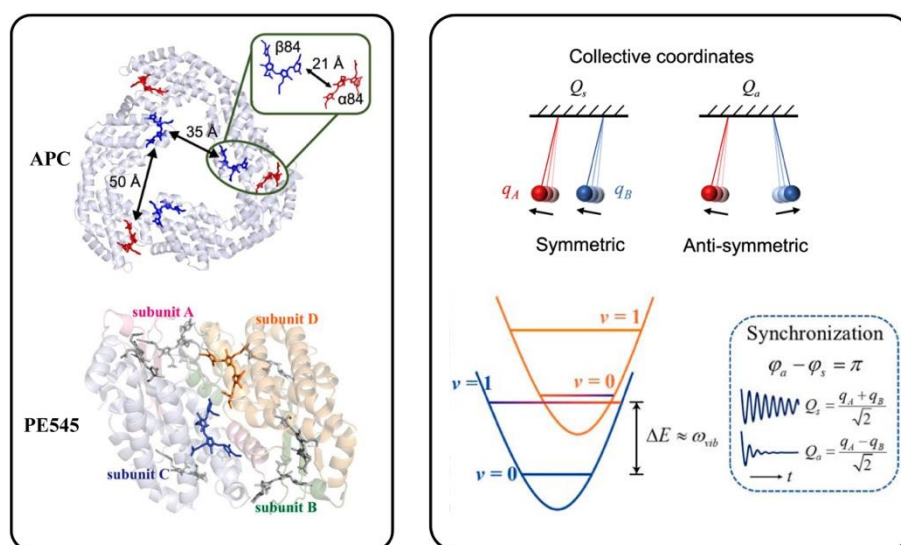


Fig. 1. Left: The crystal structure of APC and PE545. Right: The quantum phase synchronization model for photosynthetic antennas to sustain long-lived coherence.

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C14

Two-dimensional electronic spectroscopy of phycobiliproteins synthesized by recombinant technique

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Phycobilisomes in cyanobacteria and red algae achieve highly efficient unidirectional energy transfer through networks of phycobiliproteins, yet the roles of pigment number, spatial arrangement, and local pigment–protein interactions in shaping their excited-state dynamics remain unresolved. To clarify these contributions, we investigated allophycocyanin (APC) and C-phycocyanin (CPC) from *Thermosynechococcus elongatus* BP-1 expressed in *E. coli*, along with a β 153-deficient CPC mutant designed to mimic APC's pigment count and geometry [1]. As shown in Fig. 1(a) and (b), APC and CPC share homologous $\alpha\beta$ heterodimers with α 84 and β 84 phycocyanobilin (PCB) pigments, while CPC contains an additional high-energy β 153 PCB (Fig. 1(c)). If pigment arrangement governed ultrafast dynamics, the mutant CPC lacking β 153 would be expected to resemble APC.

Steady-state absorption and fluorescence, however, show that the β 153-deficient CPC retains CPC-like spectral profiles rather than shifting toward APC-like behavior. Removal of β 153 eliminates its absorption but leaves the α 84/ β 84 excitonic manifold essentially unchanged, indicating that pigment count and spatial layout do not primarily determine the energy structure. Ultrafast two-dimensional electronic spectroscopy (2D-ES) further demonstrates stark contrasts between APC and CPC. As shown in Fig. 1(d), APC undergoes rapid vibrational relaxation, funneled to the bottom of the excited-state potential within ~ 1 ps, and exhibits a well-defined 202 cm^{-1} vibrational coherence with 140–370 fs decoherence. In contrast, CPC—both wild-type (Fig. 1(e)) and β 153-deficient (Fig. 1(f))—shows broad, long-lived excited vibronic state and weak, short-lived coherence. These CPC-specific dynamics persist even when β 153 is removed, confirming that they originate not from pigment arrangement but from local pigment–protein interactions.

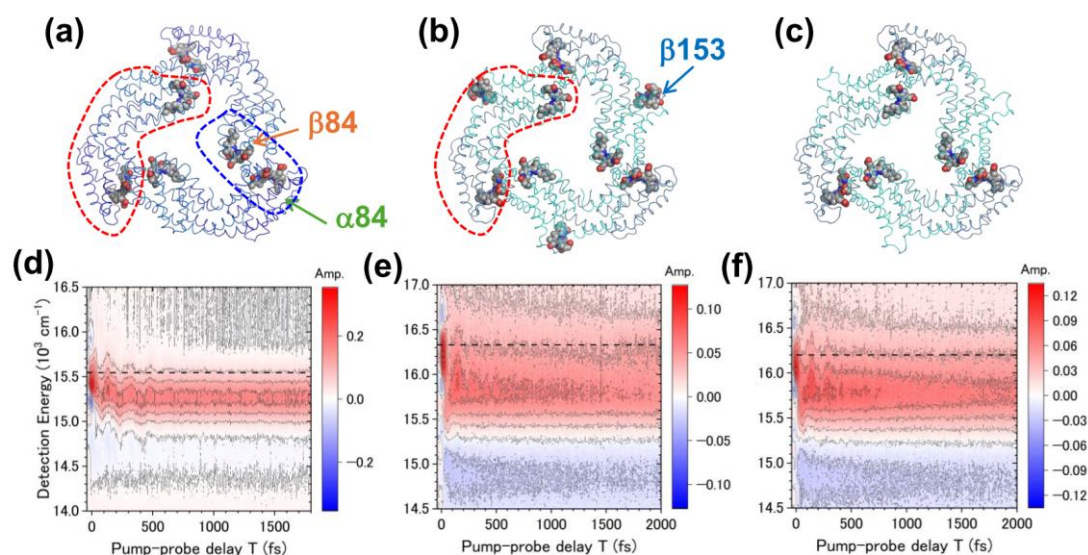


Fig. 1. Trimers structures of (a) wild-type APC, (b) wild-type CPC, and (c) β 153-deficient CPC. The red dashed lines indicate a monomer boundary. The blue dashed line surrounds a pair of pigments that form excitonic states. (d-f) Transient spectra rearranged from a series of time-resolved 2D electronic spectra at the excitation energies to the lower excitonic states. Panels (d), (e), and (f) are the spectra for wild-type APC, wild-type CPC, and the mutant CPC, respectively.

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Two-Dimensional Electronic Anisotropy to Reveal Isoenergetic Energy Transfer Dynamics in Photosynthetic Light-Harvesting Complexes

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In the photosynthetic photosystem II (PSII) of higher plants, the absorbed energy is transferred across a network of chlorophyll pigments, organized to efficiently direct excitation to the reaction center where photochemical charge separation occurs. Understanding excitation energy transfer (EET) within this highly interconnected network is essential for elucidating the structural and functional design principles underlying natural photosynthesis. However, a major difficulty lies in resolving EET pathways among numerous light-

harvesting components with structural similarity, leading to isoenergetic states across the complexes. This results in spectral congestion, making direct identification of EET pathways challenging. Here, we address this challenge using a general implementation of polarization-resolved two-dimensional electronic spectroscopy (2DES), in which the parallel and perpendicular pump-probe polarization spectra are acquired simultaneously [1]. While 2DES has proven powerful for probing EET in complex systems, polarization control provides additional information about orientational dynamics [2], useful for resolution of EET processes between isoenergetic states. The collected polarized 2D spectra can be combined to obtain the 2D relaxation of electronic anisotropy maps, from which we analyze the isoenergetic EET processes through the anisotropy decay kinetics. We demonstrate this approach on the M-CP₂ complex – an assembly the light-harvesting complex II (LHCII), CP29, and CP24 – at room temperature [3]. The anisotropy decay dynamics of M-CP₂ and isolated LHCII can be described using three (0.23 ps, 3.7 ps, and 24 ps) and two decay components (0.26 ps and 4.9 ps), respectively. Although their decay lifetimes are similar on sub-picosecond timescales, the few-ps component in M-CP₂ is faster and an extra decay component at 24 ps appears prominently (Figure 1). Structure-based calculations allow us to assign the 3.7 ps and 24-ps component to the inter-complex EET between the isoenergetic states of LHCII and CP29&CP24. Our results open up the possibility of monitoring the EET dynamics in large multichromophoric systems, such as the PSII supercomplex and even the photosystems *in vivo*.

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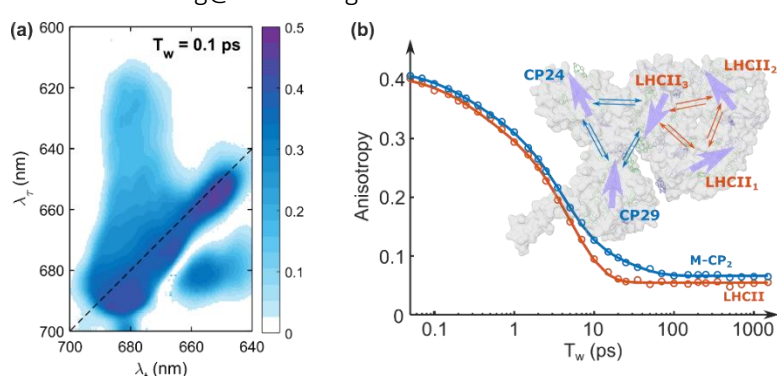


Fig. 1. (a) 2D anisotropy map of M-CP₂ at $T_w = 0.1$ ps. (b) Comparison of the anisotropy decays of M-CP₂ (blue) and LHCII (red), taken at 680/680 nm on the 2D anisotropy map in (a). The protein structure of the M-CP₂ complex is also illustrated, where the EET between the constituents are depicted by arrows.

Interchain structure around Li⁺ ion in Solid Polymer Electrolytes

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In search of better polymer material, we performed a comparative study between polyethylene (PEC) and polypropylene carbonate (PPC) based SPEs focusing Li⁺ solvation structure and Li⁺ transport properties employing FTIR and dielectric relaxation spectroscopy. Ionic conductivity enhancement with increasing lithium perchlorate (LiClO₄) concentration remained a common signature in both of these SPEs. Relatively more flexible Li⁺ solvation shell for PEC based SPEs owing to weaker Li⁺–O=C interaction compared to PPC results around one order faster polymer segmental motion and higher ionic conductivity than that of PPC based one. We used two-dimensional infrared spectroscopy to investigate the interchain structures of SPEs, poly (ethylene carbonate) and poly (propylene carbonate). We anticipate that the present experimental results with molecular understanding of Li-ion transport could guide to select potential polymer for SPEs. With the present experimental approach we infer LiClO₄-PEC can be better SPE candidate than LiClO₄-PPC.

Two-dimensional infrared (2D IR) spectroscopy has been used to study structural dynamics in various

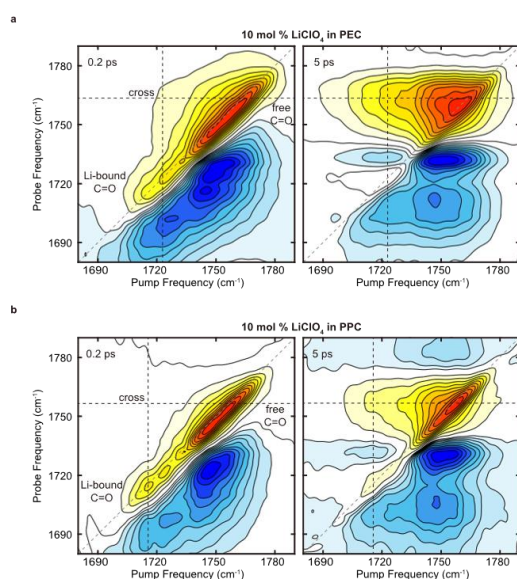


Fig. 1. 2D spectrum of polymer electrolytes. (a) LiClO₄ in PEC (b) LiClO₄ in PPC

molecular systems. In addition to the cross-peaks, the diagonal peak also provide rich structural information through the inhomogeneous frequency distribution. Despite the versatility and usefulness of the 2D IR approach, it is quite challenging to employ the technique to investigate ultrafast dynamics and relaxation processes in heterogeneous solid-state materials due to severe light scatterings.[1] In the present work, we carried out the 2D IR spectroscopy of polycarbonate-based SPEs to investigate interchain interaction between polymer chain in the SPEs consisting a series of lithium perchlorate (LiClO₄) concentration dissolved in polyethylene carbonate (PEC) (LiClO₄-PEC) and polypropylene carbonate (LiClO₄-PPC). Lithium salt concentration dependent enhancement of ionic conductivity in the polycarbonate based SPEs remain a curious phenomena. Furthermore PEC having similar carbonate interacting units like PPC differs by

addition methyl attachment in the monomer segment in PPC which is absent in PEC is reported to exhibit higher ion conductivity.[3] To provide a microscopic insight behind enhancement of ionic conductivity with increasing salt concentration along with the impact of polymer segment structure, we investigated these two interesting SPEs employing 2D-IR, FTIR, Raman and broadband dielectric relaxation spectroscopy.

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C17

Following electron transfer and confined water dynamics in Prussian Blue using electrochemical 2D IR spectroscopy

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Prussian Blue is not only the oldest synthetic pigment, but also an electrochemically active material with modern technological relevance for its electrochromic properties and applications in energy storage. We present a study of the fundamental mechanism of electron transport processes in this material and show how it varies as the material is progressively oxidized from Prussian White to Prussian Blue. Emerging methods in spectroelectrochemical 2D IR spectroscopy allow us to measure the electron transfer rate within a film of Prussian Blue deposited onto the working electrode in an electrochemical cell as a function of applied potential. The intrinsic CN stretching modes serve as a local probe of the Fe oxidation state and provide a method for studying the solvation dynamics induced by water molecules incorporated into the subcells of the zeolitic lattice. We observe a fast, ps-scale electron transfer process with a rate that varies with the state of the material, revealing the intrinsic mobility of electrons decoupled from the slow diffusion of K^+ ions. By correlating these observations to changes in the local structural distributions of Fe^{III} sites, K^+ ions and water molecules with the aid of a lattice model, we obtain insight into the mechanism of electron transport in this material. These results demonstrate a new method for observing fast electron transfer and correlating them with nuclear motions, and provide a new way to study chemical transformations at the electrochemical interface.

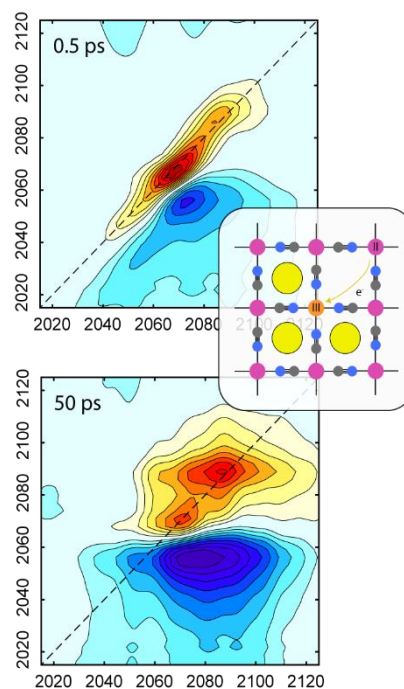


Fig. 1. Cross peaks report on fast electron transfer in electrochemically-synthesized film of Prussian Blue

Using Dual IR Reporters and 2DIR Spectroscopy to Map Interactions and Dynamics in Highly Heterogeneous Environments

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Highly selective, domain-specific infrared (IR) probes are essential for advancing vibrational spectroscopy in complex systems. However, a complete molecular picture often requires complementary molecular dynamics (MD) simulations. Here, we introduce a series of alkyne-derivatized molecules ($R-C\equiv C-R'$), with signal strengths comparable to or greater than those of

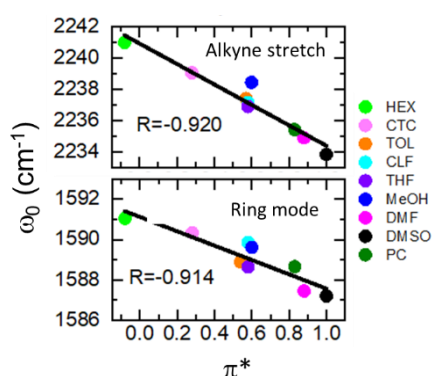


Fig. 1. Solvatochromism of the dual probe.

carbonyl probes. These molecules also contain carbonyl or phenyl groups, which serve as second IR reporters because their vibrational modes (carbonyl stretch and ring modes) are sensitive to the local environment.[1, 2] Two different dual probe molecules were designed for this purpose: one amphiphilic and one hydrophobic. These new probes demonstrate that their carbon triple bond stretch exhibits solvatochromic shifts. Moreover, the $C\equiv C$ frequency shifts are strongly correlated with solvent polarizability (π^*) as shown in **Fig. 1**. Theoretical investigations using the IFM method[3] and MD simulations demonstrate that, for both molecules, the triple bond response to the solvent environment is primarily driven by electronic polarization rather than directional interactions. Furthermore, 2DIR experiments reveal solvent dynamics-dominated viscosity rather than local interactions (**Fig. 2**). Modeling the solvation dynamics with the directly probes this dynamics. Finally, we tested the dual probes in the highly heterogeneous environment created by SDS micelles in water. Our results show that one probe preferentially localizes at the micellar interface, while the other selectively partitions into the hydrophobic core. Furthermore, the probe locations were confirmed solely through the two vibrational modes of the sample, independent of MD simulations. Two-dimensional IR measurements of the carbon triple bond stretch demonstrate that the probe in the micellar core observes faster structural fluctuations than the probe at the interface.

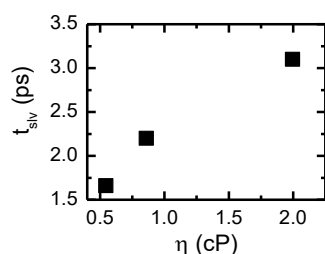


Fig. 2. Dynamics versus viscosity.

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How IR Vibrational Excitation Reshapes UV/Vis Electronic Spectra

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Vibrational pre-excitation has enabled novel spectroscopies and microscopies – such as VIPER 2D-IR, FEIR or BonFIRE – that exploit IR-induced changes in UV/Vis absorption to achieve vibrational selectivity for electronic excitation.[1-5] While these approaches rely on the perturbation of electronic spectra by IR excitation, they do not measure the induced spectral changes directly.

Here, we show that an in-depth understanding of IR pre-excitation effects can substantially enhance the performance of such methods. We demonstrate the possibilities of manipulating electronic spectra across a range of systems, including organic fluorophores, substrate–catalyst complexes, and proteins. The IR-induced spectral response is captured by broadband two-dimensional vibrational–electronic (2D-VE) spectroscopy [6] and interpreted with the support of computational spectroscopy.

Immediately after IR excitation, the effect on the electronic absorption band depends sensitively on the specific vibrational mode that is excited, with IR pumping capable of inducing either red or blue shifts of the UV/Vis absorption band. Accurate modelling of these effects requires going beyond linear vibronic coupling and accounting for Duschinsky rotation. Moreover, Fermi resonances between IR-active fundamentals and overtones of strongly vibronically coupled modes give rise to unexpected electronic responses. We further discuss how vibrational ladder climbing modulates the UV/VIS spectrum.

Allowing for variable delays between IR and UV/VIS excitation, vibrational dynamics provides additional handles for manipulating electronic spectra. Zero-quantum coherences lead to pronounced oscillations in UV/VIS cross sections, offering routes to signal enhancement and contrast generation. Intramolecular vibrational energy transfer (VET) enables indirect modulation of electronic spectra by exciting modes that are not vibronically coupled, while intermolecular VET allows selective differentiation of free versus complexed species, for example in photocatalytic systems.

By revealing and disentangling the mechanisms through which vibrational excitation reshapes electronic spectra, 2D-VE spectroscopy provides a quantitative basis for IR-controlled electronic excitation and spectroscopy. These insights open new opportunities for rational experiment design in photochemistry, optoelectronics, and biophysics.

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Quantitative Access to Molecular Excited State Properties of Conjugated Polyenes from Transient THz-Stark Spectroscopy and Theory

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The generally accepted paradigm for the excited-state ordering and dynamics of carotenoids involves a three-state model where in β -carotene the lowest-lying singlet excited state S_1 ($2A_g$) is optically forbidden from the ground state and direct excitation occurs to S_2 ($1B_u$) that is responsible for the strong absorption of blue-green light. The short-lived S_2 lifetime is associated with the rapid nonadiabatic interconversion between S_2 and S_1 .

Ultrafast THz-Stark spectroscopy [1,2] probes excited-state electronic structure via changes of the electronic absorption spectrum, which are induced by a coherent few cycle THz electric field and probed by femtosecond optical pulses on a time scale much shorter than molecular reorientation of chromophore and environment (time resolved Stark effect, Fig. 1, middle). The THz Stark spectra of β -carotene in apolar n -hexane at room temperature reveal a change of electronic polarizability $\Delta\alpha = 450 \pm 80 \text{ \AA}^3$, while changes of dipole moment are negligible [3]. The transient THz-Stark spectra suggest that structural inhomogeneities of the solvation shell play a minor role in the electric behavior of the molecular ensemble.

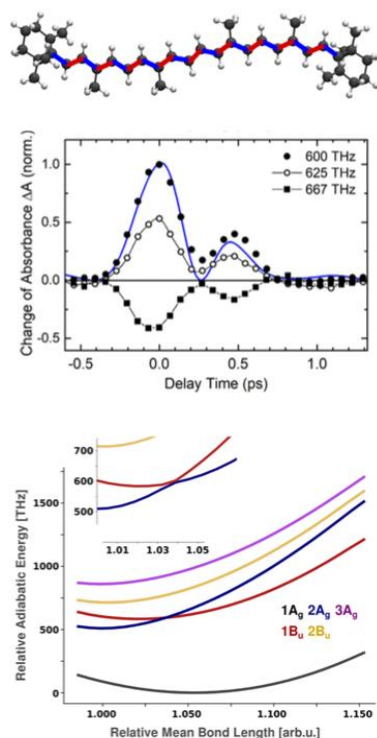


Fig. 1. Molecular structure (top) Time traces of absorbance changes ΔA as a function of pump–probe (middle) and potential energy surfaces of β -carotene (bottom).

Taking the experimental benchmark values, highest level XMS-CASPT2 ab-initio level of theory allows to quantitatively describe the excited state properties ($\Delta\alpha = 437 \text{ \AA}^3$). The simulations yield a strong dispersion of the excited state polarizability along an effective nuclear coordinate of single- and double bond bond-length-alternation in the π -system of β -carotene, i.e., strong non-Condon effects. In contrast to the established picture, the simulations yield an excited state ordering with a bright S_1 ($1B_u$) and a dark S_2 ($2A_g$) state and bond-length alternation drives S_1 relaxation via a conical interaction with S_2 (Fig 1, bottom). Detailed convergence studies show that erroneous state ordering is a result of the neglect of dynamic electron correlation, small basis sets and a particular sensitivity of the state ordering on the ground state geometry. Our results indicate that the feature of an optically accessible S_1 state is a common feature among carotenoids. E.g. in the polar and asymmetric peridinin, a commonly invoked CT state is ascribed to the large dipole moment changes upon relaxation of S_1 ($1B_u$), pointing to a common mechanism of excited state relaxation in carotenoids.

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A Versatile Mapping Program for Modelling Two-Dimensional Infrared and Electronic Spectroscopy from Molecular Dynamics Trajectories

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Coherent two-dimensional spectroscopies provide detailed insight into solvent fluctuations, intermolecular couplings, and structural dynamics. Yet, the resulting spectra are often congested and difficult to interpret, calling for computational approaches that enable molecular-level understanding and reliable spectral assignment. Simulation strategies combining molecular dynamics (MD) with nonlinear response theory are well established [1], and the CMDS community has developed robust methods to extract vibrational [2] and electronic [3] Hamiltonians from MD trajectories [4]. However, a unified and flexible framework that accommodates custom mapping schemes, different MD engines, or direct input from protein databank structures, while consistently supporting both vibrational and electronic excitations, has been lacking.

We present a new, general mapping framework that bridges this gap. The program constructs vibrational and electronic Hamiltonians from arbitrary MD trajectories using built-in or user-defined frequency and coupling maps. Implemented maps include infrared probes (amide I, OH-stretch) and electronic transitions of key photosynthetic pigments such as Chlorophyll a, Chlorophyll b, and Bacteriochlorophyll c. Standard mappings require no additional coding, while advanced schemes can be introduced through custom Python modules.

As an illustration, we simulate the 2DIR spectrum of methylthiocyanate [5] in acetonitrile (Fig. 1) by combining MD trajectories from GROMACS, Hamiltonian construction with the present framework [6], and nonlinear response calculations using NISE. Automated recognition of spectroscopic labels (e.g., thiocyanate) ensures straightforward application of the appropriate maps.

In the presentation, we will showcase additional vibrational and electronic examples, demonstrating the versatility, extensibility, and community-oriented design of the framework. The approach provides a streamlined and general route from MD simulations to predictive two-dimensional spectra, positioning it as a practical platform for next-generation CMDS studies.

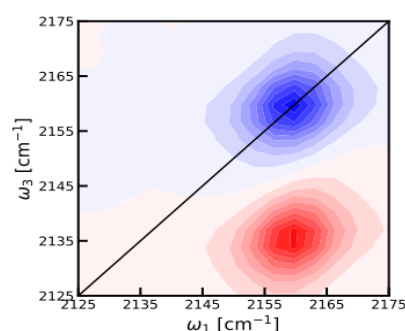


Fig. 1. Simulated 2DIR spectrum for zero waiting time and parallel polarization of methylthiocyanate in acetonitrile

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Unified System–Bath and Light–Matter Perturbation Theory for Nonlinear Optical Response

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Multi-dimensional spectra are characterized by features such as peak positions, amplitudes, and frequencies of peak evolution, which depend directly on the properties of the light-coupled degrees of freedom (DOF) of the system. These features are explicitly encoded in the double-sided Feynman diagrams (DSFD) of a given spectroscopic method. Spectral line shapes, however, reflect the dynamics of the environmental DOF (the bath), for example the dephasing between different electronic states. The bath DOF also drive irreversible processes in the light-coupled DOF that lead to redistribution of spectral features and line-shape evolution, but these processes are not captured by the diagrammatic classification.

Line-shape evolution crucially depends on the preservation of bath memory between the intervals of the nonlinear response, which is difficult to describe using traditional open quantum system (OQS) techniques [1]. The Hierarchical Equations of Motion (HEOM) enable calculation of nonlinear response while preserving bath memory, at the cost of losing the insight provided by individual Liouville pathways [2]. So far, the study of irreversible processes in terms of individual Liouville pathways has only been possible at a phenomenological level. For example, line-shape evolution accompanying rate-driven energy-transfer processes can be derived (see, e.g., [3]); however, bath (and therefore line-shape) evolution during the transfer step is neglected in these treatments. This neglect results from the

absence of an explicit representation of bath dynamics in traditional OQS methodologies.

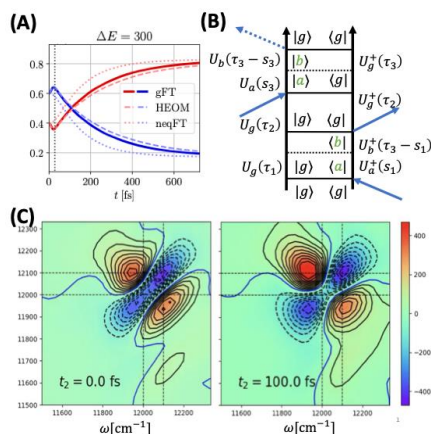


Fig. 1. (A) Transient coherent population dynamics in Förster type energy transfer simulated by HEOM (dashed), non-equilibrium Förster (dotted) and generalized Förster of Ref. [4] (full line). (B) Double-sided Feynman diagram including single inter-system perturbation. (C) 2D line shape evolution for the pathway from (B) involving transient contribution from (A).

In this contribution, we present a general methodology that consistently treats bath evolution accompanying OQS dynamics—resulting in transient coherent effects even in Förster-type energy transfer [4] (see Fig. 1)—and rigorously incorporates bath-driven relaxation into nonlinear optical response calculations. We formulate improved, spectroscopy-adapted master equations for propagation of the complete reduced density matrix in both weak and strong system–bath coupling regimes. We propose a method to consistently formulate and evaluate Liouville pathways involving zero, one, or two system–bath interactions per response interval (see Fig. 1B) and to properly weight their contributions. This approach enables evaluation of two-dimensional line shapes for unusual Liouville pathways reporting on various transient dynamical effects, estimation of the importance of different approximations in 2D line-shape calculations, and improved theoretical assignment of spectral features.

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System–Bath Partitioning and Emergent Quantum Resources in 2DES

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Here we discuss some of our recent two-dimensional electronic spectroscopy (2DES) models, based on a hierarchical equations-of-motion, phase matching approach [1].

Within an open quantum system framework, the system–bath boundary is chosen arbitrarily, and this choice may qualitatively change the resulting two-dimensional spectra. For example, in our Hamiltonian Vibration Model (HVM), vibronic states are explicitly included in the molecular Hamiltonian (the system), while in contrast, in the Bath Vibration Model (BVM), all vibrational structure is incorporated into the bath via a canonical transformation, described by an underdamped spectral density [2]. Beyond affecting the simulated spectra, this modelling choice also has important computational implications. In the HVM, the explicit vibronic structure requires a larger system Hilbert space, whereas in the BVM, the system Hilbert space is truncated; however, the introduction of an underdamped spectral density leads to a more detailed mathematical description of the bath, resulting in a significantly larger number of auxiliary density operators. Simulations show that the HVM produces 2D spectra with accurate peak positions, whereas the BVM is particularly well suited to modelling dynamic peak broadening. The optimal model therefore depends strongly on the intended application.

After introducing these models, I will discuss our latest two-bath model of a condensed-phase molecule comprising an undamped vibration coupled to an overdamped bath [3]. The dynamics are analysed using expectation values of bath coordinates to characterise dissipation and relaxation processes between multiple coupled baths. We show that information transfer occurs between the two baths even in the absence of direct bath–bath coupling. The final part of the talk will outline why bath–system–bath–mediated information transfer may be interpreted as a quantum resource.

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C24

Using Twin-Beams for an Instantaneous Detection of Quantum Coherences in 2D Electronic Spectroscopy

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Quantum coherences serve as key indicators of non-adiabatic dynamics in quantum materials and biological systems, but their clear identification continues to pose a significant challenge. Existing time-resolved spectroscopic techniques depend on temporal oscillation patterns that are typically convoluted with population-relaxation processes and experimental artifacts, which can obscure the true coherences. In this talk, I will outline a method that leverages macroscopic spectro-temporal entanglement to enable "snapshot" detection of quantum coherences within a system, bypassing the requirement of temporal scanning. We utilize twin beams generated by bright two-mode squeezing in 2D electronic spectroscopy to identify quantum coherences at a designated population time, thereby capturing the molecule's immediate coherence during its interaction with the probe pulse [1, 2]. Owing to entanglement, each coherence manifests at a unique point in a 2D frequency-dispersed signal, allowing the coherence period to be extracted directly from the spectrum without any temporal scanning. This approach provides a clear separation between population and coherence dynamics, enabling reliable identification of genuine quantum coherences in complex systems.

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Short-lived memory in multidimensional spectra encodes full signal evolution

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Ultrafast multidimensional (2D) spectroscopies are versatile tools that disentangle overlapping excitations in linear spectra and reveal the dynamics of charge and energy flow, chemical kinetics, and even many-body interactions in complex systems. However, measuring these spectra with high precision and accuracy is difficult. For example, converging the statistical noise in signals at long waiting times becomes exponentially expensive, and extended laser exposure heightens the probability of sample degradation. Overcoming these fundamental challenges has the potential to enable the investigation of systems that are too delicate for study under multidimensional spectroscopies and accelerate the progress of 2D spectroscopy-based microscopies that can offer highly resolved excitation dynamics with spatial resolution over heterogeneous environments. We address this problem by developing a new dimensionality reduction technique, the spectral generalized master equation (GME), that enables researchers to employ short-waiting time 2D spectra to determine and denoise the full evolution of 2D spectra over arbitrary waiting times with high temporal resolution (Fig. 1). We demonstrate the applicability of our approach on theoretically generated and experimentally measured 2D electronic and 2D infrared spectra. We further show that our spectral GME cuts the reduces of experiments by *multiple orders of magnitude* and accurately removes statistical noise while circumventing the increasing convergence costs with longer waiting times.

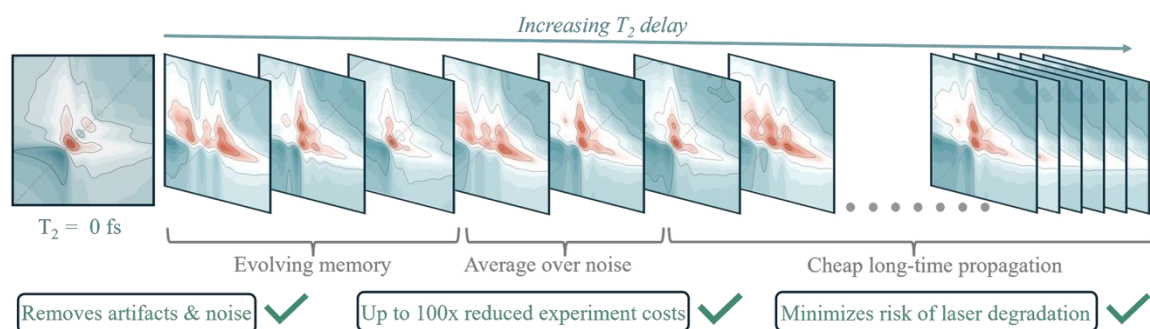


Fig. 1. Schematic of the spectral Generalized Master Equation.

Multidimensional Coherent Spectroscopy of Vibrational Correlations in Layered Organic-Inorganic Perovskites

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Layered organic-inorganic metal halide perovskites offer an attractive platform for optoelectronics, as strong excitonic effects create bright radiative emission from mobile excitons, which reside predominantly in inorganic metal-halide quasi-2D layers [1-2]. The end-groups of the organic cations nestle close to the metal halide atoms, forming hydrogen bonds between a cation (*e.g.* ammonium) and the halide ions. Here we show that a wealth of information about the local interatomic potential near the metal-halide sheets can be obtained using multidimensional infrared spectroscopy (2DIR) to study the anharmonic, coupled modes of ammonium end groups in thin films of the layered perovskite phenylethylammonium lead halide (PEA₂PbX₄) [3].

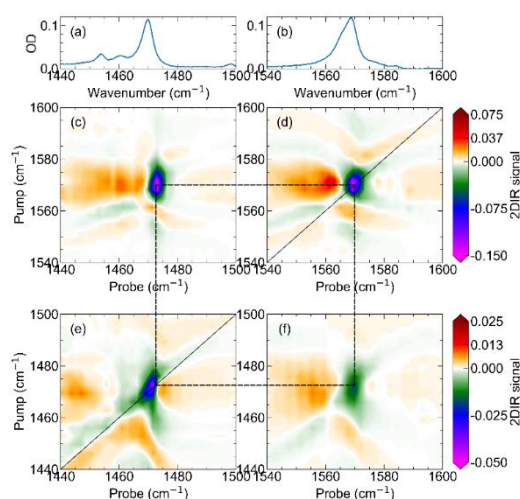


Fig. 1. 2DIR on PEA₂PbI₄ around the A₁ (1470cm⁻¹) and E modes (1570cm⁻¹) of the ammonium end groups, for perpendicular pump and probe beams. Panels (a)-(b) show FTIR absorbance; (c)-(f) show respectively +-, ++, -- and -+ pump/probe 2DIR signals at 1ps population time, where + and - refer to the upper and lower exciton. Excited state absorption peaks (ESAs) are red-orange while ground state bleaches are purple-green. anharmonicity and coupling between two (or more) modes. We stress-tested the deduced fit parameters (coupling strengths, anharmonicities, dipole orientations and magnitudes) using a Monte Carlo approach to check for unwanted correlations between parameters. The vibrational Hamiltonian's parameters were obtained for different local environments and for each halide, showing changes with crystal structure and H-bonding environment. Our findings reveal the diverse and prominent roles played by the organic cations in layered perovskites.

Samples of PEA₂PbX₄ with X=I, Br, Cl were examined via 2DIR using the “LIFETIME” experiment at the STFC ULTRA Facility, (Harwell, UK). Spectrograms were obtained with the pump tuned to either the A₁ symmetric umbrella mode of the ammonium end group at 1470cm⁻¹, or to the degenerate E modes at 1570cm⁻¹. Clear cross-peaks are evident with relatively strong intensity in the perpendicular pump/probe channel (figure) as well as for parallel pump/probe polarisations. Further, in the -Br and -Cl samples, the A₁ mode blueshifts and the E modes lift in degeneracy, while splitting into two sets of linked modes indicative of the two uniquely different local environments for ammonium. The anisotropy of the pump-probe signal's dynamics was further investigated, revealing different anisotropy lifetimes for the upper and lower excitons.

We interpreted the spectra quantitatively by fitting to predictions from the vibrational exciton model, which constructs an effective vibrational Hamiltonian including parameters (coupling strengths, anharmonicities, dipole orientations and magnitudes) using a Monte Carlo approach to check for unwanted correlations between parameters. The vibrational Hamiltonian's parameters were obtained for different local environments and for each halide, showing changes with crystal structure and H-bonding environment. Our findings reveal the diverse and prominent roles played by the organic cations in layered perovskites.

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Frequency-Resolved Quantitative Lineshape Analysis in Two-Dimensional Coherent Spectroscopy

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Lineshape analysis of the spectra measured in two-dimensional coherent spectroscopy (2DCS) has been instrumental in understanding coherent optical properties of excitons in semiconductor nanostructures [1]. A unique advantage of 2DCS vis-à-vis other methods is that the homogeneous response of an inhomogeneously broadened resonance can be measured in a single rephasing spectrum. Still, a generalized quantitative analysis of frequency-dependent homogeneous properties has not been reported. Here, we demonstrate such a method for excitons in semiconductor quantum wells (QWs).

We define the four-wave mixing (FWM) signal of the inhomogeneous distribution as a convolution of the homogeneous response with a bivariate spectral distribution function (BSDF) based on the inhomogeneous distribution. This method allows us to numerically incorporate an arbitrary inhomogeneity in the calculation of the FWM signal. We can use this calculation to perform a numerical fitting of measured 2D spectrum to obtain an arbitrary inhomogeneous linewidth [2].

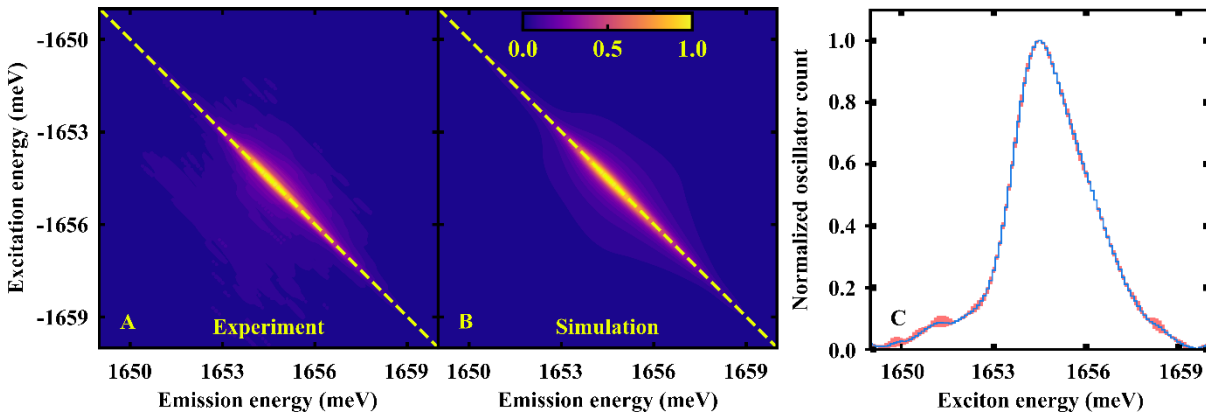


Fig. 3. A. Experimental 2D rephasing spectrum showing the QW-exciton resonance in a GaAs interfacial quantum dot at 5 K. B. Simulated rephasing 2D spectrum generated using the best-fit oscillator distribution and homogeneous linewidth. C. Extracted oscillator distribution shown in blue reveals a non-Gaussian inhomogeneous distribution. Figure adapted from [2].

Figure 1 demonstrates the excellent correspondence between the measured (Fig. 1A) and simulated (Fig. 1B) spectra based on the best-fit parameters. The obtained inhomogeneous linewidth is shown in Fig. 1C. Here, we have assumed that the homogeneous linewidth is constant across the inhomogeneous distribution. While reasonable in this case, this approximation is not generally valid. For example, coherence decay rates for localized and delocalized excitons in semiconductor QWs are different [3].

We have extended the aforementioned analysis, using machine-learning techniques, to incorporate smooth, but arbitrary variation of the homogeneous linewidth with the exciton energy. We can, thus, measure the frequency-resolved homogeneous linewidth along with the arbitrary inhomogeneous lineshape from a single 2D spectrum. This generalized procedure, may further be extended to quantify other phenomena such as spectral diffusion directly from the measured 2D spectra.

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C28

Revealing Ultrafast Excited-State Dynamics of Quantum Dots by 25 kHz Shot-to-Shot Phase-Cycling 2DES

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Two-dimensional electronic spectroscopy (2DES) has emerged as a powerful tool for disentangling excitonic energy landscapes and resolving ultrafast excited-state dynamics in quantum-confined systems. However, its application to quantum dots is often challenged by weak nonlinear signals, spectral congestion, and sensitivity to laser and sample instabilities. In this report, we present a high-repetition-rate 25 kHz shot-to-shot phase-cycling 2D electronic spectroscopy platform [1] that enables robust, background-free measurement of coherent and incoherent excitonic responses in colloidal quantum dots. By implementing rapid phase modulation and single-shot detection, this approach efficiently suppresses scatter and noise contributions while preserving femtosecond time resolution and high spectral fidelity. The resulting 2D spectra directly reveal the exciton energy structure, including spectral signatures of fine-structure splitting, and allow us to track population relaxation, coherence dephasing, and intraband energy redistribution on ultrafast timescales. These measurements provide detailed insight into the interplay between quantum confinement, Coulomb interactions, and environmental coupling in quantum dots. Our results demonstrate that high-speed shot-to-shot phase-cycling 2DES offers a versatile and sensitive route to probing complex excitonic dynamics in nanoscale semiconductors, opening new opportunities for understanding and controlling quantum dot photophysics relevant to optoelectronic and quantum technologies.

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C29

Bridging Molecular and Macroscopic Timescales: 2D IR Imaging of Copper Redox Dynamics in Ionic Liquids

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Electrochemical cells rely on redox reactions at electrode–electrolyte interfaces, where electrode composition strongly influences reaction pathways and stability. Copper is widely used as a current collector, yet its interfacial redox dynamics in ionic liquid electrolytes remain poorly understood on ultrafast timescales. Traditional electrochemical techniques provide macroscopic insight but lack the temporal resolution to capture these dynamics. Here, we combine two-dimensional infrared (2D IR) imaging with theory to bridge molecular-scale dynamics and macroscopic electrochemical measurements.

Using time-dependent 2D IR imaging, we probe ultrafast vibrational dynamics and chemical equilibria in an ionic liquid containing dicyanamide (DCA[−]). Although copper(I) is typically unstable in solution, DCA[−] stabilizes this oxidation state, enabling the formation of both copper(I) and copper(II) DCA complexes under applied potential.[1] Cross-peak analysis reveals dynamic equilibrium and migration of these species near the oxidizing electrode, permitting direct observation of reaction pathways during device operation. These results provide molecular-level insight into copper redox chemistry in ionic liquid electrolytes and demonstrate the capability of 2D IR imaging for studying operating electrochemical systems. Finally, we present a promising approach based on a spectral generalized master equation framework that dramatically reduces image acquisition time.[2]

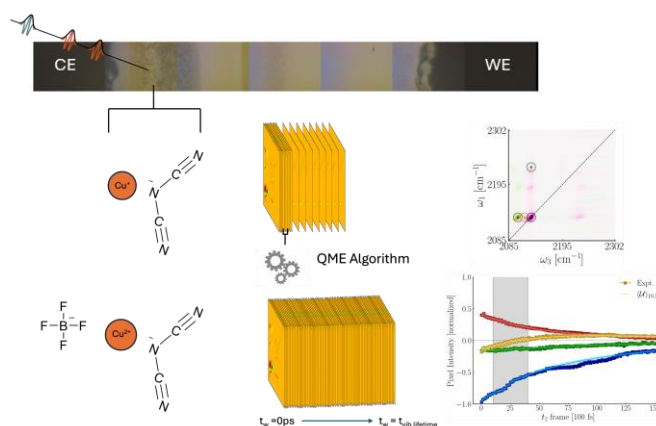


Fig. 1. 2D IR imaging across a copper electrochemical cell is performed in a point scanning approach. 2D IR spectra are collected as a function of T_w . Image acquisition times are reduced by applying a spectral generalized master equation framework along T_w .

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CO₂ Jumps Through Polymerized Ionic Liquids

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Polymerized ionic liquids are a platform to interrogate the physical processes responsible for the separation of CO₂ from other light gases and to develop chemical principles for the design of energy efficient carbon capture materials. 2D-IR spectroscopy and molecular modeling are powerful tools to interrogate CO₂ in these complex environments [1-4]. Our central hypothesis is that CO₂ is solvated in a quadrupolar solvation shell, and that by modifying this shell we can modulate the selectivity of CO₂ separation membranes. Chemical synthesis provides tunable materials with controllable mobility of ions. Polarization-resolved 2D-IR spectroscopy determines the femtosecond to hundreds of picosecond timescales of molecular motion and validates molecular models, in-situ IR spectroscopy determines the macroscopic diffusion coefficients, and the molecular models provide atomistic insights. The combination of these tools reveals that CO₂ moves through polymerized ionic liquids via rare, long-distance jumps.

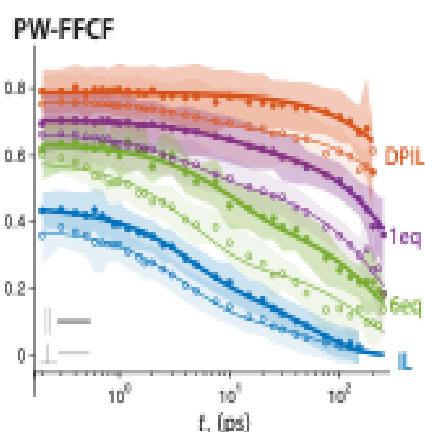


Fig. 1. PW-FFCFs of CO₂ in doubly-polymerized ionic liquid/ionic liquid blends

As the fraction content of free ionic liquid increases, the relaxation time of the polarization-weighted frequency fluctuation correlation function (PW-FFCF) gets faster (Fig. 1). Analysis of the reorientation-induced spectral diffusion (RISD) and structural spectral diffusion (SSD) capture the timescales of molecular wobbling in the local solvent cage and the cage's reorganization. A wobbling in a cone model provides estimates of the size of the solvent cavities. The observed vibrational frequency shifts report the numbers of oxygen and fluorine atoms in the first solvation shell of the CO₂.

Spectroscopic models based on polarizable and non-polarizable force fields reproduce the spectroscopic observables semi-quantitatively. Vibrational frequency shifts, orientational correlation functions, spectral diffusion timescales, and translational diffusivities are compared

critically in ionic liquids. Polarizable and non-polarizable force fields bracket the observed densities. The two models perform similarly reproducing the orientational correlation functions and frequency fluctuation correlations. The CO₂ bending mode is incorporated into the vibrational frequency through either its average or instantaneous effect. Using the average effect results in a lineshape that is too narrow and dynamics that are too slow; using the instantaneous effect results in a lineshape that is too broad and dynamics that are too fast.

The MD simulations reveal that most of the CO₂ diffusion originates in long-range jumps from one quadrupolar solvation shell to the next. Prospects for modifying the polymerized ionic liquids to increase the diffusion coefficient of CO₂ will be discussed.

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2D IR Study of Conformational Thermodynamics (ΔH and ΔS) and Fermi Resonance in the *syn*- and *anti*-Isomers of Acetic Acid in Acetonitrile

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Acetic acid in a low-temperature rare-gas matrix shows two distinctive peaks in its FTIR spectrum, corresponding to two conformational isomers, *syn*- and *anti*-isomers. In aprotic acetonitrile, however,

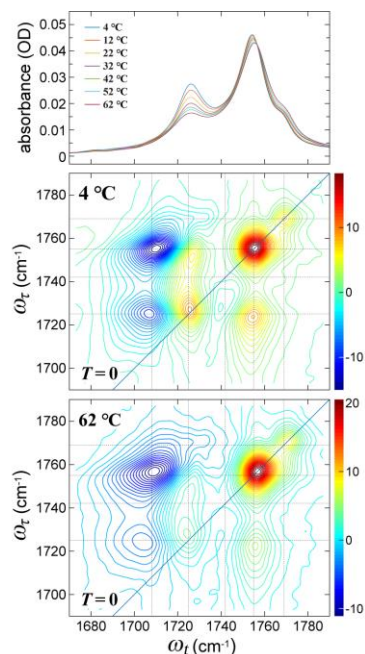


Fig. 1. (top) Temperature-dependent FTIR spectra of CH_3COOH in CH_3CN . (Middle and bottom) 2D IR spectra at 4 °C (middle) and at 62 °C (bottom).

the IR spectrum in the C=O stretch region shows a multi-band pattern that cannot be clearly understood by conventional FTIR spectroscopy. To clarify the origin of this complexity, two-dimensional infrared (2D IR) spectroscopy is applied. 2D IR studies reveal that the complicated spectral feature arises mainly from the coexistence of *syn*- and *anti*-isomers, together with Fermi resonance between the C=O stretch and the CH_3 rocking overtone in each isomer. Numerical 2D IR simulations based on a local-mode Hamiltonian model, incorporating Fermi resonance, reproduce the experimental results with high accuracy. From these simulations, the relative populations of the *syn*- and *anti*-isomers at 22 °C are determined to be 0.935 and 0.065, respectively. The Fermi coupling constants are found to be -13.5 cm^{-1} for the *syn*-isomer, and -10.5 cm^{-1} for the *anti*-isomer, and the corresponding vibrational energy levels are also quantified.

Time-dependent 2D IR investigation of deuterated acetic acid indicates that the two isomeric forms do not interchange on the 10 ps timescale. In addition, aggregation leads to the appearance of an extra spectral feature at concentrations above 20 mM.

Temperature-dependent 2D IR experiments, performed at seven temperatures between 4 and 62 °C and supported by corresponding numerical simulations, are used to determine the Gibbs free energy difference between the two isomers at each temperature. A van't Hoff plot of the temperature dependence of the Gibbs free energy difference yields thermodynamic parameters for the conformational equilibrium, giving $\Delta H = 6.3 \text{ kJ}\cdot\text{mol}^{-1}$ and $\Delta S = -0.6 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$.

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C32

Salt-pellet 2D-IR spectroscopy: a helping hand for vibrational spectroscopy of inorganic solids?

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Vibrational spectroscopy plays an important role in characterising inorganic solids in applications ranging from terrestrial and Martian geology to catalysis and materials science. For disordered materials (inaccessible to diffraction methods), IR spectroscopy is often a key investigative tool. Intermolecular interactions add substantial complications to the assignment and prediction of solid phase vibrational spectra however and for crystalline substances, it is not always clear why the IR absorption spectra look the way they do. As 2D-IR spectroscopy's forte is to reveal inter and intramolecular vibrational mode couplings, a combination of curiosity and recent progress in measuring scattering solids led us to survey salt-pellet dispersions of powdered-crystalline and disordered carbonates, nitrates and phosphates. Data for polymorphs of CaCO_3 suggest that 2D-IR may be able to contribute to the ongoing debate about why the spectrum of the asymmetric stretch band of calcite looks the way it does. In this presentation, we share some of our insights into solid state inorganic 2D-IR spectroscopy, appealing to both experiment and to Density Functional Theory simulations in harmonic and anharmonic regimes. We also discuss some issues and possible pitfalls in salt-pellet 2D-IR spectroscopy.

Polarization-controlled 2DES for extracting Q-band relaxation in chlorophyll *a*

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2DES is likely the most informative spectroscopy technique for studying molecular systems, as it provides detail information about spectral correlation and dynamics. However, it is also highly susceptible to artefacts, in particular, in the very short population time region, where excitation pulses overlap. Here the main “undesired” contributions come from self-phase modulation in the solvent and optical cell walls, also from incorrect pulse ordering signals. As a result, spectral features and dynamics are distorted, which in general prevent from measuring sub-50 fs dynamics. The notorious example is excitation energy relaxation the Q band of chlorophylls, which was predicted to proceed on a 50 fs time scale [1]. Despite the four decades of efforts (see for example [2,3]), it has never been determined with good accuracy for any of chlorophyllides.

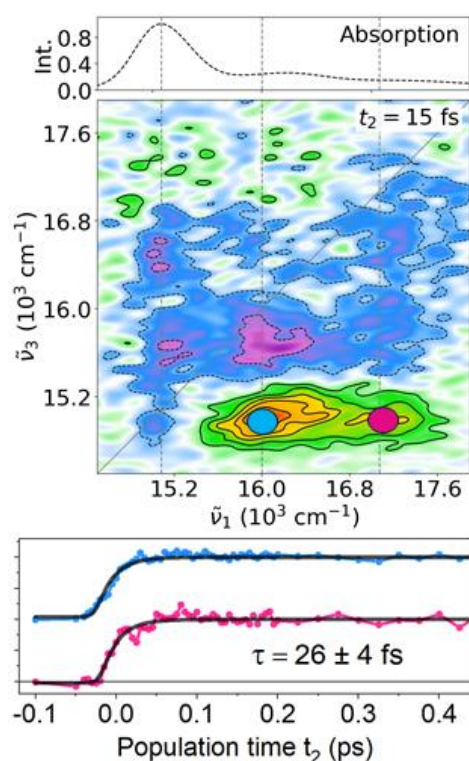


Fig. 1. CPSPD polarization-controlled 2D spectrum of Chl *a* in butanol at room temperature. The right side shows extracted cross-peaks dynamics reporting on ~25 fs Q band relaxation.

Already a quarter century ago it was realized that using polarization control in two-dimensional spectroscopy can help to disentangle congested signals [4]. In particular, the polarization scheme [60°, -60°, 0°, 0°], was suggested and demonstrated to suppress strong contributions in the 2D spectra to better reveal cross-peaks. We implemented another polarization sequence [45°, 90°, 45°, 0°], which we call “cross-peaks and short-time-dynamics specific” (CPSPD) for 2DES, which is highly effective at suppressing most of the mentioned artifacts. If implemented with care, it completely suppresses the self-phase modulation signals and dramatically reduces incorrect pulse ordering contributions. The only remaining “undesired” signals come from impulsive solvent Raman contributions, which can be mitigated by a careful choice of the solvent. CPSPD polarization scheme allows to reliably track dynamics even in the pulse overlap region, the only limiting factor being the length of the pulses, which in presented measurements were 9 fs long.

We used CPSPD polarization scheme to study chlorophyll *a* (Chl *a*), which is the most abundant pigment in oxygenic photosynthesis. This polarization scheme enabled us to reveal pure cross-peak dynamics, corresponding to the relaxation between vibronically excited and mixed Q_v and Q_x states to the lowest-energy Q_v state. Dynamics appear mostly clear from the artifacts, and the fitting procedure

allowed us to determine that the relaxation time between these states is 26±4 fs. This provides the benchmark result for various theoretical modeling approaches aiming to describe vibronic coupling and mixing in Chl *a* molecule, which are necessary for understanding very complex and rich photophysics in these biomolecules. The demonstrated method is general, and because of its selectivity can be used to extract valuable sub-100 fs dynamics information in various spectrally-congested systems.

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Conformational Switching Modulates Excited-State Pathways in a Cofacial Perylene Dimer

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Controlling excited-state pathways in supramolecular chromophore assemblies is key to designing next-generation optoelectronic and photonic materials. Here we employed half-broadband two-dimensional electronic spectroscopy (HB2DES) [1] to elucidate the conformation-dependent ultrafast photophysics of a flexible perylene diimide (PDI) dimer, **valPDI₂**, which undergoes reversible solvent-driven switching between two distinct dimer geometries (Fig. 1). [2]

Exciton-coupling calculations and ultrafast spectroscopy show that in CHCl₃ **valPDI₂** adopts an open, weakly coupled geometry that supports slow, partial excimer formation due to structural inhomogeneity within the excited state potential energy surface. In contrast, in polar 1:1 DMSO/water **valPDI₂** collapses into a cofacial stacked conformer that enables barrierless, sub-200 femtosecond excimer formation, a subset of which forms a multiexciton state over tens of picoseconds. HB2DES beatmaps reveal conformation-dependent vibrational coherences, with out-of-plane nuclear motion along the π -stacking coordinate promoting vibrationally coherent excimer formation in the closed conformer.

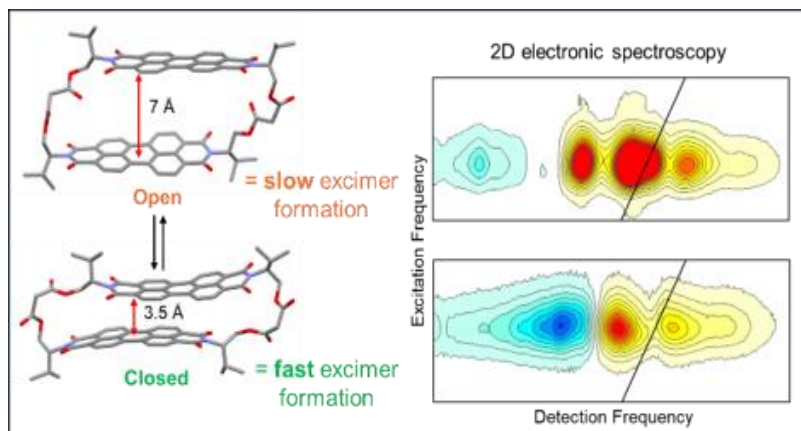


Fig. 4 (left) “open” and “closed” conformers of **valPDI₂** populated in CHCl₃ and in DMSO/water, respectively. (right) absorptive HB2DES spectra of “open” and “closed” **valPDI₂** at T = 0.2 ps.

These findings demonstrate that environmentally driven conformational control offers a powerful non-covalent strategy to modulate “mixed state” [3] dynamics in PDI assemblies. More broadly, they establish supramolecular switching as a general design principle for tuning excited-state behaviour in flexible organic chromophore arrays, with implications for the development of responsive optoelectronic and energy-conversion materials.

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Open quantum system approach to dynamics and spectroscopy in the presence of strong light-molecule coupling

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We present our recent theoretical studies of strong light–matter interaction systems in both microcavity and plasmonic nanocavities, focusing on polariton and plexciton dynamics and nonlinear spectroscopy.

In the first part, we develop path integral formulations for the Dicke and Tavis–Cummings models in the thermodynamic limit, where the collective molecular response can be effectively described by an equivalent harmonic bath. This framework allows for a unified treatment of static disorder and vibronic coupling, enabling efficient simulations of absorption spectra and polariton dynamics using both the hierarchical equations of motion (HEOM) and exact diagonalization. We also employ the Tavis–Cummings model and open system quantum dynamics approach to study the pump-probe and two-dimensional (2D) spectra of molecular polaritons. It is shown that the nonlinear response converges with a few tenth of molecules in the model Hamiltonian. Rabi oscillations are present in the strong coupling regime, and the bright state lifetime is primarily determined by the dephasing time of the molecules.

In the second part, we extend the HEOM approach to model plexciton systems, where a two-level exciton is coupled to a short-lived plasmonic mode. Our simulations reveal a transition in the absorption spectrum from an asymmetric Fano profile to a Rabi-split doublet with increasing coupling strength, while vibronic coupling acts to symmetrize the line shape. Simulated two-dimensional electronic spectra further uncover nonlinear features unique to plexciton systems, including a distinctive breathing mode pattern that serves as a spectroscopic signature of Rabi oscillations.

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Tracking relaxation dynamics of polaritons and reservoir states in organic exciton-polaritons

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Organic exciton-polaritons have garnered significant attention in optoelectronic devices and photochemistry, primarily due to their micrometer-scale diffusion lengths and tunable energy levels^[1]. Recent studies^[2-4] have underscored the critical role of reservoir states in the dynamical processes following polaritonic relaxation. However, questions remain regarding the peak assignments of reservoir states and their specific functions during relaxation. In this study, we employed pump-probe and two-dimensional electronic spectroscopy (2DES) coupled with model simulations to investigate photoexcited dynamics of organic exciton-polaritons^[5]. By comparing experimental data with simulations, we identified that reservoir states consist of dark states and excited-state molecules uncoupled to the cavity mode, each exhibiting distinct spectral characteristics. Analyses of the spectral evolution in 2DES reveal that reservoir states are generated through three pathways: direct photoexcitation, rapid relaxation from middle polaritons, and entropy-driven relaxation from lower polaritons. Moreover, we find that equilibrium is established between dark states and uncoupled excited-state molecules following polaritonic relaxation, regardless of the excitation wavelength. This equilibrium facilitates subsequent polaritonic regeneration. Our recent efforts have focused on modulating the equilibrium constant via engineered light-matter interactions, an approach that constitutes a novel strategy for controlling post-relaxation polaritonic dynamics. These insights advance our understanding of relaxation dynamics in organic exciton-polaritons that can benefit the design of next-generation optoelectronic devices.

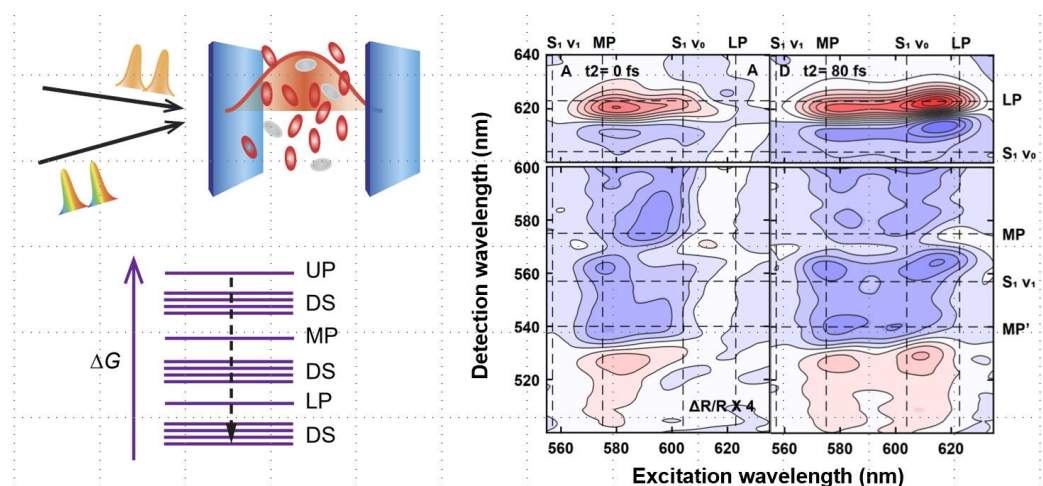


Fig. 1. Two-dimensional electronic spectroscopy probes relaxation dynamics of polaritons and reservoir states

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Ultrafast collective multimode Rabi oscillations in an open cavity probed by two-dimensional electronic spectroscopy

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Ultrafast two-dimensional electronic spectroscopy (2DES) has proven to be a powerful tool for unraveling coherent polariton dynamics [1,2], opening up new opportunities for ultrafast switching of strong light-matter interactions [3]. So far, however, short, tens-of-fs polariton dephasing times at room temperature set serious limitations for such applications. Here, we seek to overcome these challenges by strongly coupling excitonic emitters to the Fabry-Pérot (FP) modes of open cavities [4] which provide flexible control over the cavity length and thus mode number and detuning.

We report the first 10-fs 2DES experiments on a tunable, open FP cavity (Fig. 1a) hosting few-nm thick layer of molecular J-aggregates. We uncover rich multimode Rabi oscillations persisting for hundreds of fs at room temperature. Exemplarily, Fig. 2b shows a 2DES map recorded in the multimode regime for a cavity length of $\sim 6.5 \mu\text{m}$ and waiting time of 40 fs. The 2DES map shows a rich, square-grid pattern of multiple interacting nonlinear polariton peaks with line widths of $\sim 5 \text{ meV}$. The central FP mode is tuned in resonance with the J-aggregate excitons ($\sim 1.6 \text{ eV}$) and displays a $\sim 25 \text{ meV}$ normal mode splitting. A dispersive lineshape of the resulting polariton peaks suggests that the dominant nonlinearity arises from the two-exciton blue shift of the aggregates [1]. Coherent multimode couplings are reflected in pronounced amplitude oscillations of the individual 2DES peaks, persisting for several 100 fs (Fig. 1c), much longer than the bare exciton dephasing time [1]. The coherent Rabi oscillations, often appearing with periods that are an integer fraction of the cavity round trip time, are superimposed on slower incoherent polariton relaxation dynamics. Experiments for various cavity lengths are presented and interpreted in terms of multimode Tavis-Cummings Hamiltonians.

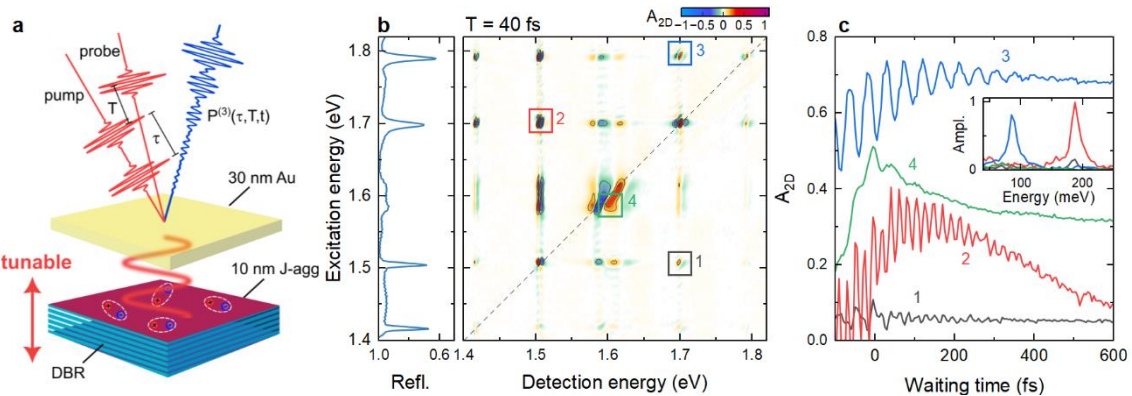


Fig. 1. Open cavity 2DES. a) Scheme of tunable Fabry-Pérot cavity with a bottom mirror (DBR) coated with a J-aggregate thin film and a top gold mirror. 2DES is performed in partially collinear reflection geometry close to normal incidence. b) 2DES spectrum at $T=40 \text{ fs}$ (right) with polariton diagonal and cross peaks and linear reflectivity (left). c) Dynamics at a few selected points showing long-lived multimode Rabi oscillations with different oscillation periods (inset for Fourier transforms of the oscillatory components) on top of incoherent inter-mode energy transfer. Traces are shifted for clarity.

In essence, the demonstrated combination of ultrafast 2DES and tunable open cavity spectroscopy uncovers an intriguing platform for exploring and exploiting multimode Rabi oscillations and Floquet physics in nonlinear polaritonic systems with long coherence times at room temperature.

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Correlating Vibrational Dynamics with Vibrational Strong Coupling Reaction Modification

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Polaritonic chemistry offers a fundamentally new paradigm to modify chemical reaction landscapes, from altering rate constants to tuning equilibrium constants, simply by embedding a reaction within a vibrationally resonant optical cavity capable of establishing vibrational strong coupling (VSC) [1]. Despite several reports of polaritonic chemistry, the lack of a theoretical explanation limits the rational design of new polaritonic reactions. Simpkins *et al.* studied the reaction of phenyl isocyanate (PHI) with cyclohexanol (CH) in THF solution (Fig. 1), finding that vibrational strong coupling can slow the reaction rate significantly [2]. Notably, that work emphasized experimental reproducibility, which has been a challenge in the field of VSC-modified chemistry.

Following Simpkins *et al.*, we have taken an approach inspired by traditional physical organic chemistry to investigate the consequences of modifying functional groups on a molecular scaffold, as well as the surrounding solvent. We compare various substitutions on the phenyl ring, which we find to influence the degree of VSC-modification. The goal of this work is to correlate the vibrational dynamics of the strong ν_{NCO} transition, which is the most effective mode for VSC, with the urethane formation reaction's VSC susceptibility. Among the available theoretical treatments of VSC-modified chemistry, there is a role for vibrational redistribution through the collective coupling in the polaritonic states. Alternatively, a statistical mechanics perspective suggests that the renormalization of the vibrational partition function is the origin of the VSC modification. We anticipate that detailed structure-function relationships coupled with ultrafast vibrational spectroscopy will enable progress in advancing a theoretical foundation of polaritonic chemistry.

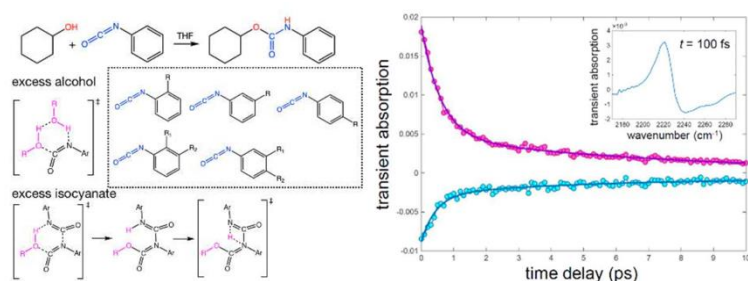


Fig. 1. (left) Reaction found to exhibit VSC modification proceeds by two different reaction mechanisms depending on the local microsolvation. (right) Transient IR absorption of the isocyanate transition provides access to IVR and vibrational relaxation dynamics.

In addition to assessing VSC-controllability, we use 2D-IR spectroscopy to address key mechanistic questions that have been identified in the broader chemical community but that have not been addressed in the context of the polaritonic reaction. The reaction between PHI and CH proceeds through two different transition states depending on the local concentration of PHI and CH [3]. When PHI is in excess, PHI essentially catalyzes the reaction, whereas when CH is in excess, the CH catalyzes the reaction. Thus, microsolvation plays an essential role in the reaction mechanism. Based on our molecular dynamics simulations, over a rather wide range of relative PHI/CH concentrations, there are no conditions where only one mechanism is likely to predominate. Spectral diffusion, as revealed by 2D-IR, enables a dynamical perspective on the pre-reactive complexes in solution. This information is essential in advancing a rigorous theoretical interpretation because it is likely the two mechanisms are not modified in the same way by VSC.

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Operando CMDS of Carrier Cooling in Perovskite Solar Cells

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Perovskite solar cells have been at the forefront of intensive research for over a decade [1]. A deeper understanding of carrier cooling and many-body interactions in metal-halide perovskites can provide new strategies to further enhance photovoltaic performance. Ultrafast spectroscopy has revealed sub-picosecond carrier relaxation dynamics in thin films; however, direct operando insight within fully functional devices remains scarce. Here, we implement action-detected two-dimensional electronic

spectroscopy (A-2DES) under photovoltaic operating conditions, enabling coherent multidimensional spectroscopy (CMDS) to be performed directly on working perovskite solar cells [2].

We investigate devices based on FA, FAMA, and MA perovskite absorber compositions and record 2D spectra within initial 2 ps population time. The multidimensional maps reveal pronounced composition-dependent differences in carrier cooling and spectral evolution. The spectral reshaping reflects carrier thermalization toward band-edge states. We model the 2D lineshapes using a semiconductor Bloch framework including many-body interactions and carrier-phonon scattering. The simulations well reproduce the experimentally observed dynamics, enabling quantitative comparison of cooling dynamics and dephasing across compositions.

These results demonstrate that CMDS can be applied under realistic operating conditions and establish operando A-2DES as a powerful approach for correlating ultrafast carrier physics with photovoltaic device behavior in perovskite solar cells.

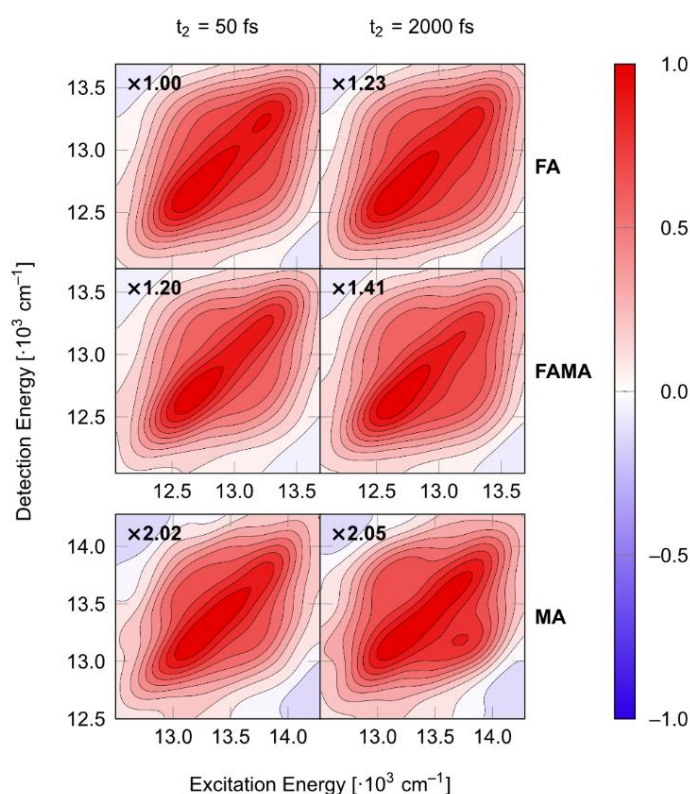


Fig. 1. A-2DES maps of solar cell devices with different perovskite absorber compositions. For each device, spectra recorded at 50 fs and 2 ps waiting times are shown. The first, second, and third rows correspond to FA, FAMA, and MA compositions, respectively. To facilitate direct comparison, all 2D maps are scaled relative to the FA device at 50 fs, using the scaling factors indicated in the respective panels.

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Unravelling the Origin of Ultrafast Vibronic coupling in Hybrid Perovskite Nanosystems

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The mechanically soft lattice of hybrid perovskites gives rise to strong entanglement between structural fluctuations and electronic dynamics, making vibronic coupling a central factor governing their optoelectronic behaviour. Here, we investigate the role of organic cation vibrations in modulating electronic states of hybrid FAPbBr₃ perovskite nanocrystals using an ultrafast vibrational-control

approach based on vibrationally promoted electronic resonance (VIPER). Mode-selective excitation of molecular vibrations reveals an ultrafast (~300 fs) coupling between the N–C=N stretching modes of the formamidinium cation and the electronic response, demonstrating that specific high-frequency vibrations actively influence optoelectronic properties [1].

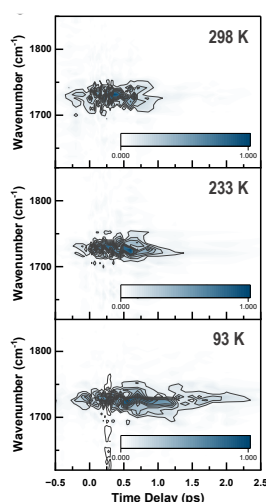


Fig. 1. 2D VIPER contour maps from 298 K to 93 K.

To uncover the structural origin of this vibronic coupling, multidimensional vibrational spectroscopies are combined with molecular dynamics simulations. Temperature-dependent VIPER measurements show a substantial prolongation of vibronic coupling lifetimes upon cooling (0.3 ps at 300K to 1.25 ps at 93K, Fig. 1), accompanied by immobilization of formamidinium cation rotational dynamics as revealed by polarization-dependent two-dimensional infrared spectroscopy. The correlated slowdown of vibronic and structural dynamics identifies hydrogen bonding as the key mediator of carrier–lattice interactions, whereby cation motion induces distortions in the PbBr₆ framework that governs the electronic states. Ab initio molecular dynamics further resolve the distinct contributions of wobbling-in-cone motion and jump-like reorientation of the cation to

hydrogen bond fluctuations and vibronic coupling. Together, these findings establish a direct structural–dynamical pathway linking organic cation motion to electronic modulation in hybrid perovskite nanocrystals.

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Posters

In Situ 2DES of Hierarchical Supramolecular Self-Assembly

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Molecular self-assembly, the spontaneous organization of disorganized molecules into ordered structures, offers a powerful route to fabricate complex nanomaterials, yet the molecular pathways under non-equilibrium conditions remain poorly understood. This is particularly true for hierarchical architectures, where intermediates are difficult to access experimentally due to their structural heterogeneity and short lifetimes. Here, two-dimensional electronic spectroscopy (2DES) serves as the central probe to resolve, in real time, the self-assembly of the amphiphilic cyanine dye C8S3, culminating in the formation of double-walled nanotubes (DWNTs) [1], a synthetic analogue of the light-harvesting antenna structures found in green sulfur bacteria [2]. The fully formed DWNTs comprise two concentric nanotubes (NTs) with inner and outer diameters of ~ 7 nm and ~ 14 nm, lengths up to $10\ \mu\text{m}$, and exhibit remarkably low static disorder [3]. How such highly ordered structures emerge from disordered molecular precursors in an aqueous solution, however, has remained unclear.

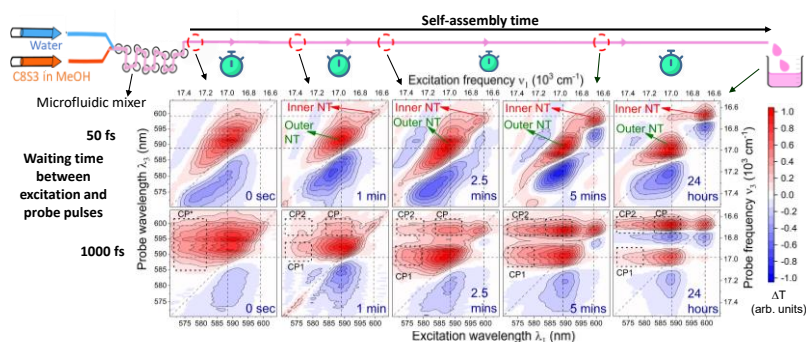


Fig. 1. C8S3 in methanol is mixed with water in a microfluidic mixer (top row), and DWNT self-assembly is monitored in real time using 2DES at different times after mixing, as shown in each panel. Indicated in 2D spectra are the GSB/SE contributions from the lowest-energy exciton transitions in the inner and outer NTs (middle row), and from cross-peaks (bottom row) that indicate exciton transfer. Cross-peak naming - at 0 seconds, CP* - from higher energy states; at later times, CP - from lower energy states of the outer to the inner NT and CP1, CP2 - from higher to lower energy states of outer and inner NTs.

The key advantage of 2DES is its ability to resolve transient heterogeneity and overlapping excitonic features, and to track inter-nanotube coupling via cross-peaks, thereby providing key indications on the underlying supramolecular structure. Additionally, integrating 2DES with a microfluidic platform converts reaction time into a spatial coordinate, enabling direct monitoring of short-lived intermediates as self-assembly proceeds (Fig. 1, top row). Immediately downstream of the microfluidic mixer, 2DES reveals a single, broad excitonic feature characteristic of highly disordered J-aggregated bilayers. Over the next few minutes, this feature splits into two diagonal peaks (Fig. 1, middle row) and increasingly intense cross-peaks (Fig. 1, bottom row), indicating the formation of two spatially adjacent, electronically coupled exciton manifolds. Based on their spectral positions, spectral features, and temporal order, the outer nanotube is shown to be structured prior to the inner nanotube. While the principal features of DWNTs arise within 5 minutes, the 2D peaks exhibit significant broadening, indicating considerable molecular disorder. Only after 24 hours, the distinct 2D peaks characteristic of well-structured DWNTs emerge.

This research establishes 2DES, combined with microfluidic control, as a powerful platform for in situ pathway-resolved studies of supramolecular self-assembly and provides mechanistic insight into how disordered precursors evolve into highly ordered, biomimetic light-harvesting nanostructures.

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Multiexcitonic triplet state maintains significant singlet character throughout singlet fission

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Singlet Fission is an ultrafast internal conversion of an excited single state S_1 into a correlated triplet state, $(TT)^1$ that may be useful for photovoltaics and quantum technologies. In this work, we examine the formation of triplet pair state within a library of conformationally flexible pentacenic (Pc) dimers¹ that exhibit a distinct near-infrared (NIR) spectral signature for $(TT)^1$. This NIR excited state absorption (ESA) signal has been previously demonstrated to only appear² when the triplets on the monomeric units are strongly bound to each other and thus provides a unique marker for the strongly bound $(TT)^1$ state for this study. By employing a combination of polarization-controlled spectroscopies, we demonstrate that the formation of the $(TT)^1$ state is highly specific to planar, lower-energy conformations, is excitation wavelength independent and is vibrationally coherent with large stabilization energy of the $(TT)^1$ photoproduct. Leveraging polarization control³ to track the electronic character during $S_1 - (TT)^1$ conversion, we demonstrate that substantial singlet-triplet electronic mixing persists throughout this process. This observation is generalized across variety of bridging motifs and suggests that while structural fluctuations in conformationally flexible dimers are often expected to decorrelate the triplet pair, they are evidently insufficient in reducing singlet-triplet electronic mixing even on tens of picoseconds timescales. Synthetic strategies for singlet fission must aim to prevent the singlet-triplet electronic mixing.

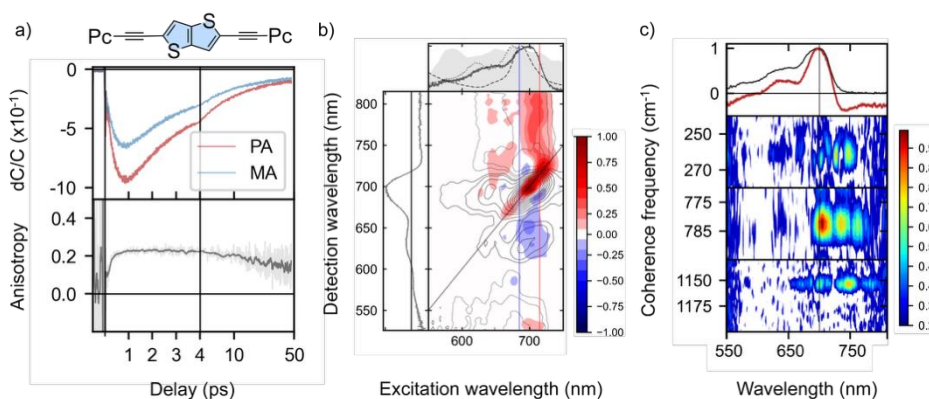


Fig. 1. a) Top: Transients in the $(TT)^1$ specific excited state absorption band. Bottom: Corresponding pump-probe polarization anisotropy. b) Kinetic rate map (colour map) for the $(TT)^1$ formation of $\tau = 516$ (1.5) fs timescale overlaid with absorptive two-dimensional spectrum (contour lines) at 0.5 ps. Left: pump-probe spectrum at the same time-delay. Top: steady-state absorption spectrum (black solid) overlaid with the pump spectrum (filled grey). c) Quantum beat map plotted as a function of detection wavelength. Top: pump-probe spectrum (red) and steady-state absorption spectrum (black).

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Trimming Excitonic Landscape with Light

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Selective control of excitonic states in supramolecular assemblies remains a central challenge due to spectral congestion and the intimate coupling between electronic structure and molecular organization^{1,2}. Here we demonstrate that controlled optical illumination can selectively reconfigure excitonic functionality in double-walled supramolecular nanotubes (DWNTs) self-assembled from cyanine dyes in aqueous solution (Fig. 1a,b)³. Using ultrafast two-dimensional electronic spectroscopy (2DES), we show that light selectively suppresses outer wall (OW) excitons while preserving inner wall (IW) excitons and their delocalized character. Atmospheric molecular oxygen acts as a photooxidative agent, enabling controlled attenuation of the OW exciton without chemical additives or post-synthetic treatment (Fig. 1c). Structural analysis reveals that photooxidative disorder remains confined to the outer wall, preserving the cylindrical scaffold and excitonic order of the IW. This mechanism operates in any light-accessible environment, including solid-state matrices, making it suitable for precise manipulation of individual DWNTs. Polarized single-object photoluminescence (PL) microscopy reinforces these findings, showing that individual DWNTs maintain the highest degree of excitonic PL polarization upon full OW silencing, which was recently linked to the exciton robustness in a 2D supramolecular assemblies².

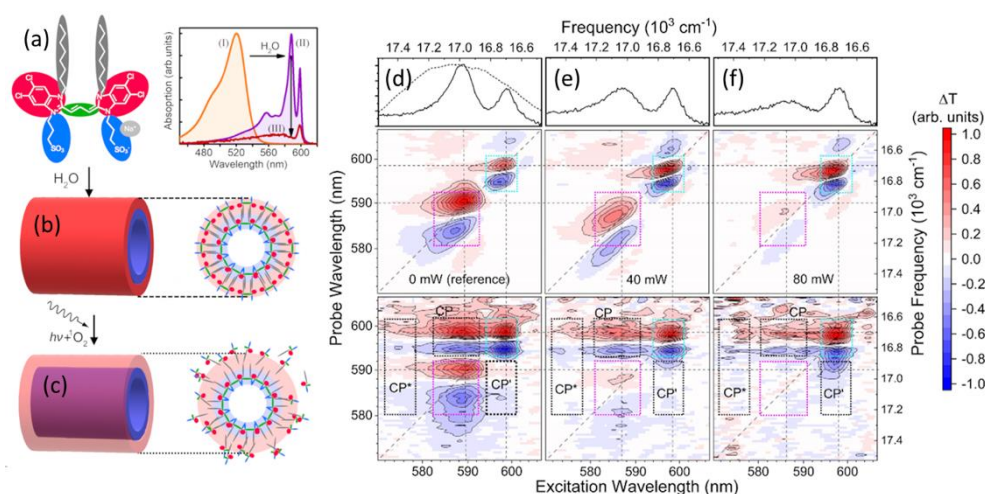


Fig. 1. (a-c) DWNT formation from cyanine dyes and selective OW photorestructuring; (d) 2DES mapping of reference DWNTs displays diagonal peaks corresponding to OW (pink) and IW (cyan) excitons, and OW-IW coupling cross-peaks (CP) that report on inter-wall energy transfer; (e-f) 2DES tracking the photorestructuring process, revealing progressive suppression of the OW diagonal feature in the 2D maps, accompanied by disappearance of the CP, indicating interwall decoupling. Top panels in (d-f) show corresponding steady-state absorption spectra; middle panels show 2DES at 25 fs waiting times; bottom panels show 2DES at 1000 fs waiting times.

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Anisotropic quantum beats report room temperature functional vibronic couplings in a light-harvesting aggregate

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Photosynthetic bacteria and plants employ highly organized light-harvesting complexes to absorb solar energy and funnel excitations efficiently to reaction center, where long-lived charge-separated states are formed. Cryogenic experiments and theoretical proposals have suggested a functional role of mixed vibrational-electronic (vibronic) states in the above process. Yet, at room temperature, decoherence can destroy coherent vibrational-electronic admixtures thus bringing their functionality into question. In this work, we have studied porphyrin nanotubes at room temperature as a model system for large photosynthetic aggregates. Using polarization-controlled 2DES experiments on the Q band to selectively probe the Qx-Qy coupled states, we demonstrate suppression of the prominent, but isotropic, quantum beat contributions from the observed non-linear signal. The anisotropic quantum beats survive the polarization sequence and serve as unique reporters of functional vibronic couplings. Supporting vibronic-exciton model for the nanotubes suggests that disorder may be the vital ingredient that enables electronic delocalization across overlapping vibrational-electronic manifold – vibronic couplings are not only robust to energetic disorder, but are in fact, enhanced by it across the spectral range of Q band.

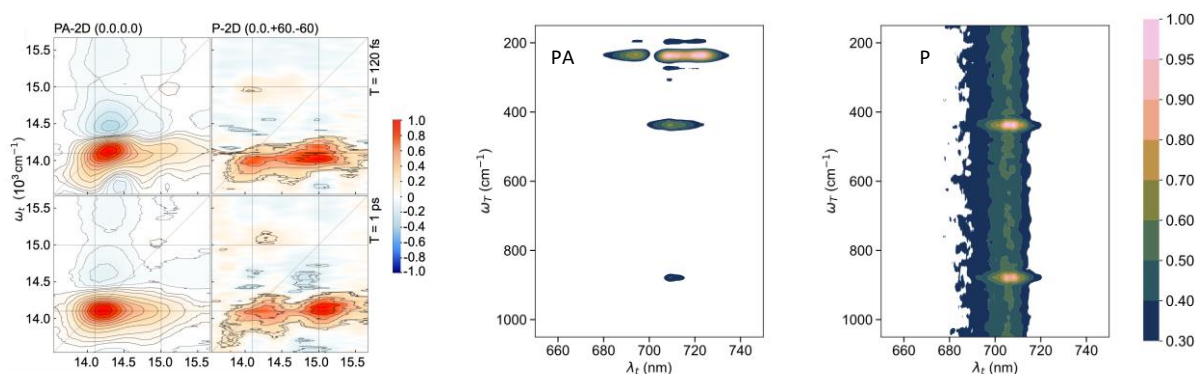


Fig. 1. (left) Polarization controlled 2D spectra for all PA and P-2D pulse-sequences at $T = 120$ fs and 1ps. Surviving Qx-Qy amplitude in P-2D reveals coupled Qx-Qy states. (right) Quantum Beat pump-probe spectra from PA (left) and P (right) pulse sequences reveals anisotropic quantum beats that survive the latter sequence, providing direct evidence of their participation in Qx-Qy coupling. Adapted from ref. [1].

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Excitation-Intensity-Driven Quantum Pathway Control In Two-Dimensional Coherent Spectroscopy

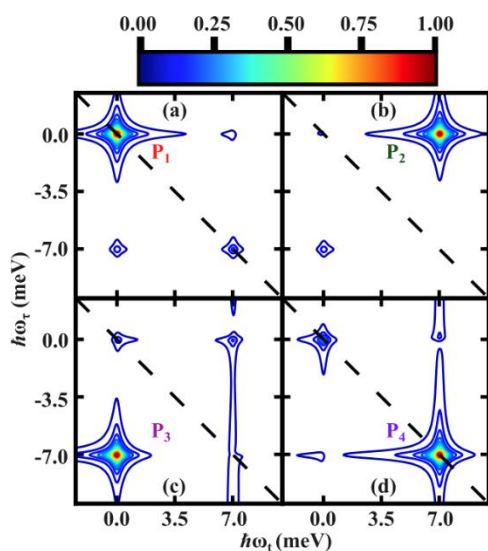
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A V-type three-level system extends a two-level system by incorporating two excited states coupled to a common ground state and serves as a prototypical platform for studying quantum interference between multiple excitation pathways. When both nondegenerate transitions are driven simultaneously, coupled-state interactions produce intricate coherent dynamics, making selective pathway control challenging. Developing simple and experimentally accessible strategies for isolating specific quantum pathways in multilevel systems is therefore of central importance in coherent multidimensional spectroscopy.



We present a nonperturbative theoretical study demonstrating that excitation-pulse intensity alone enables deterministic quantum pathway control in a V-type system. By numerically solving the extended optical Bloch equations and simulating rephasing two-dimensional coherent spectra using a collinear four-pulse sequence with phase cycling, we explore the nonlinear response beyond the perturbative regime.

Our results show that appropriate combinations of excitation pulse areas isolate specific quantum pathways. By tuning the intensities of the first three pulses, each of the four characteristic peaks in the 2D spectrum, two diagonal and two cross peaks, can be selectively enhanced while the remaining contributions are suppressed. In each case, constructive interference amplifies the desired

pathways, whereas destructive interference suppresses the others. The nonlinear signal becomes dominated by the coherence associated with the selected transition, effectively turning on a specific response channel. The accompanying figure presents four representative spectra, each exhibiting a single pronounced peak with visibility exceeding 90%, clearly demonstrating intensity-driven pathway isolation.

Beyond amplitude selectivity, we further demonstrate continuous phase control of individual peaks over a full 2π range while maintaining high visibility. This simultaneous amplitude and phase control establishes excitation-pulse intensity as a powerful and experimentally accessible knob for coherent manipulation.

The proposed scheme requires no polarization selectivity, chirping, or complex pulse shaping and is directly transferable to experimental two-dimensional coherent spectroscopy of multilevel excitonic and other strongly driven quantum systems.

A Kinetic Monte Carlo Study of Equilibration, Anisotropy Decay, Exciton-Exciton Annihilation and First Passage Dynamics in Photosynthetic Antenna Complexes

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Understanding the characteristic timescales governing excitation energy dynamics in photosynthetic antenna complexes is essential for interpreting ultrafast spectroscopic measurements. In this work, we investigate equilibration, anisotropy decay, exciton–exciton annihilation (EEA), and first-passage energy transfer dynamics in the LHCII monomer, trimer and the larger LHCII(M)–CP24–CP29 pentamer using kinetic Monte Carlo (KMC) simulations.

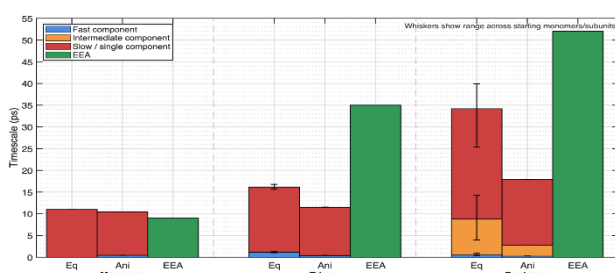


Fig. 1. Characteristic timescales of the transport dynamics within photosynthetic antenna complexes.

Equilibration was quantified through the decay of the total variation distance (TVD) between the time-dependent population distribution and its stationary distribution. The equilibration for the LHCII monomer was found to be 11 ps with a single exponential decay whereas for the LHCII trimer, the equilibration dynamics are well described by a biexponential decay with a dominant slow component of approximately 14–15 ps and a fast component of 1.07–1.25 ps, largely independent of

the initial excitation site due to the approximate structural symmetry of the complex. In contrast, the LHCII(M)–CP24–CP29 pentamer exhibits more complex kinetics requiring a triexponential fit, with slow components in the range of 16–31 ps, intermediate components of 3–14 ps, and sub-picosecond fast components. Faster equilibration is observed when excitations originate in the LHCII-C monomer consistent with its central position and strong inter-subunit connections.

Simulations of anisotropy decay show qualitatively similar multiexponential behavior, with slow components that partially track the population equilibration timescales. However, simulation of different models shows that anisotropy does not generically probe all equilibration processes but instead detects only those population redistributions that produce significant changes in transition-dipole orientation. Thus, anisotropy decay provides an orientationally weighted probe of equilibration dynamics rather than a direct proxy.

EEA dynamics were investigated both through direct KMC simulations of two-exciton encounters and through fits to the corresponding mean-excitation ODE model. The resulting annihilation times are approximately 7–9 ps for the monomer, 35–38 ps for the trimer and 52–57 ps for the pentamer complex. Notably, the latter two values are roughly two times longer than the equilibration timescales, contradicting the common assumption that annihilation occurs on half the equilibration timescale.

By analyzing first-passage (hitting) times and inter-monomer transfer probabilities, we show that the factor-of-two relationship instead emerges when comparing EEA times with excitation energy transfer (EET) hitting times rather than equilibration times. Directional asymmetries in inter-monomer energy transfer are also observed, including a dominant pathway linking CP29 and LHCII-C. These findings hope to clarify the relationship between experimentally observable timescales and equilibration.

New Insights on Dynamics in Chlorophyll-f Containing Far-red Photosystem I

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Structure-function relationships in cyanobacterial far-red photosystem I (far-red PSI) of *Chroococcidiopsis thermalis* PCC7203 isolated in detergent were studied using a combination of transient absorption spectroscopy and two-dimensional electronic spectroscopy at 77 K. By comparing partially pre-reduced and fully pre-oxidized final donor (P_{700}) conditions and variable excitation pulse energies, chlorophyll-f bands could be compared with assignments in a recently published structural paper by Consoli et al. [1] on far-red PSI of the same organism. Three major observations resulted from the study. Firstly, a ring of 6 chlorophyll-f molecules located at the center of the far-red PSI trimers might play a role in the distribution of excitations between monomers according to availability of open reaction centers, as observed from pulse energy-dependent residual populations of chlorophyll-f pools at 740–750 nm and 800 nm on the longest decay timescale. Secondly, one chlorophyll-f molecule assigned by Consoli et al. [1] to an absorption band at 756 nm and the A_{-1B} position in the PSI reaction center (Fig. 1) is likely involved in photochemistry, as evidenced by a transient band shift signal peaking at 688 nm, the decay of which correlates with recovery of 756 nm bleaching in about 150 ps. The proposed mechanism for the origin of this band shift is the formation of a radical pair $A_{-1B}^+A_{0B}^-$ that is

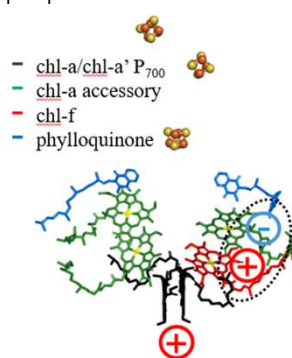


Fig. 1. Cofactors in far-red PSI with a proposed “metastable” radical involving a chlorophyll-f molecule.

is not stable in the presence of P_{700}^+ , but induces electrochromic shifts on accessory chlorophyll-a molecules in the A branch of the reaction center. Thirdly, the possibility of pseudo-photochemical quenching [2] occurring via the decay of the same radical pair that is suspected to induce a transient band shift is identified. This represents an alternative (or complementary) explanation to the P_{700}^+ channel that is often named as the cause for a low fluorescence quantum yield in closed PSI. In conjunction with a complete mapping of energy transfer pathways via 2DES data that is presented in this work, the findings provide important insights into the evolutionary role of chlorophyll-f incorporation into pigment-protein complexes. Future studies on far-red PSI by other organisms for which conflicting structural assignments were made in the past hold great promise for a better understanding of far-red light photoacclimation in cyanobacteria.

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Non-Monotonic Temperature Dependence of Energy Transfer in a Photosynthetic Antenna Complex

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Photosynthetic organisms direct electronic excitation energy through pigment-protein antenna complexes with remarkable efficiency [1,2]. For many decades, this energy flow has been understood primarily within Forster resonance energy transfer (FRET) theory [3]. Using two-dimensional electronic spectroscopy from 10 to 300 K, we investigate ultrafast energy transfer in the allophycocyanin (APC) trimer. We observe a pronounced non-monotonic temperature dependence of the transfer between the α and β pigments: the transfer time decreases from ~ 400 fs at 10 K to ~ 200 fs near 30-40 K, then increases to ~ 500 fs at 300 K. This trend contradicts classical Förster theory [4]. By modeling APC as a vibronic dimer coupled to its environment and solved using the quasi-adiabatic path-integral approach, we find that only a super-ohmic spectral density reproduces the experimental data. This reveals that specific low-frequency environmental modes can suppress energy transfer at cryogenic temperatures, demonstrating that environmental dynamics significantly influence excitonic energy flow even in a minimal antenna complex.

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Photosystem II Subunit Design Revealed by 2DEV Spectroscopy

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Photosystem II (PSII) is the main light harvesting system in oxygenic photosynthesis. It can achieve a near-unity quantum efficiency of light harvesting in ideal light conditions, and is the only natural system that can split water to evolve oxygen. The PSII light harvesting antenna is composed of distinct pigment-protein complexes – six in the C₂S₂-type supercomplex [1]. However, it is not well understood why PSII contains distinct subunits, rather than repeated subunits which would be metabolically less taxing to transcribe and translate. One hypothesis is that it is important for PSII's photoprotective capabilities, which is accomplished by activatable quenchers dissipating excess absorbed light energy as heat in the supercomplex's periphery, preventing the formation of dangerous reactive oxygen species in light

stress conditions. This is made possible through PSII's unique lack of energetic funneling of excitations to the reaction center, where a charge separation takes place in the first step of the photochemical reactions of photosynthesis. This lack of energetic funnel allows bidirectional energy transfer both toward and away from the reaction center [2], which is likely facilitated by the inhomogeneity of energy transfer pathways available to excitations at different exciton states in the complex [3], in other words, unique protein subunits. However, some of the subunits are structurally very similar, so the advantage of encoding distinct proteins is unclear based on structure alone. For example, light harvesting complex II (LHCII) and chlorophyll-protein 29 (CP29) PSII subunits have almost identical protein structures (Fig. 1), but are encoded by unique genes and occupy distinct locations in the PSII supercomplex.

In this work, we use two-dimensional electronic-vibrational (2DEV) spectroscopy to investigate the energy transfer pathways in CP29 and monomeric LHCII subunits. This method uses two axes that are correlated with independent degrees of freedom to improve spectral resolution of excited-state dynamics over the more common two-dimensional electronic spectroscopy (2DES). The differences identified in the CP29 and monomeric LHCII spectra can be correlated to energy transfer pathways with high spatial resolution, due to the relatively subtle differences between the CP29 and LHCII structures, as well as the ability of 2DEV to probe differences in chromophores' local environments. This work sheds light on the PSII design principles that confer its ability to accomplish contradictory goals of efficient light harvesting and effective photoprotection.

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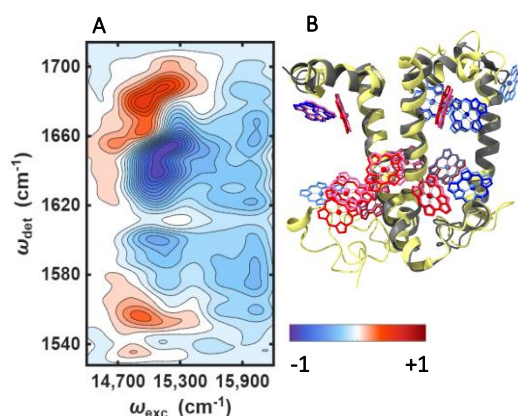


Fig. 1. (A) 2DEV spectrum of monomeric LHCII at 250 fs. (B) Structural overlap of CP29 (3PL9 – yellow backbone, dark red chlorophyll *a*, dark blue chlorophyll *b* and LHCII (1RWT – grey backbone, pink chlorophyll *a*, light blue chlorophyll *b*).

Ultrafast Excitation-Energy Dynamics in Intact Cells of *Gemmatimonas Phototrophica* Probed by Two-Dimensional Electronic Spectroscopy

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Two-dimensional electronic spectroscopy (2DES) correlates excitation and detection frequencies and enables direct visualization of energy-transfer pathways in complex molecular systems. Ultrafast multidimensional spectroscopy has played a central role in uncovering mechanisms of light harvesting and charge separation in photosynthetic systems [1]. While the technique is routinely applied to isolated pigment-protein complexes, extending 2DES to intact photosynthetic cells presents substantial challenges due to optical scattering and excitation-density-dependent annihilation. Here, we apply broadband 2DES to intact cells of *Gemmatimonas phototrophica*, whose photosystem consists of a reaction center enclosed by concentric LH1 and Lh antenna rings forming a distinctive double-ring architecture [2].

Measurements were performed using a home-built 2DES setup employing a sequence of three femtosecond excitation pulses followed by heterodyne detection of the emitted third-order signal. Fourier transformation with respect to the coherence time t_1 yields frequency-frequency maps that resolve correlations between excitation and detection bands. To address strong scattering from whole cells, we implemented double modulation lock-in detection, enabling background suppression and stable signal acquisition under low excitation fluence [3]. Broadband excitation spanning 750-900 nm covers the bacteriochlorophyll absorption bands associated with the outer Lh and inner LH1 subsystems, and pulse compression provided ~15 fs temporal resolution.

The resulting 2D spectra display distinct diagonal features corresponding to the spectrally separated antenna bands and evolving off-diagonal correlations during the population time t_2 . Early population times reveal pronounced changes in peak shape and orientation consistent with ultrafast intra-antenna relaxation. On longer timescales, the growth and redistribution of cross peaks resolve a network of interconnected energy-transfer pathways linking the outer Lh antenna, the inner LH1 ring, and the reaction center within the intact double-ring complex. The observed dynamics span femtosecond to nanosecond regimes, reflecting a multistep cascade of excitation redistribution under near-native conditions. Importantly, we were able to resolve spectral signatures associated with the reaction center itself. Accessing these contributions is particularly challenging in large pigment-protein assemblies, and even more so in intact cells, because the comparatively weak reaction center signals are partially masked by the stronger responses of the surrounding antenna complexes.

In addition, polarization-controlled measurements were implemented to enhance sensitivity to specific relaxation pathways and to assist in separating overlapping spectral contributions. These results demonstrate that coherent multidimensional spectroscopy can resolve interconnected energy-transfer processes directly in living photosynthetic cells and provide a functional perspective complementary to structural characterization of the *G. phototrophica* photosystem.

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Excitonic Energy Transfer in Red Algal Photosystem I Reveals an Evolutionary Bridge between Cyanobacteria and Plants

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Photosystem I (PSI) achieves near-unity quantum efficiency, yet its energy-transfer mechanisms remain debated[1][2]. The red alga *Cyanidioschyzon merolae* represents an evolutionary intermediate, featuring a monomeric core surrounded by three to five LHCR subunits[3]. We applied two-dimensional electronic spectroscopy at 8 K and 80 K to disentangle excitation pathways in *C. merolae* PSI. Cryogenic measurements suppressed thermal broadening, resolving five dynamical components: sub-picosecond equilibration (0.3–0.8 ps) across the core–LHCR interface, population transfer (2.6–4 ps) into lower-energy manifolds, and slower feeding (18–53 ps) into red pools distributed across both core and antenna. Persistent ground-state bleach on long timescales indicates exciton stabilization in terminal sinks. Comparison of 8 K and 80 K spectra reveals heterogeneous exciton partitioning among multiple sinks at low temperature, whereas modest thermal activation promotes convergence into core-associated red chlorophylls. Atomistic excitonic modeling with time-nonlocal master equations provided a quantitative framework for these dynamics. Our results demonstrate that *C. merolae* PSI broadens the kinetic funnel by distributing sinks across core and antenna—an adaptation that extends spectral coverage while maintaining trapping efficiency, bridging cyanobacterial and plant paradigms in photosynthetic evolution.

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Exploring Chromophore-Environment Interactions of Metalloporphyrins in *De Novo* Proteins

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The remarkable ascent of protein design means that artificial proteins can be developed rapidly, with a high-degree of structural control. A remaining major challenge is understanding how to fine-tune the environment around bound cofactors to achieve the desired functionality, such as photocatalysis. As established from studies of natural photosynthetic light harvesting complexes,[1] the protein plays a pivotal role in tuning the energy landscape and function of cofactors, and thus characterization of these interactions in *de novo* protein counterparts is important to enhance function.

The 4D2 family of *de novo* tetrahelical proteins are highly stable and well physically characterised, naturally binding hexacoordinated metalloporphyrins through axial ligation to two histidine residues.[2] We have used ultrabroadband two-dimensional electronic spectroscopy[3] to probe the photophysics and the pigment-protein interaction dynamics of heme-b or zinc protoporphyrin IX bound to variants of the model 4D2 proteins.

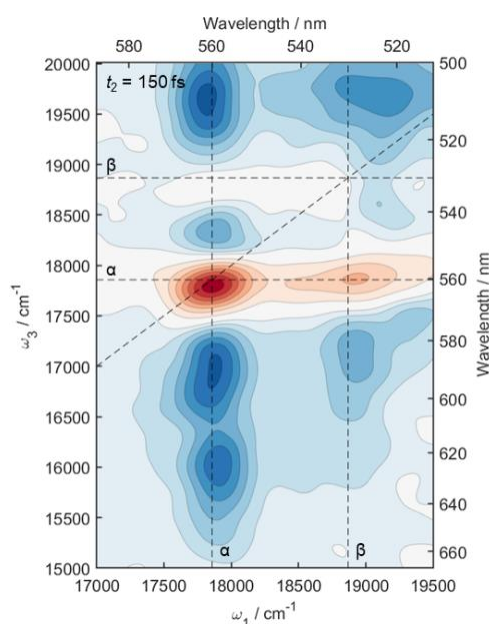


Fig. 1. 2DES spectrum of heme(II)-4D2 for a waiting time of 150 fs.

The native form of the 4D2 protein binds ferrous heme b. Using 2DES, we show photoexcitation of the Q-bands leads to ultrafast dissociation of the iron-histidine bond, generating a transient 5-coordinated form of heme b. The iron-histidine bond recombines on a picosecond timescale, in excellent accord with the analogous natural cytochrome *bc₁* protein. Complementary spectroscopies show no evidence of photooxidation of the iron centers, contrary to the conclusions of previous studies of some natural cytochrome systems.

The time-evolving 2DES line shapes provide key information on the dynamical interactions between cofactors and their immediate environment, and thus report on the strength of chromophore-protein interactions. The cofactor modularity of 4D2 has shown that the coupling strength and dynamics are sensitive to the inserted chromophore.

Our studies of these novel model proteins will shape their future iterative design and development of functional control. As for when heme b is bound, it has provided an accessible platform for disentangling the dynamics of specific binding sites in complex natural systems, such as in cytochrome *bc₁*.

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Multi-Exciton Processes in Ultrafast Nonlinear Optical Spectroscopy

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Excitation Energy Transfer (EET) in photosynthetic complexes and organic solar cells is typically rationalized within a single-exciton picture, where energy is transferred between the lowest-excited states of individual chromophores. At high photon fluence, however, multiple excitons can be simultaneously generated, whose encounter may result in the net loss of an exciton through a process known as Exciton-Exciton Annihilation (EEA) [1] (Fig. 1a). Multi-exciton EET and EEA processes are commonly described using phenomenological models, which are limited by various underlying assumptions, e.g., normal exciton diffusion, uniform random distribution, and instantaneous EEA within the so-called annihilation radius [2,3]. Instead, recent advances in ultrafast nonlinear optical spectroscopy have revealed that exciton diffusion is anomalous rather than normal [4], annihilation can be suppressed by quantum interference [5], and excitons can meet several times before eventually annihilating [6]. Although several microscopic theories have been proposed [7,8], a comprehensive understanding of these processes is still lacking.

To fill this gap, we develop a multi-exciton framework for EET and EEA processes in molecular systems based on a generalized quantum master equation formulated in an arbitrary orthonormal basis. Based on the degree of delocalization between multi-exciton and higher-excited states, this approach provides a unified description of different microscopic theories, i.e., (generalized) Förster and (modified) Redfield theories. Furthermore, by adopting the product basis of one-exciton eigenstates, the method offers both interpretational and numerical advantages for the treatment of weakly-correlated systems. Finally, the method is applied to molecular chains of varying lengths to systematically investigate how static disorder, many-body interactions, energy mismatch, and excitonic coupling influence the ultrafast dynamics and spectral features observed in higher-order optical responses (Fig. 1b).

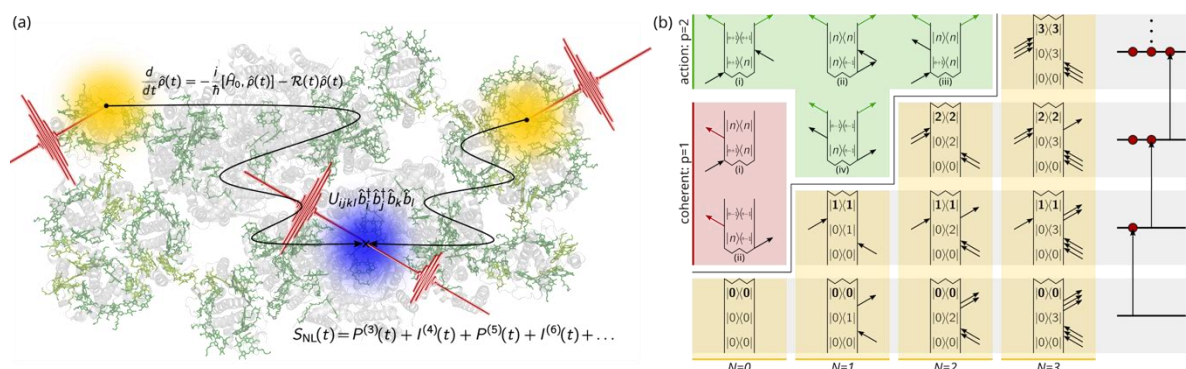


Fig. 1. (a) Pictorial representation of multi-exciton EET and EEA processes in molecular aggregates and (b) double-sided Feynman diagrams contributing to the N-th order signal in coherent- and action-detected spectroscopy.

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P14

Multiple Quantum Coherences of Electronic States and Vibrational Modes in Cyanocobalamin

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Cyanocobalamin is an inorganic chromophore widely known for its role as the cofactor in certain enzymes. The primary role of cobalamin is to aid in the catalytic activity of the enzymes. The vibrational modes and the electronic states together forming the potential energy surface responsible for the catalytic activity of cobalamins needs to be identified. To investigate the relationship between ground state vibrations and excited state potential energy surfaces, triple sum frequency spectroscopy (TSF) was used. Different vibrational modes associated with the cyanocobalamin corrin ring were found to be enhanced as the excited electronic state changes. A shift in the frequency of these vibrational modes were observed when time delay was introduced between the infrared and the visible/UV pulses. This shift in frequency of the vibrational mode may be attributed to the electronic excitation preceding the vibrational excitation, during temporal overlap of the pulses, leading to the excitation of vibrations of the excited electronic state. This work provides insights on characterizing the potential energy surfaces of different electronic states particularly those that may be involved in reactivity.

Energy Transfer in a Vibronically Coupled System: Transient Anisotropy of Chlorophyll c

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Chlorophyll (Chl) *c* pigments are found in the light-harvesting complexes (LHCs) of various marine photosynthetic organisms, alongside Chl *a* and carotenoids. Despite their widespread distribution, the number of spectroscopic studies on Chl *c* remains scarce. Their distinguishing feature within the Chl family is that they are porphyrins rather than chlorins; their fully unsaturated tetrapyrrole macrocycle imposes near- D_{4h} symmetry (see Fig. 1A vs. 1B). One controversial aspect of Chls is their electronic energy level structure. Reimers *et al.* proposed a band assignment in the Q-band region based on vibronic coupling for different Chls, excluding Chl *c* [1]. Preliminary fluorescence measurements comparing excitation anisotropies of Chl *c* to Chl *a* show a loss of purely polarized features, indicating a higher vibronic coupling in Chl *c*. A spectroscopic method that has previously been used to successfully characterize the Q_x and Q_y mixing in Chl *a* is transient absorption anisotropy (TAA) [2]. Additionally, TAA also enables observation of population transfer between isoenergetic energy levels in a porphyrin molecule [3].

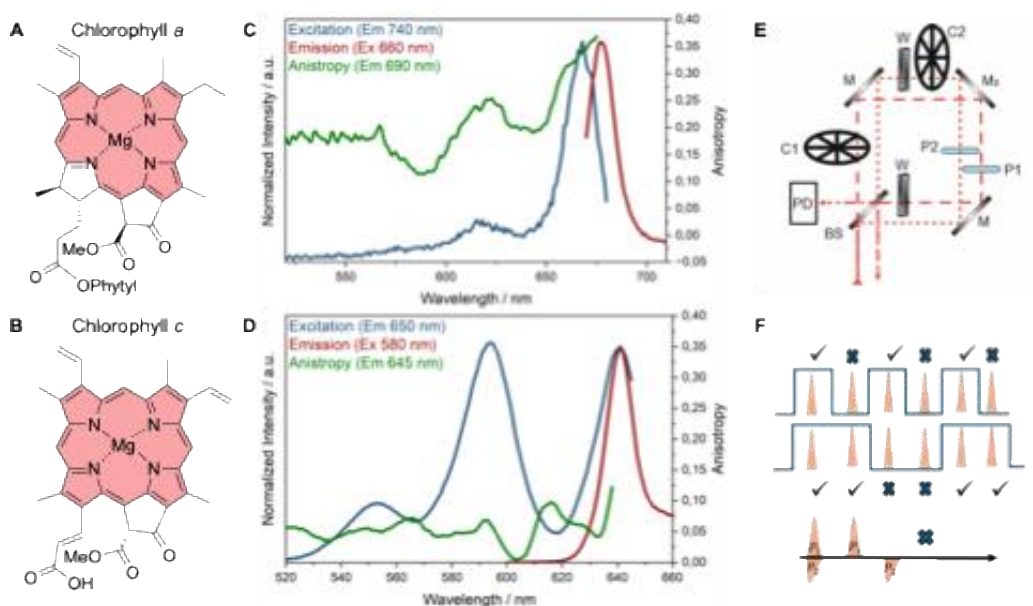


Fig. 1. Molecular Structure of Chl *a* **A** and Chl *c* **B**, emphasizing the differences in ring structures. The fluorescence spectra of Chl *a* **C** and Chl *c* **D** [2]. While the excitation anisotropy of Chl *a* reaches a value of $r = 0.35$ in the peak of the Q_y band, the anisotropy of Chl *c* is limited to a maximum value of $r = 0.125$, indicating a loss of a purely polarized feature. Panels **E** and **F** show a sketch of the Sagnac interferometer and the corresponding pulse sequence [3].

Based on a modified Sagnac interferometer [3], we measure transient absorption with parallel and orthogonal pump- and probe-polarizations on a shot-to-shot basis. This reduces the data acquisition time for averaged, broadband TAA spectra to 30 minutes. In this work, we use our Sagnac-based TAA experiment to measure TAA spectra of Chl *c* to elucidate the high degree of vibronic mixing in this system. This approach reveals the ultrafast $Q_x \rightarrow Q_y$ transfer time, which is hidden in conventional TA spectroscopy.

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Revealing Anti-Correlated Vibrational Coherences in Zinc Phthalocyanine via Polarization-Resolved Two-Dimensional Electronic Spectroscopy

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Zinc phthalocyanine (ZnPc) is a promising macrocyclic molecule for applications in solar energy harvesting and artificial photosynthesis systems due to its intense Q-band absorption [1]. Its high D_{4h} molecular symmetry complicates the investigation of vibronic coupling because of the spectral convolution of degenerate Q_x and Q_y states. In this work, we utilize polarization-resolved 2DES (p-2DES) to disentangle these interactions by generating and comparing amplitude beatings extracted from magic angle (MA), diagonal-peak (DP) and cross-peak (CP) specific spectra [2]. Oscillatory residuals were extracted from a 40 x 40 nm window centered on the Q-band and Fourier transformed to yield the power spectra for MA, DP and CP polarization schemes (Fig. 1a). While the MA power spectrum reveals totally symmetric a_{1g} modes (258, 578, 673 and 816 cm^{-1}), it fails to capture the 224, 476 and 741 cm^{-1} modes (Fig. 1c). These modes are clearly resolved in DP and CP power spectra but vanish in MA due to an anti-correlated phase relationship, a characteristic signature of asymmetric b_{1g} and b_{2g} vibrations [3]. To elucidate the structural deformations associated with these modes, Density Functional Theory (DFT) calculations were performed using Gaussian 09W [4]. The results indicate that these modes induce opposing energy shifts in the Q_x/Q_y manifold, lifting the doubly excited state degeneracy by up to 0.4 eV. This energetic splitting may suggest the Jahn-Teller effect, while the selective cancellation of these asymmetric coherences in the MA scheme provide direct experimental evidence of electronic equilibration through a conical funnel. By correlating experimental power spectra with theoretical molecular displacements, we demonstrate that p-2DES are essential for revealing the symmetry-breaking coordinates that govern the ultrafast dynamics of phthalocyanines.

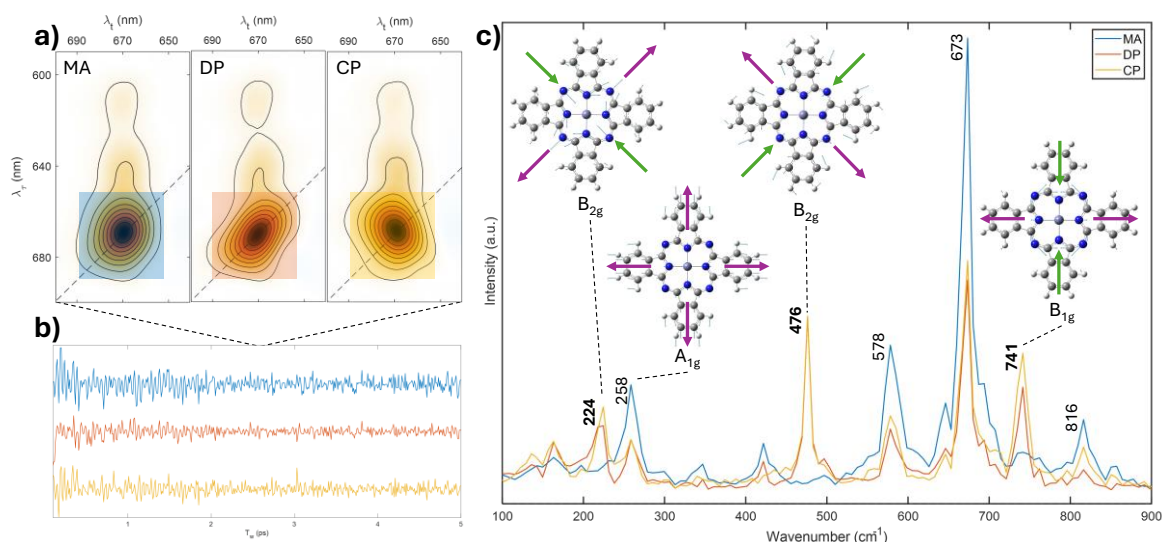


Fig. 1. Polarization-resolved 2DES and amplitude beating analysis of ZnPc in DMF. **(a)** Representative 2D spectra at $T_w = 500$ fs. **(b)** Oscillatory residuals extracted from each 2D spectra. **(c)** Fourier power spectra for each polarization configuration, overlaid with DFT-calculated displacement vectors.

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Selective Q-Band Excitation Modulates Exciton–Exciton Annihilation in H₂TPPS J-Aggregates

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Porphyrin supramolecular assemblies offer a versatile framework for emulating key structural and functional motifs of natural photosynthetic pigments.[1] Intermolecular coupling established upon self-assembly promotes exciton delocalization, leading to distinct electronic properties and enhanced linear and nonlinear optical responses. Within this context, transitions in the Q-band region are particularly relevant, as they regulate energy flow in biological light-harvesting complexes and underpin

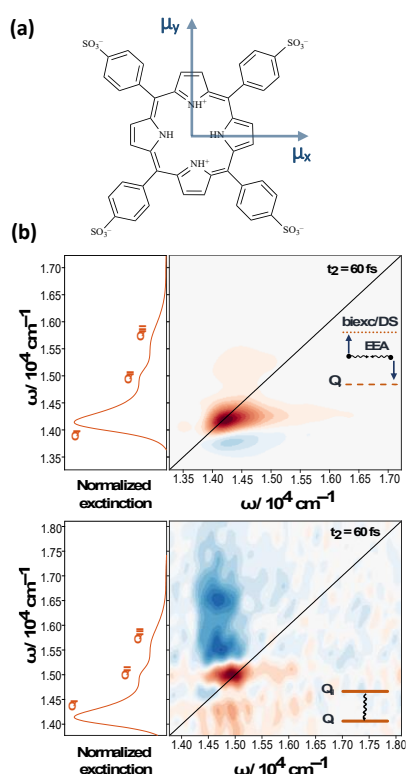


Fig. 1. (a) Molecular structure of H₂TPPS with x- and y-oriented transition dipoles coupled in the J-aggregate. (b) 2DES maps of the J-aggregate under two excitation profiles represented in green on the left panels overlaid with the UV-Vis spectrum (orange trace).

applications ranging from photovoltaics and optoelectronics to coherent energy transport and photodynamic therapy. Achieving directional control over energy transport in these aggregates requires a detailed understanding of exciton-exciton interactions, namely how monomeric units communicate under external stimuli and collectively generate emergent excitonic phenomena. Among the processes governing exciton diffusion and relaxation, exciton-exciton annihilation (EEA) plays a pivotal nonlinear role: while it represents a loss channel that limits transport efficiency, it may also contribute functionally to charge-separation pathways. Here, we investigate the ultrafast photophysics of the Q-band manifold (500–700 nm) of tetra-(*p*-sulphonatophenyl)porphyrin (H₂TPPS) by fully non-collinear two-dimensional electronic spectroscopy (2DES),[2] comparing monomeric and J-aggregated forms. The aggregate Q-band structure originates from distinct excitonic subbands rather than purely vibronic effects, with the lower-energy Q_I band exhibiting predominantly Q_x J-type character and the higher-energy Q_{II} band mainly Q_y H-type character.[3]

This mixed J/H excitonic landscape dictates excitation-dependent relaxation pathways and directly controls EEA through the sign of intermolecular dipole–dipole couplings: destructive interference between two-exciton wavefunctions in H-type configurations suppresses annihilation, whereas J-type excitation promotes it.[4] Accordingly, selective excitation of Q_I (J-type, x-polarized) activates coherent EEA and strong interband coupling, while excitation of Q_{II} (H-type, y-polarized) ion. The coexistence of J- and H-type subbands within the same system thus enables direct probing of interference-controlled EEA, while low-frequency out-of-plane modes further modulate the electronic dynamics via strong vibronic coupling. This provides design principles to tailor energy flow and annihilation mechanisms in functional porphyrin-based materials.

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Substitution and Environmental Effects on Ultrafast Relaxation Dynamics of Red-absorbing Squaraine Dyes

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In recent years, squaraine dyes have attracted significant attention for their application in solution-processed photovoltaic devices, such as dye-sensitized solar cells (DSSCs). This interest stems from their strong absorption in the red and near-infrared (NIR) regions of the solar spectrum, which remain largely underutilized by current technologies [1]. Despite extensive experimental effort, their complex photophysical behavior, particularly on the femtosecond timescale, remains poorly understood. Studies on structurally related dyes suggest that non-adiabatic pathways involving conical intersections (CIs) may play a central role in the relaxation dynamics [2]. In particular, the presence of a sloped topography CI between the first excited state and the ground state has been reported to drive ultrafast non-radiative decay, a detrimental process to the photovoltaic performance [3].

In this work, we present an investigation of the ultrafast photophysics of three commercially available squaraine dyes by means of two-dimensional electronic spectroscopy (2DES): 4-bis[4-(N,N-diisobutylamino)-2,6-dihydroxyphenyl] squaraine (SQ), 2,4-bis[4-(N,N-diphenylamino)-2,6-dihydroxyphenyl] squaraine (DPSQ), and 2,4-bis[4-(N,N-dibenzylamino)-2,6-dihydroxyphenyl] squaraine (DBSQ), whose general structure is depicted in Figure 1. Starting from a previously characterized benchmark dye to establish a reference relaxation mechanism [3], the influence of N-substitution at the donor moieties is examined to evaluate its influence on CI-mediated decay. In parallel, the excited-state dynamics are explored across a selected set of solvents with varying solute-solvent interactions, allowing to assess the role of the environment in modulating relaxation pathways. Particular attention is given to solvent-induced perturbations of the intramolecular hydrogen-bond network, which governs structural stability and conformational motions [4]. These insights aim to provide guidelines to the rational design of red-absorbing squaraine dyes and operating conditions that suppress ultrafast CI-mediated non-radiative losses and improve DSSCs' performance in the red-NIR region.

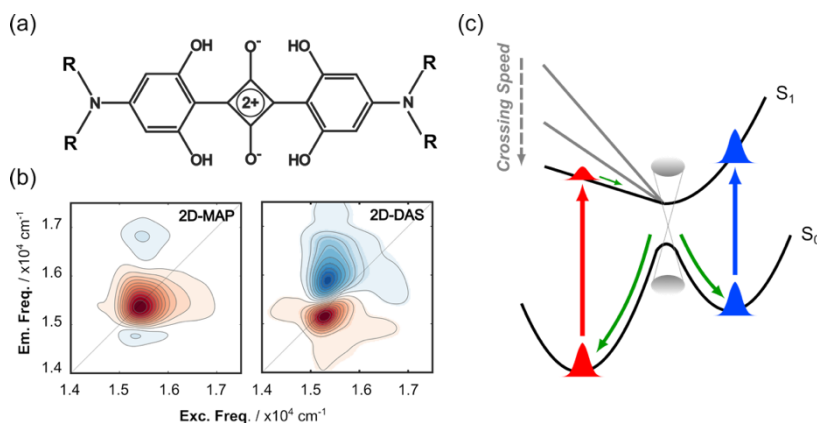


Fig. 1. General molecular structure of the studied squaraines; (b) example of a 2DES map (left, $t_2=150$ fs) with extracted 2D-DAS (right, $\tau=170$ fs); (c) pictorial scheme of conical intersection slope variation.

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Two-Dimensional Electronic Spectroscopy of a System with a Double-Well Excited-State Potential: A Theoretical Study

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Photochemical problems of contemporary interest involve excited-state relaxation processes, such as excited-state proton transfer (ESPT) and twisted intramolecular charge transfer (TICT), which involve potential energy surfaces (PESs) with multiple local minima that drive the course of photochemistry. Polyatomic molecules that exhibit ESPT or TICT dynamics possess double-minimum structures in both ground/excited electronic states, corresponding to protonated/deprotonated forms and planar/twisted forms, respectively. At thermal equilibrium, since only the lowest vibrational state of the ground electronic state in one of the forms (usually, the protonated form and the planar form, respectively) is predominantly populated, the ground state essentially behaves like a single harmonic potential well, from which the photoexcitation process is initiated, while the dynamics of excited electronic states involve PESs with two minima that can be faithfully represented by double-well structures.

In a recent study, we theoretically explored the spatiotemporal and information-entropic aspects of excited-state linear wavepacket interferometry (WPI) upon excitation by time-delayed phase-locked ultrafast pulse pairs [1], modelling the excited-state potential as both symmetric and asymmetric double wells, demonstrating prospective applications in quantum control and quantum computation/information [2]. Such a linear WPI is a time-domain realization of the overall one-photon absorption in a ‘pump-pump’ measurement, which underlies the machinery of population-detected coherent spectroscopy (widely known as coherent ‘action’ spectroscopy), and eventually led to the development of coherent multidimensional spectroscopy.

To gain deeper insights into the excited-state dynamics, we numerically simulate two-dimensional electronic spectra of a model system with a double-well PES in the excited electronic state. We show that the spectra contain rich information due to quantum tunnelling and reflection. Since two-dimensional electronic spectroscopy correlates excitation and detection frequencies, enabling direct identification of the populations of vibronic states and their coupling, it disentangles vibronic coherence and population transfer. We show that the spectra contain rich information about vibrational wavepacket splitting (due to tunnelling and reflection), and demonstrate that the resulting spectra encode a detailed signature of the double-well potential energy landscape.

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Probing Quantum Indeterminacy in Ultrafast Excited State Intramolecular Proton Transfer

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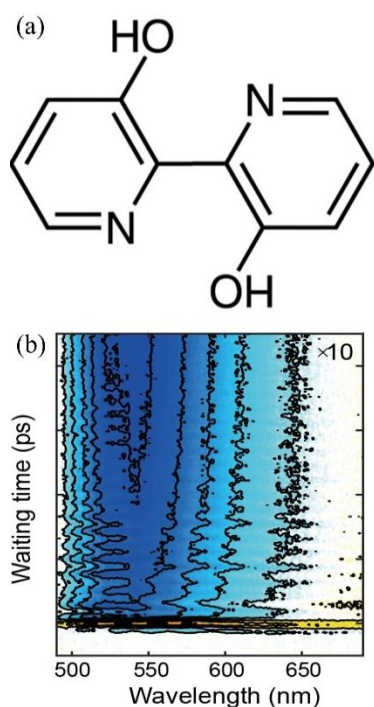


Fig. 1. (a) Molecular structure of BP(OH)₂, and (b) TA spectrum of BP(OH)₂ dissolved in cyclohexane. Positive and negative signals are shown in red and blue, respectively. The amplitude is multiplied by 10.

Excited state intramolecular proton transfer (ESIPT) in the condensed phase is typically described within the semiclassical molecular dynamics framework [1]. However, certain excited state dynamics remain difficult to rationalize without accounting for quantum nuclear behavior [2]. To explore these aspects, we investigate the double proton transfer dynamics of [2,2'-bipyridyl]-3,3'-diol (BP(OH)₂; Fig. 1a) using femtosecond transient absorption (TA) spectroscopy with <15 fs time resolution. The broadband TA spectrum of BP(OH)₂ in cyclohexane (Fig. 1b) exhibits periodic oscillations. Considering that the ESIPT of BP(OH)₂ is completed within tens of femtoseconds, these observed wavepackets represent vibrational coherences surviving in the reaction product. Fourier transform analysis identifies primary modes at 200, 295, 334, 533, and 555 cm⁻¹, matching theoretical predictions for the di-keto state, confirming that the system reaches the double-proton-transferred state on an ultrafast timescale.

A significant physical paradox arises when analyzing the specific reaction pathways through which the observed vibrational coherences are formed. According to simulations based on quantum chemical calculations, a semiclassical sequential proton transfer mechanism should produce a coherent vibrational peak at 238 cm⁻¹ due to the recoil originating from the proton transfer. However, this coherent vibrational mode does not manifest in the TA spectra of BP(OH)₂. Given that the period of this missing mode (140 fs) is significantly longer than the reaction timescale (<100 fs), its absence cannot be attributed to rapid dephasing due to the reaction [1]. Thus,

the absence of such classical signatures indicates the presence of non-classical dynamics beyond the semiclassical models under these ultrafast conditions. We attribute this to the structural flexibility of BP(OH)₂, which facilitates "quantum recognition" by the electronic environment, where the first proton resides in a quantum-indeterminate position and is instantaneously recognized as "moved" upon photoexcitation.

To further validate this mechanism, we have synthesized ESIPT systems with nearly identical electronic structures but varying donor–acceptor distances to modulate the reaction barrier. By tuning this barrier, we aim to induce delocalization between the enol and keto wells, clarifying the non-classical pathways that cannot be explained by the semiclassical model alone. This approach offers a new perspective beyond conventional interpretations based on electronegativity and classical thermodynamics, recognizing that nuclear positional uncertainty can also play a crucial role in reaction dynamics.

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P21

Remote Control of Fluorescent Protein Photoswitching

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Several mutants of fluorescent proteins (FPs) can be switched by light between emissive and dark states many hundreds of times. These reversibly switchable (rs) FPs are essential tools in the development and application of superresolution microscopy, allowing microscopic visualization with spatial resolution down to a few tens of nm, well below the Abbe limit. Within rsFPs three different mechanisms operate - negative, positive and decoupled rsFPs - each of which has its own specific application. We investigated the underlying photomolecular dynamics of these different families of rsFPs, and here report some unexpected long-range effects of mutations on their various behaviours.

An example of a 'decoupled' rsFP is Dreiklang. Dreiklang derives its name from the three distinct wavelength which are used to respectively switch on, off and excite fluorescence. This decoupling gives Dreiklang potential applications beyond bioimaging, such as in optical read-write memories. The on-state is an equilibrium between a 510 nm absorbing anionic chromophore (the emissive state) and a 400 nm neutral form, which can be turned off by irradiation at 400 nm to form a photohydrated product which itself absorbs at 340 nm (Fig. 1). 'Off' photoswitching was studied by ultrafast TA and IR. These show that excited state decay is followed by pico- to microsecond dynamics in the ground state.[1]

The on to off switching reveals intriguing and unexpected kinetics. The driving force for photohydration is electron transfer from a tyrosine to the chromophore in 100 fs, creating ground state radical ions, which either react (the off-switching photohydration) or recombine. The IR spectra are unusual, in that they reveal formation over several picoseconds of bleach bands (i.e. population disappearing) at 1500 and 1568 cm^{-1} which subsequently decay in hundreds of picoseconds (Fig 1). Neither mode can be assigned to the reactants (tyrosine and chromophore) so must arise from adjacent amino acid residues. Mutations were made in residues remote from the site of hydration but connected to it by a hydrogen bonded proton wire (Fig 1). The mutations cause modifications to the amplitude of the bleach modes but do not significantly shift their frequencies. Further, the overall rate of off-switching is a function of mutation, indicating a functional role.

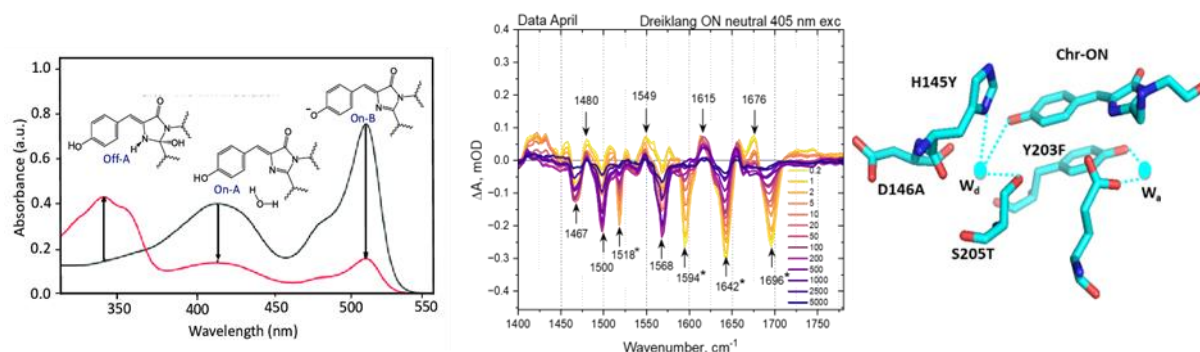


Fig. 1. The transient IR spectra show depletion at 1500 and 1568 wavenumber in picoseconds followed by slower recovery. The dynamics of these modes are controlled by mutations at H145 and S205 but D146 has little effect.

Further examples of mutations which modify the photochemical mechanism in the positive photoswitching FP 'Kohinoor' will be presented in the poster.

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Solvent-Driven Spectral Diffusion in IR-125 Probed by Full and Reduced Fluorescence-Detected Two-Dimensional Electronic Spectroscopy

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Fluorescence-detected two-dimensional electronic spectroscopy (F-2DES) provides direct access to frequency-frequency correlations and bath-induced spectral diffusion dynamics but typically requires experimentally demanding three-dimensional datasets $S(\omega_1, \omega_3; T)$. Here, we investigate whether key signatures of system-bath interactions traditionally extracted from full F-2DES measurements can be accessed using reduced, spectrally resolved pump-probe scans acquired at significantly lower experimental cost. Using the organic dye IR-125 dissolved in Methanol as a model system, we record full F-2DES spectra at selected pump-probe waiting times (T) and extract the center-line slope (CLS) as a function of T , which serves as a measure of spectral diffusion [1]. The CLS exhibits a decay on the sub-picosecond timescale, consistent with ultrafast solvent reorganization around the polymethine chain of the dye and the concomitant loss of frequency memory.

In Parallel, we acquire reduced measurements in which either the excitation coherence time (t_1) or detection time (t_3) is fixed at 0 fs, yielding excitation frequency-resolved $S(\omega_1; T)$ and emission-frequency-resolved $S(\omega_3; T)$ datasets, respectively [2]. From these reduced datasets, we track the evolution of spectral peak positions and linewidths as a function of waiting time T . The emission-resolved peak positions show a clear time dependence associated with the dynamic Stokes shift [3], reflecting solvent relaxation following photoexcitation. Notably, a simple metric derived from the relative evolution of excitation- and emission-resolved peak positions reproduces the initial decay trend of the CLS obtained from the full 2D spectra. The approach naturally extends to solvent-dependent studies: slower CLS decay and larger dynamic Stokes shifts are anticipated in less polar solvents isopropanol due to hindered dielectric relaxation, while linewidth evolution provides insight into static and dynamic disorder. These results demonstrate that reduced F-2DES measurements retain essential information on early-time spectral diffusion and solvation dynamics, offering an experimentally efficient and complementary route to probing environmental coupling in molecular systems.

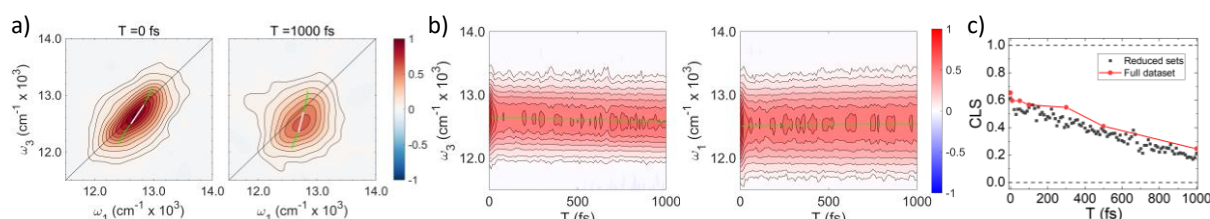


Fig. 1. (a) Absorptive F-2DES spectra of IR-125 in methanol at waiting times $T = 0$ fs and $T = 1$ ps. The lines indicate the extracted center-line slope, showing the loss of excitation–detection frequency correlation with increasing population time. (b) Emission frequency–resolved $S(\omega_3; T)$ and excitation frequency–resolved $S(\omega_1; T)$ contour maps obtained from reduced scans with fixed excitation coherence time ($t_1 = 0$ fs) and fixed detection time ($t_3 = 0$ fs), respectively. Symbols mark the extracted spectral peak maxima at each T . (c) Comparison of the CLS decay extracted from full F-2DES spectra (solid line) with a reduced dataset metric derived from the relative evolution of excitation- and emission-resolved peak positions (scatter points). The initial agreement highlights that reduced measurements capture the dominant early-time spectral diffusion dynamics.

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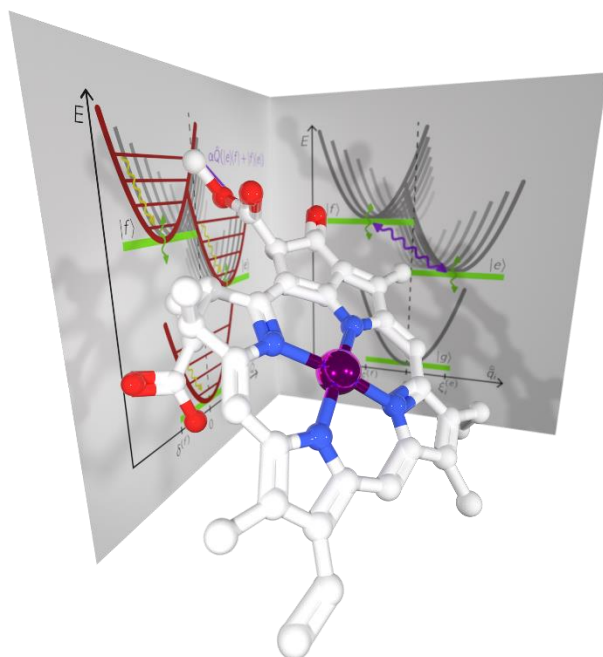
Ultrafast Non-Adiabatic Dissipative Energy Dynamics of Porphyrins: from Explicit to Implicit Model

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Theoretical spectroscopic models involving diabatic coupling between excited electronic states have proven to be successful in describing the static and time-resolved optical spectra of porphyrin molecule derivatives—notably of various chlorophylls which have a vital role in pigment-protein complexes of the majority of photosynthetic organisms [1]. Common modeling approach is based on the addition of selected vibrational modes that facilitate the diabatic coupling into the system Hamiltonian using the number basis representation. However, this approach has severe scaling limitations. The system Hamiltonian of a single molecule increases exponentially in size with the number of vibrational modes explicitly added and linearly with the number of vibrational levels included, i.e. the level of lattice truncation. Propagating such a model in time and optimizing its physical parameters to correspond to the experimental data (e.g. absorption spectra, 2DES) is a challenging task even in the case of a single molecule [2] or a dimer [3]. Large molecular complexes composed of dozens of molecules are intractable.



We propose a new approach to alleviate this limitation by transferring the explicit vibronic modes to the thermodynamic bath and performing normal mode analysis [4]. As a result, a purely electronic system Hamiltonian with a set of new spectral densities/bath correlation functions is obtained. This procedure substantially reduces the system Hamiltonian size while preserving all non-adiabatic effects present in the original model. Besides, it also lowers the overall coupling strength between the system and the bath rendering perturbative approaches—like the Redfield theory that operates in the limit of weak system-bath coupling—more suitable.

We compare the new implicit approach using derived spectral densities with the explicit one.

We investigate its validity across a range of physical parameters such as diabatic coupling strength and forms, energy gap sizes, number of vibronic modes, and system-bath interaction strengths. In the long run, the new model should enable us to study the dynamical properties of large molecular complexes in greater detail and shed light on the role of non-adiabatic effects in photosynthetic complexes as well as in other systems. Under the assumption of uncorrelated site environments in molecular complexes, this method can be straightforwardly used to simulate multidimensional spectroscopic experiments on the level of theory that effectively goes beyond the Born-Oppenheimer approximation.

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Using Polarization Dependent 2D Electronic Spectroscopy to Reveal Hidden Vibronic Mixing in Photosynthetic Systems: A Theoretical Perspective

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Photosynthetic light harvesting in the phycobilisomes of cyanobacteria and red algae has attracted considerable attention due to its exceptionally high energy transfer efficiency. A key step is within the allophycocyanin (APC) units of the phycobilisomes, where $\alpha 84$ (donor) and $\beta 84$ (acceptor) form a strongly coupled dimer, and the dimer exhibits ultrafast downhill energy transfer from the $\alpha 84$ to $\beta 84$. Previous studies have indicated that vibronic mixing involving coherently coupled electronic and vibrational states plays an important role in $\alpha 84 \rightarrow \beta 84$ energy transfer.[1][2] Two-dimensional electronic spectroscopy (2DES) experiments are well suited for probing couplings between electronic states; however, in near degenerate cases, spectral congestion makes it difficult to resolve these couplings. Furthermore, the existence of coherent coupling between $\alpha 84$ excited state to $\beta 84$ vibrationally excited state has never been clearly demonstrated. In this theoretical study, we apply the quantum Schrödinger-Langevin equation (QSLE) that effectively describes the influence of a dissipative environment on wave packet dynamics and treats strongly vibronic coupled systems to investigate the vibronic mixing in the $\alpha 84$ and $\beta 84$ system.[3] We show that conventional 2DES cannot unambiguously probe vibronic mixing, whereas the polarization dependence dependent 2DES could highlight spectral features. Electronic–vibrational coupling between chromophores leads to a modification of the original transition dipole moments. As such, a cross-peak-specific 2DES could be performed to increase cross-peaks due to vibronic mixing between $\alpha 84$ and $\beta 84$. In conclusion, this theoretical work provides a generalized approach that demonstrates the usefulness of the QSLE framework for strongly coupled electronic–vibrational systems, and offers insights into general vibronic mixing, as well as its role in excitation energy transfer that is specific to vibronic mixing, enabled by the existence of molecular aggregates and molecular crystals.

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2D-ES Analysis of Colloidal Plexcitonic Nanohybrids

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Polaritonic materials arise from strong coupling between the confined electric field of an optical cavity and the matter states of a quantum emitter (e.g., organic or inorganic excitons). They have attracted increasing interest for potential applications in photocatalysis, nanoscale energy transfer, and optoelectronics [1]. Plexcitonic materials are a subclass in which plasmonic nanoparticles act as optical cavities through near-field confinement [2]. These nanoparticles, synthesized in diverse morphologies via scalable wet-chemical methods, can be functionalized with selected emitters to couple to the enhanced plasmonic field.

The coupling strength depends on plasmon–exciton detuning, dephasing rates, emitter–surface distance, and the number of coupled emitters. Regimes from weak to strong coupling can thus be accessed, exhibiting distinct dynamical behavior. Strong coupling occurs when coherent energy exchange between hybrid plexcitonic states exceeds dissipative losses [3]. Understanding relaxation dynamics in this regime is essential for rational material design. Two-dimensional electronic spectroscopy (2D-ES) offers femtosecond resolution and excitation-frequency selectivity [4], making it a powerful tool for resolving ultrafast processes in strongly coupled systems.

Here, plexcitonic systems were prepared using J-aggregates of the cyanine dye TDBC as quantum emitters and gold nanorods (AuNRs) as plasmonic nanocavities, whose longitudinal plasmon resonance was tuned to match the excitonic transition. The resulting strongly coupled systems were investigated by two-dimensional electronic spectroscopy (2D-ES) in the boxcars geometry, enabling separate acquisition of rephasing and non-rephasing spectra.

2D-ES measurements were performed on plexcitonic samples spanning different coupling regimes. For each sample, data were collected at varying pulse energies to probe potential power-dependent dynamical processes within the plexcitonic nanohybrids. In addition, the influence of the excitation laser spectrum was examined by conducting 2D-ES experiments at different excitation wavelengths,

resonant with either the exciton reservoir or the lower plexciton branch. This selective excitation allowed different initial states to be preferentially populated by the first pulses, enabling us to investigate how the nature of the initially prepared state affects the subsequent ultrafast dynamics.

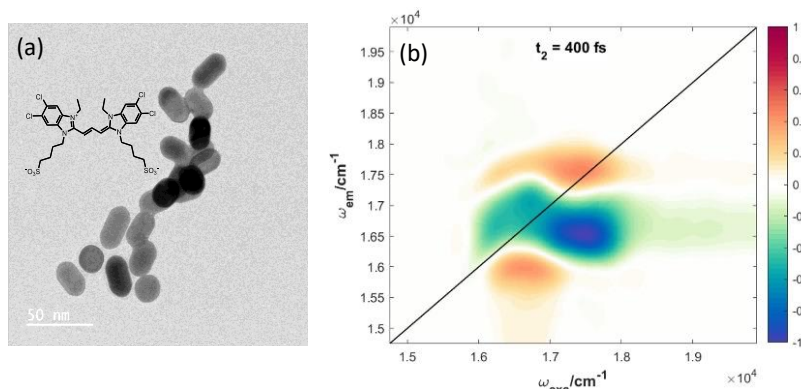


Fig. 1: (a) TEM image of AuNRs and molecular structure of TDBC; **(b)** 2D-ES map of the AuNR/TDBC plexciton at waiting time 400 fs.

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Plasmon-Exciton Hybridization and Helicity-Controlled Ultrafast Dynamics in Chiral Plexcitons

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We study plasmon–exciton hybridization and helicity-controlled dynamics in a chiral plexcitonic system consisting of intrinsically chiral gold helicoid nanoparticles [1] coupled to TDBC J-aggregate excitons. The steady-state optical response is described using a non-Hermitian Hamiltonian that explicitly incorporates electric and magnetic dipole plasmonic modes, radiative and non-radiative decay channels, and their coherent coupling to a narrow excitonic resonance. Rather than forming well-separated Rabi-split polaritonic branches, the observed helicity-dependent spectral splitting arises from interference between electric and magnetic dipole channels, producing Fano-like hybrid lineshapes.[2]

To probe the dynamical consequences of this interference-mediated hybridization, we perform polarization-resolved femtosecond transient absorption spectroscopy. The time-domain response reveals polarization-selective relaxation kinetics, indicating that switching light polarization addresses distinct hybrid channels with different dissipation pathways on the plexcitonic system. The combined non-Hermitian modelling and femtosecond measurements establish optical helicity as a control parameter that selects between interference-mediated hybrid eigenchannels and explains energy flow in lossy nanophotonic systems.

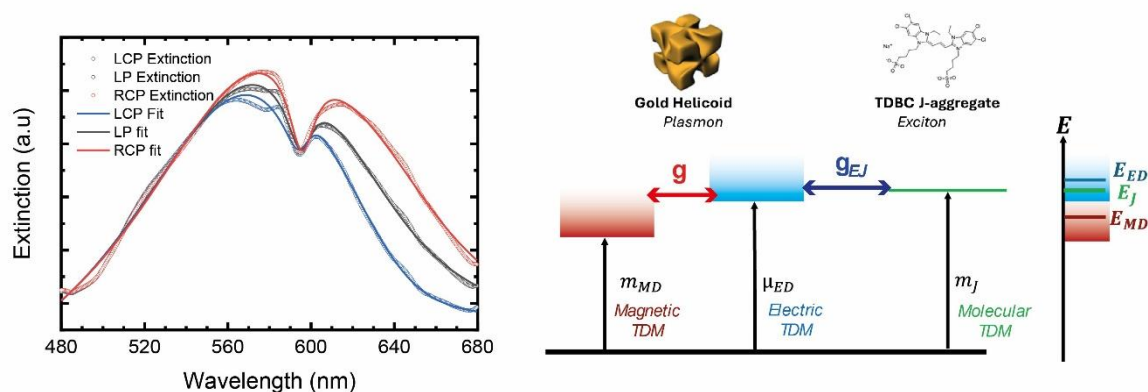


Fig. 1. Left: Steady-state Optical Response of Plexciton. Right: Schematic illustrating the coupling between electric dipole and magnetic dipole plasmonic fields on the helicoid and a discrete excitonic state in the TDBC J-aggregate.

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Coarse-Graining F-2DES: Back in Action

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Two-dimensional electronic spectroscopy (2DES) is a widely used nonlinear spectroscopic technique that allows measuring ultrafast dynamics of complex molecular systems, especially those involved in biological light harvesting. Simulating 2DES for large systems, however, remains computationally demanding. To address the computational challenges, a coarse-grained simulation approach was recently developed [1] using the existing spectral simulation framework based on the Numerical Integration of the Schrödinger Equation (NISE) [2]. This approach relies on defining weakly coupled segments from a system with multiple chromophores. While this method effectively captures excitation energy transfer, it does not fully resolve relaxation processes, which remain less accessible through conventional 2DES techniques.

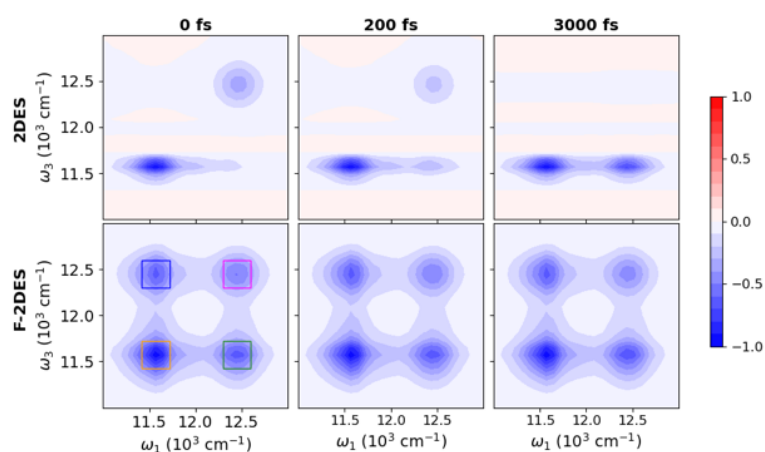


Figure 1. Coarse-grained simulations of 2D spectra for LH2 at 0, 200 and 3000 fs.
Top: Conventional 2DES. Bottom: Fluorescence-detected 2DES.

In this work, we extended this coarse-grained simulation method to model fluorescence detected two-dimensional electronic spectroscopy (F-2DES). This is a type of action detected spectroscopic technique in which the addition of a fourth pulse leads to the measurement of experimental observables such as photocurrent, or fluorescence. We achieved the implementation of this technique by considering a rate model that includes exciton-exciton annihilation, emission and non-radiative decay pathways [3].

To account for the photons emitted, each of the contributions to the signal are weighted by the one or two exciton quantum yields, Q_1 and Q_2 , respectively [4]. An advantage of the method compared to previous F-2DES simulation methods is that it allows for different parts of a system to have distinct quantum yields. The method was tested in the LH2 light-harvesting complex of purple bacteria (see Figure 1), as a well-studied guinea pig system. We found that key experimental features [5] are well predicted using this methodology. The simulations also showed to recover standard 2DES signatures when selectively disabling annihilation effects, thus confirming the hypothesis that cross-peaks in F-2DES, for the system under study, stem from annihilation processes.

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Fourth Order Extension of Multichromophoric FRET

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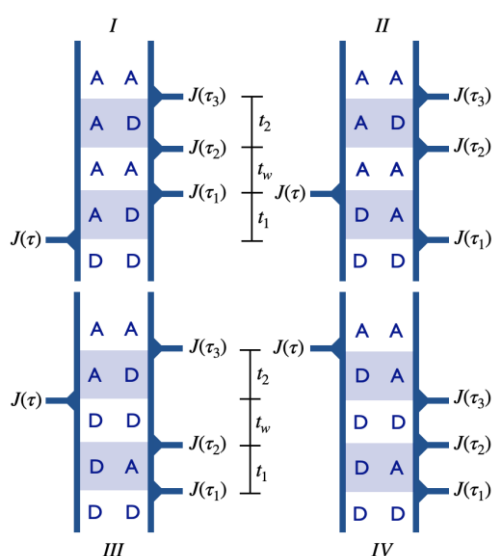
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Understanding the excited state energy transfer (EET) timescales between different molecular aggregates in photosynthetic systems is a key pursuit of current spectroscopic studies. In conjunction with spectroscopy experiments, models of two-dimensional electronic spectroscopy (2DES), and its fluorescence-detected counterpart (F-2DES) provide a powerful way to understand how specific system properties manifest in those spectra. In recent work, exciton population transfer rates are explicitly

used to simulate coarse grained (F-)2DES spectra [1, 2]. Therefore, it is important to have accurate and computationally efficient methods to obtain these transfer rates. Inspired by recent chlorosome model investigations [3], we developed a fourth-order [4] coupling extension to the second-order coupling Time Domain Multichromophoric Förster Resonant Energy Transfer (TD-MCFRET) method [5,6]. This new method allows us to compute inter-aggregate transfer rates in more intermediate parameter regimes than the lowest-order theory. This allows us to account for intra-aggregate relaxation timescales and (auto-)correlations in the Frenkel exciton Hamiltonian. With this new method we expand our options to compute population transfer rates in large systems and to improve coarse grained simulations of (F-)2DES spectra.



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Theory and Simulations of Fluorescence-Detected Two-Dimensional Electronic Spectroscopy: From Rigorous Quantum Mechanics to Simple Kinetic Models and Polarization Schemes

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Action-detected two-dimensional electronic spectroscopy is a collection of emerging nonlinear spectroscopic techniques that offer notable advantages over coherently detected two-dimensional electronic spectroscopy. Fluorescence-detected 2D electronic spectroscopy has been used to study molecular aggregates with very high detection sensitivity and spatial resolution. However, the information content that is extracted from action-2D electronic spectroscopy is not fully understood.

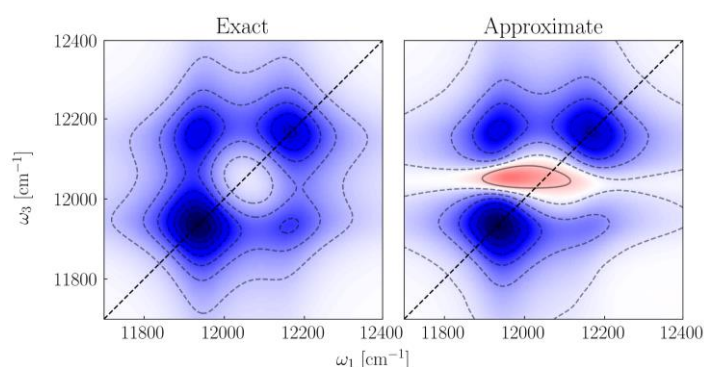


Fig. 1 F-2DES spectra at $t_2 = 0$ for a molecular dimer. Left: quantum-mechanically rigorous approach (exact observable). Right: approximate kinetic-model method.

Simulations have proven indispensable in interpreting the spectral features of various processes such as exciton-exciton annihilation, exciton dynamics, and incoherent population mixing. In this poster, I present our recent work in which we benchmarked commonly used approximate methods for simulating fluorescence-detected 2D electronic spectroscopy of a molecular dimer across a wide range of system parameters [1]. We compared

approximate spectra to the rigorously derived approach (Fig. 1), which only requires the knowledge of the molecular Hamiltonian and transition dipole moments. This work established the regimes of validity of approximate and practical methods for simulating action-detection 2D signals in relation to the time scales of competing processes in typical molecular aggregates. In addition, I will present our latest work in the derivation and simulation of polarization schemes to selectively suppress the coherent signal measured in 2D action spectra and isolate the incoherent mixing signals. Our results present an opportunity to disentangle coherent and multiexciton responses, maximizing the informational content and interpretability of action 2D spectroscopy.

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Prominent Signatures of Energy Transfer in Action-Detected Spectra of a Photosynthetic Protein

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Compared to conventional two-dimensional electronic spectroscopy (2DES), which detects a coherently radiated signal field, Action-detected 2DES (A-2DES) detects incoherent observables [1], such as photocurrent, photoionization, and fluorescence, created by the combined action of pump and probe. The unprecedented detection sensitivity of A-2DES is promising for probing large aggregates and assemblies such as photosynthetic cells, where Mie scattering can render the traditional coherent 2DES quite challenging. However, recent studies have revealed [2] that the excited state dynamics of the LH2 antenna protein, an archetypal large photosynthetic aggregate, while prominently seen in 2DES, are nearly absent in A-2DES. In such cases, the prominence of excited state dynamics is limited by a $1/N$ ratio[3], where N represents the size of the aggregate. This has led to questions regarding the effectiveness of A-2DES.

In this study, we found comparable dynamics of cyanobacterial photosynthetic protein APC probed by 2DES versus A-2DES. Simulations with a recently developed coarse-grained computational method for A-2DES [4] reveal that the slower annihilation rate in cyanobacterial proteins modifies the $1/N$ limit assumption of infinitely fast electronic equilibration across all the sites of the aggregate, leading to a reduced effective N for the system.

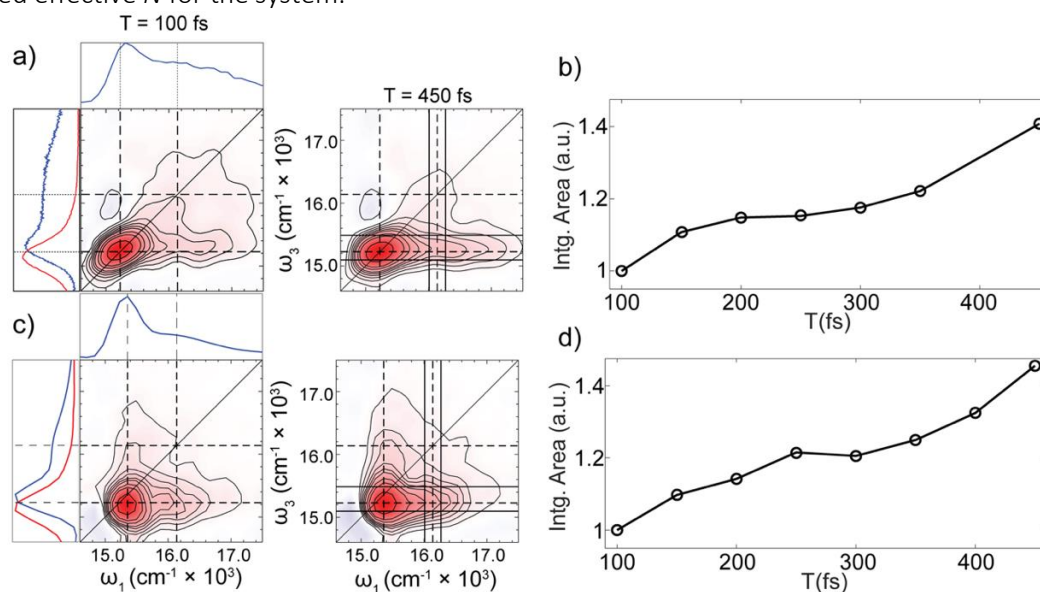


Fig. 1. Real absorptive (a) 2DES and (c) A-2DES spectra of the allophycocyanin trimer (APC) at T = 100 fs. Top panel overlays absorption for 2DES and fluorescence excitation spectrum for A-2DES while the side panel overlays the emission spectrum as well. (b, d) Integrated area enclosed by the horizontal and the vertical bands in the lower cross peak region is plotted for different T. This work is currently in preparation.

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P31

P32

Directly observing many-body correlations with two-dimensional electronic x-ray spectroscopy

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Nonadiabatic effects like spin-vibronic coupling and many-body electron-electron and electron-phonon correlations dictate charge transport efficiency in molecules and materials. Despite their esteem as drivers of excited state chemistry, these effects have consistently evaded direct experimental observation. The work presented here aims to uncover many-body correlations in the excited state with two-dimensional electronic x-ray spectroscopy (2D EX). This novel multidimensional spectroscopy combines coherent two-dimensional spectroscopy with tabletop attosecond x-ray spectroscopy. 2D EX fully correlates the valence shell electronic excitation with a core-level transition through an x-ray probe pulse generated by high harmonic generation (HHG). The attosecond probe (energy range ~30-100 eV) detects molecular and material dynamics with elemental, oxidation state, and spin state specificity through the inherent localization of core electrons, which are sensitive to the atom's time-dependent spin and charge. Resolving the electronic excitation frequency through two visible pump pulses maps the many-body interactions of the valence excited sector (i.e., the excited state dynamics) onto the many-body x-ray transition, resolving new spectral features and lineshapes that report on correlations. The dynamics can be measured across the waiting time with ~4-5 femtosecond precision.

I will discuss development of the first 2D EX platform and initial experimental data. This includes: 1) experimental design of the attosecond beamline and pump pulse pair creation; 2) preliminary 2D EX spectra of iron-containing semiconductor materials; 3) deconvolution algorithms for removing pump spectral features, which allow us to reach ~single cycle time resolution; and 4) modeling correlations through Feynman pathways and lineshape dynamics, which encode many-body interactions through measurable correlation functions.

Extracting high-order signals in two-dimensional spectroscopy via the mouse-tracking algorithm

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Contemporary spectroscopy is predominantly grounded in perturbative theory, with foundational techniques spanning from absorption spectroscopy (first order linear response) to transient absorption and two-dimensional spectroscopy (third order nonlinear response). In practice, improving the signal-to-noise ratio often requires increasing excitation intensity, which unavoidably activates higher-order nonlinear processes. While these high-order contributions encode rich physics such as collective excitations and many-body interactions. Consequently, the development of a robust spectroscopic methodology capable of efficiently isolating or identifying high order nonlinear responses has become an urgent and unmet need.

Two-dimensional (2D) spectroscopy achieves the successful isolation of third-order nonlinear signals through coherent control [1-3], non-collinear beam geometries provide spatial isolation via phase-matching [4], while collinear phase-cycling schemes decode individual quantum pathways through programmed phase variations [3]. However, once the experiment requires the resolution or exclusion of higher-order nonlinear responses, both approaches face challenges, often necessitating a significant increase in the number of phase-cycling steps or more complex data processing, thereby substantially raising experimental and computational complexity.

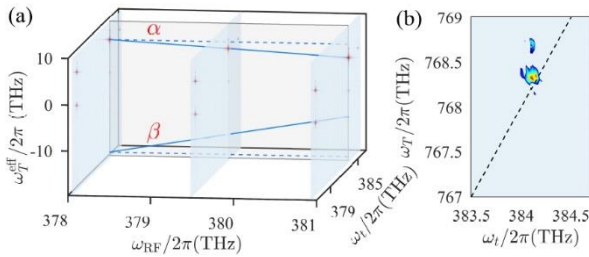


Fig. 1. Mouse tracking separation algorithm for two-dimensional spectroscopy. (a) Three spectra under different rotation frame frequency ω_{RF} . The blue solid lines correspond to slopes $k_{\alpha} = -1$ and $k_{\beta} = 3$ for the frequency shift $\omega_{\text{T}}^{\text{eff}}$ of the 3rd- and 7th-order signals, respectively. (b) The sorted 7th-order signals.

In this work, we present a computationally assisted spectroscopic protocol for isolating and reconstructing high-order nonlinear signals using a partially collinear two-dimensional spectroscopic geometry. The method employs a rotating frame to induce a spectral shift, extending beyond conventional approaches for reducing the sampling rate. A "mouse tracking" algorithm is then used to identify the appropriate shift rates corresponding to the perturbation order, enabling the isolation of high-order signal contributions. Notably, the technique does not require prior knowledge of the nonlinear order; for any high-order component, it achieves signal

recovery with only a fixed, minimal number of phase-cycling steps and just two measurement repetitions.

Our results establish an effective framework for isolating and identifying high order signals in two-dimensional spectroscopy, and we demonstrate its power by cleanly extracting the 7th order nonlinear response from a rubidium atomic vapor. This approach opens a practical route for probing ultrafast many body dynamics in complex quantum and materials systems.

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Generation of few-cycle pulses from industrial grade Yb:laser through cascaded post-compression

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Two-dimensional electronic spectroscopy (2DES) often requires broadband ultrashort pulses with pulse energy of tens to hundreds of micro-Joules, excellent temporal pulse fidelity, spatial homogeneous mode, and the possibility to control polarization to resolve ultrafast electronic dynamics in complex systems [1–4]. While modern lasers can meet individual requirements, combining bandwidth, energy, stability, and usability in a single commercially available system remains challenging [5]. We address this by a cascaded two-stage hollow-core-fiber (HCF)-based post-compression platform designed specifically for spectroscopic applications.

The system employs two sequential noble-gas-filled HCF spectral-broadening stages with independent dispersion compensation, enabling large compression factors (≥ 30) while preserving pulse quality.

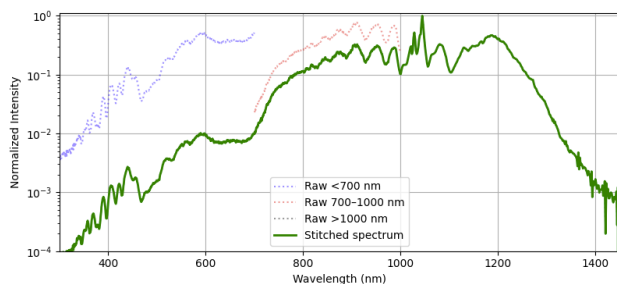


Fig. 7. Spectrum after the second post-compression stage (before compressor)

Custom dispersive mirrors and thin fused-silica wedges provide broadband phase control, with the second stage pre-compensating dispersion from air and downstream optics.

The enclosed system (SAVANNA², UltraFast Innovations) has a footprint of $60 \times 320 \text{ cm}^2$ and is driven by an industrial Yb laser delivering $500 \mu\text{J}$, 170 fs pulses at 1030 nm and 10 kHz. Polarization control of the few-

cycle output is implemented by an all-reflective, low-dispersion quarter-waveplate (KAUAI, UltraFast Innovations), between 650 and 1400 nm.

After the first HCF stage, the 170 fs pulses are compressed to 43 fs with $\sim 80\%$ transmission. The second stage broadens the spectrum to 700–1350 nm (1% level), extending to 400–1400 nm at 0.1% (see Fig.1). Following compression, we obtain 5.7 fs pulses, corresponding to sub-two optical cycles at 1030 nm. The transmission of the second stage is $\sim 70\%$, resulting in an overall system throughput exceeding 55%.

The demonstrated cascaded HCF platform delivers sub-two-cycle, octave-spanning pulses with high energy, stability, and broadband polarization control in a compact, user-friendly system. It provides a practical light source for advanced 2DES and related ultrafast spectroscopy techniques.

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Spatially resolving energy transfer in carbon nanotube films with ultrafast 2D confocal microscopy

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Exciton transfer and diffusion are intimately related phenomena controlled by the microscopic aspects of materials in next generation devices[1-3]. Typically, one does not happen without the other, although how they are related is not always directly measured, but inferred. Transfer and relaxation are most often studied using transient absorption (TA) and ultrafast 2D spectroscopies. These techniques measure energy flow with time, but do not spatially map the flow of energy[4-5]. Exciton diffusion is often measured using TA microscopy, but typically with narrow band lasers that are insensitive to energy relaxation[6]. Thus, most studies measure one or the other, but do not directly measure both energy transfer and diffusion simultaneously.

We have built a next-generation ultrafast scanning microscope called 2D-Scanning Microscope (2D-SM) that measures broadband 2D spectra with spatial resolution that can excite and interrogate exciton dynamics at spatially distinct locations. The pump pulses are created using a pulse shaper, which can be narrowed for precise frequency resolution or shaped into a pulse train for generating a 2D spectra. As a result, our instrument gives the temporal information provided by multidimensional spectroscopy via lineshape and cross peaks with the spatial dynamics of excitons provided by TA microscopy.

Using our new apparatus, we directly imaged the exciton dynamics of a thin filmed single-walled carbon nanotubes (CNTs), a semiconducting material with exceptional charge transport and optical properties. We find that as the excitons diffuse, they hop between nanotubes, producing a spatially dependent energy pattern. This information is fundamental for understanding the way in which energy flows through these thin films and is useful in optimizing the design of photovoltaic devices.

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Two-Dimensional Ultraviolet Spectroscopy with Conventional Optics

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Two-dimensional coherent spectroscopy has been widely used in the infrared and visible regimes to investigate the complicated and ultrafast dynamics. Expanding this technique to ultraviolet (UV) regime is challenging, due to the limit of generating broadband UV light source and preparing phase-stabilized pulse pairs. Here, we introduce an approach to perform two-dimensional ultraviolet spectroscopy with high time resolution and phase stability, which using only the conventional optics.

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CNN-Based Full Lineshape Analysis in Two-Dimensional Coherent Spectroscopy

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Two-dimensional coherent spectroscopy (2DCS) has emerged as a powerful tool for resolving ultrafast dynamics with combined time and frequency resolution. In rephasing 2D spectrum, homogeneous and inhomogeneous broadening are commonly quantified through diagonal and cross-diagonal slice analysis, implicitly assuming a single homogeneous linewidth across the whole inhomogeneity [1]. When the homogeneous linewidth varies with energy across the inhomogeneous ensemble, conventional methods lead to ambiguous conclusions, as cross-diagonal slices extracted at different regions of the peak can yield inconsistent linewidth estimates. Consequently, information encoded in subtle energy-dependent lineshape variations remain largely intractable.

We present a physics-guided, machine-learning-assisted framework that enables direct extraction of frequency-dependent homogeneous linewidths from experimental 2DCS data in the presence of arbitrary inhomogeneous broadening. A forward model based on the bivariate spectral distribution function (BSDF) formalism [2] is used to generate large ensembles of realistic synthetic 2D spectra, spanning broad classes of inhomogeneous oscillator distributions and energy-dependent homogeneous dephasing rates. These spectra are used to train a convolutional neural network (CNN) for direct inference of (i) the underlying inhomogeneous oscillator distribution $g_{1D}(\Omega)$, and (ii) the frequency-dependent homogeneous linewidth $\gamma(\Omega)$. By learning the nonlinear mapping between full 2D spectrum and these parameters, the approach bypasses reliance on orthogonal slices. This enables systematic extraction of linewidth variations that appear as energy-dependent cross-diagonal changes [3,4], which cannot be reliably studied using traditional analysis.

A schematic in Fig. 1 highlights (A) a rephasing 2D spectrum exhibiting frequency-dependent cross-diagonal broadening across the inhomogeneous distribution, (B) representative cross-diagonal slices extracted at different energies, and (C) the corresponding oscillator distribution $g_{1D}(\Omega)$ and homogeneous linewidth $\gamma(\Omega)$ as functions of energy obtained from our CNN-based inference.

The framework is designed for complete lineshape analysis of 2D spectra, in which all features of the diagonal peak contribute to parameter inference, making it well suited for complex systems with non-trivial linewidth behavior. The robustness of the method is demonstrated by applying it to previously published datasets [2,3] acquired under different experimental conditions across multiple sample systems. Importantly, no prior assumption of the functional form of $\gamma(\Omega)$ is required, allowing spectroscopic insight to emerge directly from the data. This approach extends the interpretive power of multidimensional spectroscopy in complex condensed-matter systems and provides a pathway to accessing energy-dependent dephasing phenomena, including mobility-edge-like behavior [4].

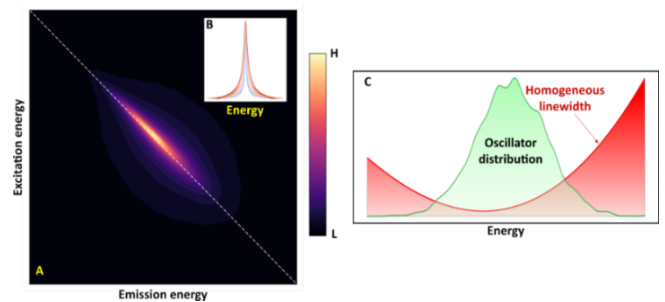


Fig. 1. Schematic of a rephasing 2D spectrum exhibiting energy-dependent cross-diagonal slices and the extracted homogeneous linewidth and oscillator distribution.

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Incorporating population-to-population, coherence-to-population, and coherence-to-coherence transfer processes in the response function expressions for multi-level systems in four-wave mixing experiments

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Typically, the third-order nonlinear optical response functions of multi-level systems does not include any relaxation¹. Such relaxation processes include incoherent population transfer (population-to-population conversion), coherence-to-population transfer, or coherence-to-coherence transfer. In this work, we derive the third-order nonlinear optical response functions, $\{\sum_{i=1}^6 R_i(t_1, t_2, t_3)\}$, for a multi-level system while phenomenologically incorporating all such processes. The derivation incorporates full complex-valued time correlation functions to describe the transition frequency fluctuation correlation dynamics of the states and also allows for the coupling of multiple states to common bath (environment) modes^{2,3}.

Considering a minimal model comprising two excitations, a simulation of the spectral manifestations of population-to-population and coherence-to-population transfer will be shown on two-dimensional (2D) electronic spectra⁴. In the incoherent transfer case, the spectral positioning of the signals at the diagonal peaks and cross peaks will be critically analyzed under different regimes of transfer rates and correlation timescales (see Fig. 1). For the coherence-to-population transfer process, we will show that strongly correlated states, that is, states coupled to common bath modes, and excitonic coupling in the regime $V \sim \frac{\Delta}{2}$ will be suited. Here, Δ represents the energy-gap between the two states participating in the relaxation. A fast relaxation rate and smaller energy-gap between the states are likely to generate stronger signals on the 2D spectra, and we will clearly show that large energy-gaps (in the range of thousands of wavenumbers) are unlikely to generate observable coherence-to-population transfer signals, albeit they have been proposed in previous works⁵.

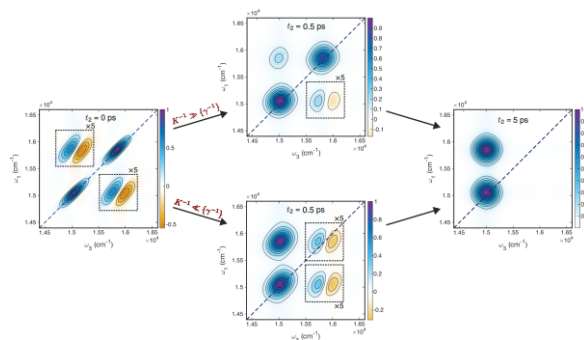


Fig. 1. Simulation of the 2D electronic spectra of two excitations with zero excitonic coupling, and a complex-valued frequency fluctuation cross-correlation function, which implies the states are coupled to common bath modes. The incoherent hopping signatures have been shown here. At $t_2 = 0$, the spectrum comprises two diagonal peaks and two quantum correlation (QC) cross peaks^{2,3}. The panels in the center describe the two asymptotic limits: (1) $K^{-1} \gg \{\gamma^{-1}\}$, and (2) $K^{-1} \ll \{\gamma^{-1}\}$, where K^{-1} describes the transfer timescale, and $\{\gamma^{-1}\}$ describes the set of system correlation timescales. Case (2) is interesting, where the fast transfer rate results in QC-type cross peak structure even at the diagonal. The panel to the right describes the spectra post transfer.

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Multiple Davydov Ansatz For Accurate Non-Markovian Dynamics In Multiple Thermal Baths

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Open quantum systems coupled to multiple thermal reservoirs are central to nanoscale heat transport, quantum thermodynamics, and energy-transfer processes in condensed-phase and light–matter systems. However, accurate real-time simulation of such systems remains challenging because bath memory, finite-temperature fluctuations, and nonequilibrium effects between reservoirs can strongly influence the system dynamics. Standard Markovian master-equation approaches, such as Lindblad or Redfield theories, often neglect these memory effects, while numerically exact methods such as hierarchical equations of motion and quasi-adiabatic path integrals may become computationally demanding for multi-reservoir systems, low-temperature regimes, and broad parameter scans.

Here, we present a multiple Davydov Ansatz framework for simulating non-Markovian dynamics in a two-bath spin–boson model with distinct bath temperatures. The method combines the multi-D2 variational Ansatz, the time-dependent variational principle, thermal field dynamics, and spectral-density discretization. In this formulation, the bath degrees of freedom are not traced out. Instead, each reservoir is represented explicitly by time-dependent coherent-state configurations, allowing bath memory and multimode system–bath correlations to be retained within a wavefunction-based formalism. Thermal field dynamics further maps finite-temperature reservoirs into an enlarged Hilbert space, enabling pure-state propagation while preserving thermal fluctuations.

The method is benchmarked over a broad range of bath coupling strengths, cutoff frequencies, and temperatures. Population dynamics obtained from the multiple Davydov Ansatz are compared with hierarchical equations of motion and quasi-adiabatic path integral calculations, showing accurate and convergent behavior across weak, intermediate, and strong coupling regimes. These results demonstrate that the multiple Davydov Ansatz provides a flexible and scalable route to real-time non-Markovian multi-bath dynamics. The framework is therefore promising for future studies of nonequilibrium heat transport, thermal rectification, quantum thermodynamics, and more complex multi-level or multi-qubit open quantum systems. [1]

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Q01

2D Electronic-Vibrational Spectroelectrochemistry: Vibronic Coupling in Organic Mixed Conductors.

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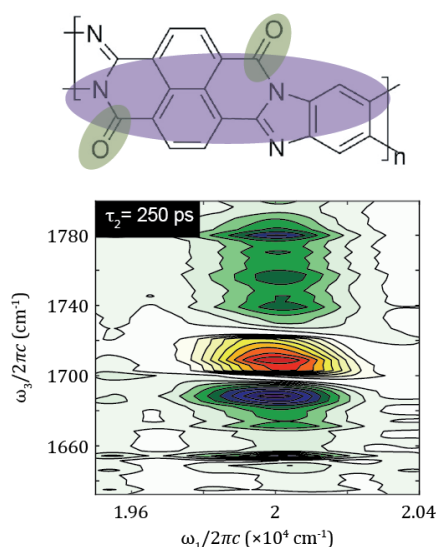


Fig 1. Chemical structure of BBL and a representative 2D EV spectrum of the carbonyl stretching region at $\tau_2 = 250$ ps.

Organic mixed ionic-electronic conductors (OMIECs) are an emerging class of materials with applications ranging from energy storage to neuromorphic computing [1]. The ladder polymer poly(benzimidazobenzophenanthroline) (BBL) exhibits high electron mobility and volumetric capacitance. However, the microscopic mechanism of charge transport, driven by electron delocalization and structural reorganization, has remained ambiguous from conventional structural and spectroscopic techniques [2].

To directly investigate the ion-electron correlations that drive charge transport in BBL, we employ a broadband two-dimensional electronic-vibrational (2D EV) spectrometer with a spectroelectrochemical cell to enable *in situ* electrochemical control [3]. Our apparatus utilizes few-cycle excitation pulses (500-1000 nm bandwidth), providing simultaneous resolution of the neutral and charged (polaron/bipolaron) electronic manifolds in BBL. The broadband excitation is enabled by a free-space Mach-Zehnder interferometer with attosecond phase stability, which accommodates interferometry for few-cycle sources

from visible to near-IR wavelengths. The ensuing dynamics are probed using femtosecond mid-IR pulses ($1100\text{--}4000\text{ cm}^{-1}$) to correlate the photoexcited dynamics with IR-active vibrational modes.

By systematically varying the electrochemical potential, we selectively populate the n-doped states, moving BBL through its distinct reduction transitions (insulator $\rightarrow$ conductor $\rightarrow$ insulator). By photoexciting the polaron and bipolaron bands and monitoring how the backbone vibrational modes (e.g., C-N, C=C, and C=O stretches) respond, we evaluate vibrational relaxation times and spectral diffusion as a function of applied potential. These experiments allow us to directly correlate BBL's conductivity with microscopic vibronic coupling strength, which maps how the electronic, structural, and ionic degrees of freedom drive efficiency in OMIECs.

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Same Same but Different: Vibronic Coupling to Selectively Target Subpopulations

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Selective electronic excitation of specific molecules or subpopulations within a mixture is challenging when their ultraviolet/visible (UV/VIS) spectra are nearly identical. **V**ibrationally **P**romoted **E**lectronic **R**esonance (VIPER) excitation achieves this by combining the selectivity of infrared (IR) excitation, which targets specific molecular subpopulations, with UV/VIS excitation, which triggers the desired processes. The IR excitation modulates the UV/VIS spectrum of the selected subpopulation via vibronic couplings, shifting it into resonance with a subsequent off-resonant UV/VIS pump pulse (Fig. 1).^[1-3] We demonstrate that 2D Vibrational Electronic (2D-VE) spectroscopy is an ideal tool to investigate these IR-induced UV/VIS modulations. In photocatalysis, control over product formation and stereochemistry is often compromised when multiple subpopulations engage in competing reaction pathways. Here, VIPER excitation allows for selective electronic excitation discriminating between molecular subpopulations involved in different interactions. For our study we investigate the dynamic equilibrium of Schreiner's thiourea catalyst^[4] binding a coumarin substrate (Fig. 1). We identify the CO stretching vibration of the substrate to be vibronically coupled and also to differ in wavenumber between bound and free state, rendering it an ideal handle for our VIPER 2D-IR studies. Beyond direct vibronic coupling, we demonstrate that vibrational energy transfer (VET) from the catalyst to the substrate via hydrogen bonds can also effectively modulate the UV/VIS spectrum of the substrate in the complex, offering an alternative path for selective electronic excitation. Furthermore, we explore how IR-induced modulations of UV/VIS bands can be used to discriminate between isomers of a photoswitch or between different leaving groups from the same photocages via VET.

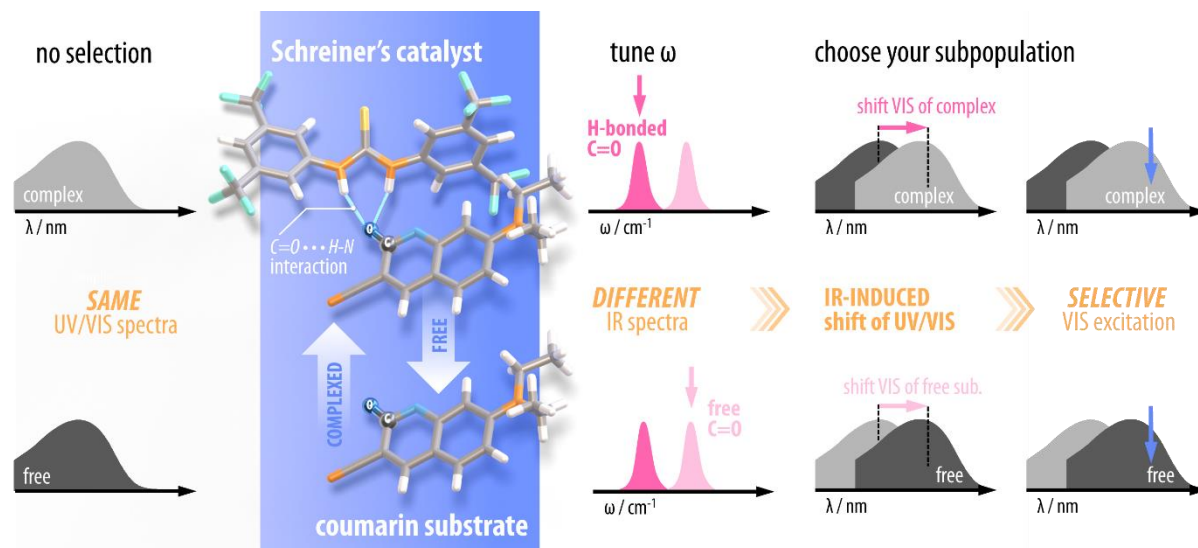


Fig. 1. Complex motif in photocatalysis, the substrate can either be free or H-bonded to the catalyst. Both subpopulations absorb in the same UV/VIS range, making selective electronic excitation challenging. IR excitation can selectively address different subpopulations and typically induces a red-shift of the UV/VIS absorption of the targeted subpopulation, enabling selective electronic excitation.

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Intra-Defect and Defect-Lattice Coupling in Diamond Revealed by Two-Dimensional Infrared Spectroscopy

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Atomic-scale defects in diamond underpin emerging technologies in quantum information, nanoscale sensing and photonics. Knowledge of the coupling between vibrational modes in these defects is essential for controlling their optical, magnetic and spin-related properties. We used two-dimensional infrared (2DIR) spectroscopy to probe the ultrafast vibrational dynamics of two diamond defects with different local structures, revealing coupling between local vibrational modes and the presence of hot ground states.

The H1a defect contains two IR-active orthogonal C-N-C stretch modes, with lifted degeneracy in mixed-N-isotope samples. 2DIR spectroscopy reveals ultrafast vibrational coupling between these modes

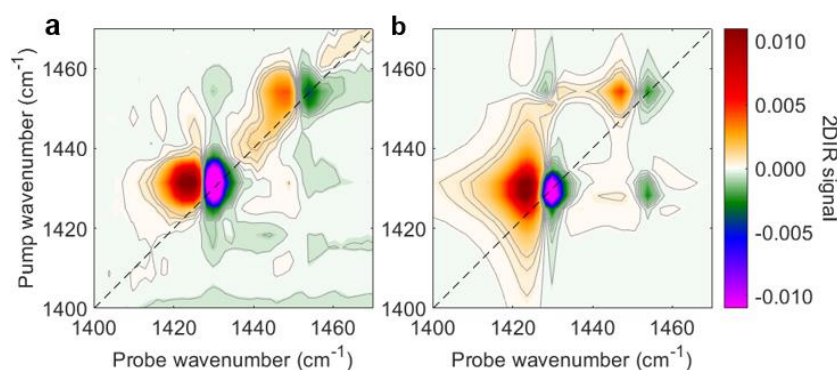


Fig. 1: **a** Experimental and **b** simulated 2DIR spectrum of the C-N-C stretch modes of the H1a defect in a mixed isotope N_{14} - N_{15} diamond sample, with parallel pump and probe polarizations.

which were previously thought to be energetically isolated (Figure 1a), and additionally resolves low-intensity cross-peaks in the perpendicular pump-probe channel belonging to a lower-wavenumber vibrational mode. These features enable a more complete vibrational energy structure of the defect system to be established. To interpret the 2DIR spectra quantitatively, we model the defect system using a vibrational exciton Hamiltonian in which the local C-N-C stretch modes are treated as coupled anharmonic oscillators whose delocalised modes arise from coherent mixing. The model generates the full set of third-order Liouville pathways for both the parallel and perpendicular pump-probe polarization channels, and reproduces peak amplitudes, diagonal widths and spectral diffusion effects (Figure 1b). From these simulations we extract the coupling strength, anharmonicities, transition dipole magnitudes, and dephasing timescales, thus providing a self-consistent description of the defect's vibrational structure and dynamics.

We also investigated ultrafast defect-lattice dynamics in diamond using the N_5 :C-H⁰ defect, an analogue of bond-centred hydrogen in semiconductors [1]. We demonstrate that pumping the defect's stretch mode generates localized non-equilibrium phonons which persist for great enough times in a hot ground state to transiently alter the potential energy surface of the defect. This work challenges the assumption of a rigid diamond lattice during defect relaxation, especially surprising due to diamond's high thermal conductivity, with implications for optical manipulation of defects in crystalline semiconductors and quantum defect engineering.

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Q04

Tracking reaction kinetics with 2D-IR spectroscopy

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Understanding the underlying mechanisms of chemical reactions is an important part of chemistry. Since IR spectroscopy is sensitive to structural changes, it provides a direct means to monitor reaction dynamics. However, tracking reactions via FTIR spectroscopy is often difficult due to overlapping contributions and unclear band assignment of the various intermediates.

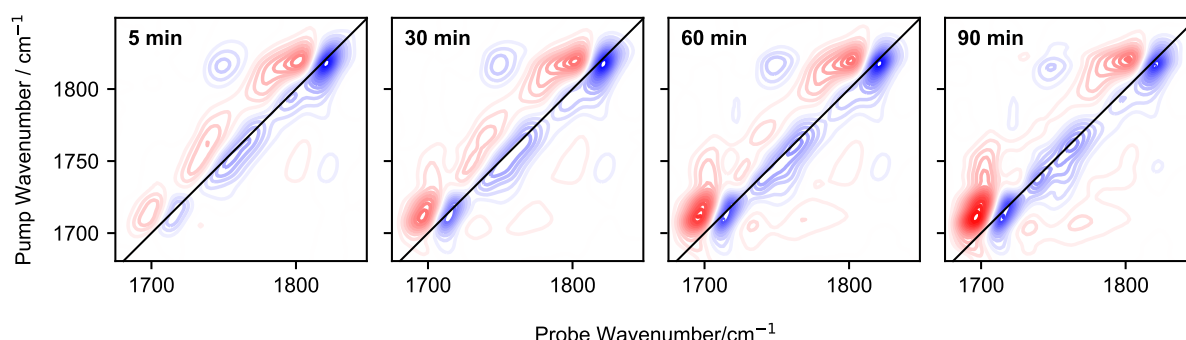


Fig. 1. 2D-IR spectra at $t_2=200$ fs for different times after initiation of the reaction.

In contrast, two-dimensional infrared spectroscopy (2D-IR) contains a plethora of information regarding the involved vibrations, couplings, structural correlations and lifetimes, making it possible to obtain detailed mechanistic insights superior to conventional FTIR spectroscopy. By monitoring reaction kinetics from minutes to hours with 2D-IR it is possible to achieve a better understanding of the different intermediates of the reaction, by extraction of their corresponding 2D-IR spectra.

We have recently demonstrated a first proof-of concept experiment monitoring enzymatic kinetics via 2D-IR [1]. Here, we apply the same to an organic reaction, the decomposition of acetic anhydride in DMSO, which is surprisingly complex. We also discuss difficulties of the approach, as the non-linear concentration dependency of the signal. By monitoring the reaction kinetics in situ with both FTIR and 2D-IR, we obtain both, the vibrational dynamics and the reaction kinetics along the reaction pathway and can resolve transient vibrational interactions that are inaccessible in linear spectroscopy, providing deeper insight into the reaction mechanism.

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Q06

Ultrafast Vibrational Dynamics in Concentrated Electrolyte Solutions

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In a typical electrolyte solution for electrochemical reactions, where concentrations of ions are often in the range of 1 M, ions are separated by only a few solvent molecules. Depending on the charge densities, fluctuations in local electric fields can be enormous, exerting significant influence on the redox processes central to electrochemical reactions. We have used two-dimensional infrared spectroscopy to probe vibrational dynamics in various concentrated electrolyte solutions such as LiSCN/N,N-dimethylformamide and tetrabutylammonium hexafluorophosphate in acetonitrile probing the CO₂ reduction catalyst Re(bpy)(CO)₃Cl. Ultimately, motional narrowing limits the sensitivity of nearly any spectroscopy method to probe the largest amplitude electric field fluctuations. Nevertheless, there is exciting promise to combine experimental measurements, computational simulations and updated theoretical treatments to rationally advance and leverage electric field effects in driving chemical outcomes.

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Q07

Vibrational Population Relaxation and Spectral Diffusion of the CN Stretching Mode of $[\text{Ru}(\text{bpy})(\text{CN})_4]^{2-}$ Investigated by Nonlinear Infrared Spectroscopy

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The vibrational dynamics of a solution contain detailed information about the intermolecular interactions between the solute and solvent molecules. The transition metal complexes in aqueous solutions have been examined due to the strong IR activity of the ligand modes and the symmetry of the molecular configuration. So far we have investigated the vibrational dynamics of metal complexes in water such as $[\text{Fe}(\text{CN})_6]^{4-}$,¹ $[\text{Ru}(\text{CN})_6]^{4-}$,² $\text{Fe}(\text{CN})_5(\text{NO})^{2-}$,³ and $[\text{RuCl}_5(\text{NO})]^{2-}$.⁴ In this work ultrafast vibrational population relaxation and spectral diffusion dynamics of the CN stretching mode of $[\text{Ru}(\text{bpy})(\text{CN})_4]^{2-}$ in D_2O were investigated by infrared pump-probe and two-dimensional infrared (2D-IR) spectroscopies. The vibrational couplings among the slightly different non-degenerated CN stretching modes were visualized in the 2D-IR spectra, which explains the remarkably short vibrational lifetime of one of the modes because of the overlap with the ground state bleach and the transient absorption signals of the different modes. The bimodal decay of the frequency-frequency time correlation function suggests that hydrogen-bond reorientation surrounding the solute and the vibrational energy redistribution in several tens of picosecond affect the vibrational dynamics.

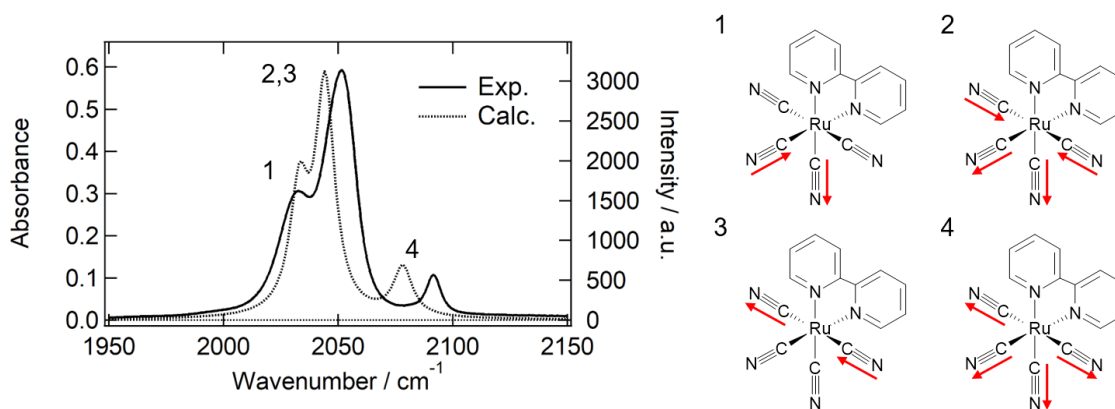


Fig. 1. IR absorption spectra of the CN stretching mode of $[\text{Ru}(\text{bpy})(\text{CN})_4]^{2-}$ in D_2O . The solid line represents the experimental result and the dashed line represents the simulated one, respectively. The oscillating atoms of the four modes are schematically depicted in the right side.

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Q08

Bringing the bulk to the surface: towards the application of surface enhanced 2D-IR spectroscopy for the study of solvation dynamics

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Vibrational spectroscopy is able to provide structural information about chemical systems and how they are influenced by their environments. The development of time-resolved techniques, such as pump-probe and two-dimensional infrared (2D-IR) spectroscopy, made possible mapping the structural dynamics of these systems, yielding information about the interactions between molecules and the solvent bath and species dissolved therein. However, infrared spectroscopy is limited by the low oscillator strength of vibrational transitions, which requires measurements to be conducted with high concentrations of the molecules of interest. This is a significant limitation when it comes to study solvent effects on chemical reactions: low solubility and aggregation make the comparison with the reaction conditions not always possible. One approach to circumvent this issue is to use surface enhanced spectroscopy, in which the molecule of interest is attached to a plasmonic material (typically gold) that is able to significantly enhance the oscillator strength of the vibrational transition, achieving monolayer sensitivity.

Surface enhanced 2D-IR has been applied to study the dynamics of monolayer, but studies in which the solvation dynamics of surface-bound species is compared to the one in the bulk are scarce. [1] In this study, we aim to systematically investigate under which conditions the surface bound species mimics more closely the bulk behavior. As model system we chose benzoic acid: its good solubility in multiple solvents allows to test a wide range of solvation environments. Moreover, despite the widespread presence of carboxylic and carboxylate groups in chemical systems the information about their dynamics when bound to monolayers are scarce, making their characterization an additional point of interest of this study.

Multiple conditions will be tested for the monolayer: (I) surface density, (II) linking geometry and (III) linker length. Difference surface density will be tested to assess the influence of neighboring molecules on each other solvation shell; benzoic acid will be bound to gold via a thiol linker either in the para and meta position to test which one yields the better solvent exposure, as they do not affect the overall orientation of the molecule with respect to the surface. [2] The influence of the length of the linker will be assessed as well as previous studies have shown that longer linkers tend to increase the interactions between neighboring molecules of the monolayer and dominate over the spectral diffusion observed in 2D-IR experiments. [3-4]

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Q09

Identification of Reactive Mode via Polarization Dependent Coherent Vibrational Spectroscopy

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Elucidating the reaction coordinate is fundamental to understanding chemical reaction mechanisms. Coherent vibrational spectroscopy (CVS) is a powerful tool for identifying reaction coordinates in photo-excited state. However, since CVS characterizes the reaction coordinate by analyzing the propagation of a coherent vibrational wavepacket, its application is typically limited to ultrafast dynamics occurring within the vibrational coherence time. Unveiling reaction coordinates in systems with weak reactivity or slow reactions remains a challenge.

To address these limitations, we proposed a method to identify reactive modes by analyzing the anisotropic response of the coherent vibrational wavepacket (CVW) obtained via polarization-dependent transient absorption (PD-TA) spectroscopy [1]. This approach utilizes the transition dipole moment angle rotation, which depends on the wavepacket position on the adiabatic potential energy surface. Unlike non-reactive modes the reactive mode is identified through its detection frequency dependent anisotropy. While these theoretical insights are available, they are yet to be experimentally confirmed. by solvent viscosity, but its reactive modes are also well established [2]. PD-TA measurements were performed by a home-built experimental setup with a pump pulse duration of approximately 16 fs.

Consistent with theoretical predictions, reactive modes exhibited anisotropy with a dependence on the detection frequency, whereas non-reactive modes showed weak such dependence. This work experimentally demonstrates the identification of reactive modes by observing the anisotropic response associated with CVW propagation, thereby suggesting the method to systems with slow reactions and weak reactivity. Such a method is expected to broaden the investigation of complex photochemical and photophysical processes beyond the conventional limits of ultrafast spectroscopy.

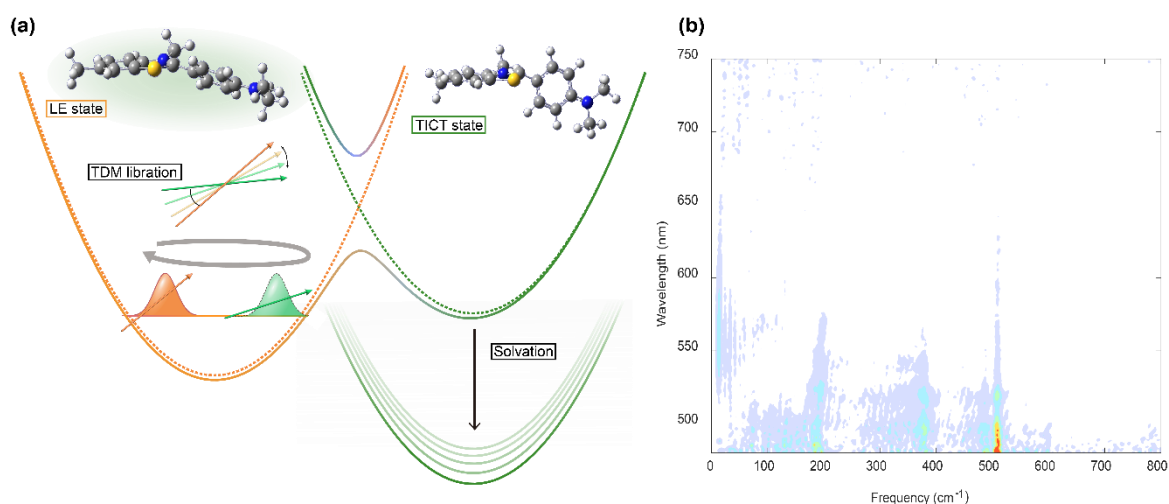


Fig. 1. (a) Energy diagram representing the photoexcited TICT dynamics of ThT (b) Parallel-polarized CVS of ThT/1-propanol solution

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Q10

Investigating Pre-Catalyst Solvation in Ionic Liquids using Ultrafast 2D-IR Spectroscopy

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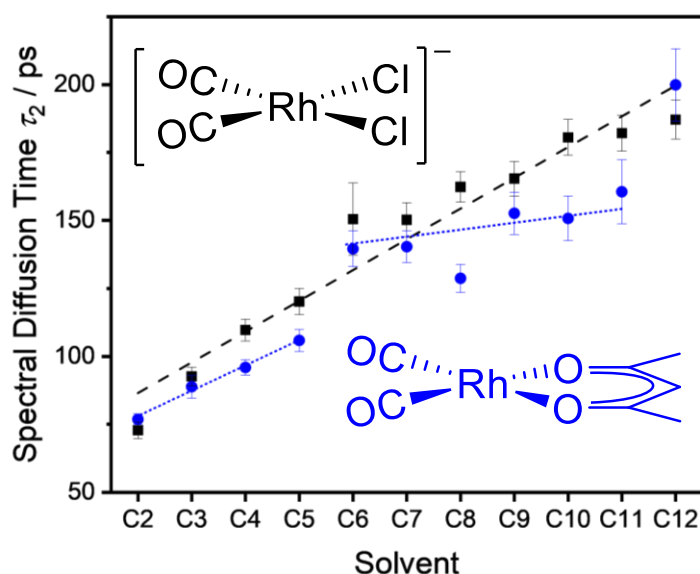


Fig. 1. Longer spectral diffusion time with IL chain length for the neutral compound **1** and the anion **2**, showcasing the different trend each exhibits.

Ultrafast 2D-IR spectroscopy was employed to investigate the solvation environment of catalytically relevant rhodium complexes in ionic liquids (ILs) *via* spectral diffusion dynamics. ILs have gained interest as media for catalysis due to their high chemical and thermal stability [1], and low volatility [2], specifically in industrial processes such as hydroformylation, for which $[\text{Rh}(\text{acac})(\text{CO})_2]$ (**1**) acts as a pre-catalyst [3]. However, relatively little is known about the solvation of metal complexes in these systems, due to the fact that ILs can display nanostructure [4], often leading to multiple solvation environments within the same solvation bath. Comparison between the neutral complex **1** and the two charged

species $[\text{Rh}(\text{CO})_2\text{Cl}_2]^-$ (**2**), and $[\text{Rh}(\text{bipy})(\text{CO})_2]^+$ (**3**), has yielded a detailed picture of the different solvation environments for charged and neutral compounds.

In this work, 11 different IL chain lengths ($n = 2 - 12$) of the type $[\text{C}_n\text{mim}][\text{Tf}_2\text{N}]$ were investigated. The measured timescales for spectral diffusion of the carbonyl asymmetric stretching mode of **1** showed an initial linear increase with alkyl chain length followed by a plateau coinciding with the formation of a non-polar subphase in the IL. This is a clear indication for preferential solvation by the alkyl chains. In contrast, the anion **2** showed carbonyl mode spectral diffusion dynamics that increased linearly across the whole series of IL, consistent with a different solvation environment dominated by charged species. Additionally, trends in peak position, peak width, vibrational relaxation, and anharmonicities as a function of IL chain length were found to be consistent with differences in preferred solvation environment of the metal complexes according to charge. All this taken together gives valuable insights into catalyst solvation and informs the solvent choice in these industrially important processes.

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Q11

Making Proteins Transparent: Selective FEIR Spectroscopy of a Chromophore in Complex Protein Environments

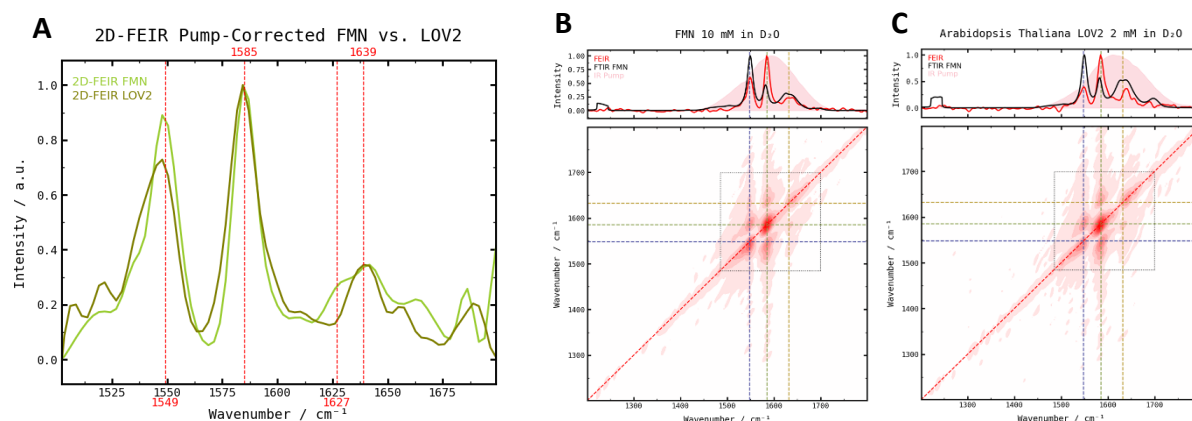
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Fluorescence-encoded infrared (FEIR) spectroscopy is a technique that uses the sensitivity of fluorescence to investigate vibrational dynamics in solution. By encoding IR-induced vibrational information into an electronically excited state and detecting the resulting fluorescence, FEIR enables detection down to the single molecule level. [1–4]

By applying Fourier transformation along the interferometric (IR-IR) delay and the encoding (UV-IR) delay, 2D-type FEIR maps were obtained which separate population and coherence contributions. Datasets recorded under different experimental conditions could be reliably compared with the help of a two-dimensional scaling function. We demonstrate the selective detection of the chromophore flavin mononucleotide (FMN) within the LOV2 domain of *Arabidopsis thaliana* phototropin 2, where FEIR isolates the chromophore signal from the strong background absorption of the protein backbone.



The demonstrated selectivity, combined with the scalability of the 2D-type FEIR maps, enables direct comparison between free FMN and protein-bound FMN, providing a probe of binding-pocket-induced spectral shifts and environmental effects. More generally, this approach allows the possibility to compare chromophore environments across structurally related systems, such as different LOV domains, offering a new route to systematically investigate protein–chromophore interactions.

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Q12

A single GLFG repeat which serves as a model system for NuPs forms hydrogel with reversible fibers

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The Nuclear Pore Complex (NPC) is a complex assembly of protein structures which allow regulated entry and exit of materials from the nucleus to the cytoplasm and vice versa. In the NPC, cargo entry and exit are specifically managed by the Nucleoporins (NuPs); intrinsically disordered gel-like motifs

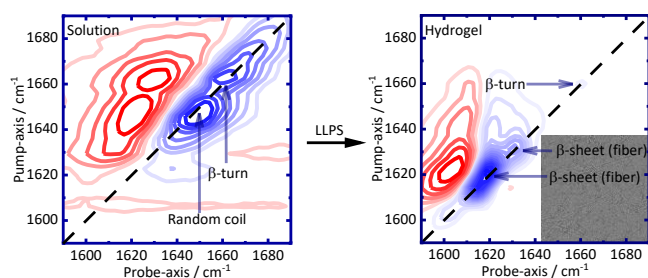


Fig. 1. 2D-IR spectrum of solution (left) and hydrogel (right) of GQ-12 in D₂O with parallel polarization between pump and probe pulses at 300 fs waiting time. The inset shows TEM image of fibers stained with uranyl acetate.

which are composed of repeated FG/GLFG/GFxG (x: other amino acid sequence) units and are formed by liquid-liquid phase separation from hydrophobic collapse of the repeating units. Interestingly, how the NPC machinery conducts itself in such a highly regulated fashion is still an open question in the field of nucleocytoplasmic transport. Addressing this conceptual gap will open avenues in health care such as in targeted gene and drug delivery, where it is quintessential to

tweak properties of delivery vehicles to negotiate the nucleocytoplasmic barrier, allowing targeted and maximum payload delivery. In that regard, the following question holds precedence: What kind of secondary protein structures contribute to the NuP's hierarchical assembly and how does this arrangement define the hydrogel's properties?

In this contribution, using two-dimensional infrared (2D-IR) spectroscopy, we provide a cohesive picture on LLPS of a peptide, GQ-12 (GGGGLFGNTQTQ), a model system which serves as the smallest building block of the NuPs. With 2D-IR spectroscopy we investigate the peptide's global secondary structural changes due to LLPS in the Amide-I' region, its physical properties namely the hydrogel's response to temperature and concentration, and residue-specific structural information by employing isotope labelling. We show that monomeric GQ-12 in solution is composed of random coil (RC) with β -turn which on self-assembly forms a hydrogel composed of parallel β -sheets fibers with cross- β arrangement (confirmed by negative staining transmission electron microscopy) and β -turn. By recording "on the fly" 2D-IR spectra of GQ-12 when it is undergoing LLPS, we show that the two distinct signatures at 1617 cm⁻¹ and 1632 cm⁻¹ (associated with fibers) increase at the expense of RC signature at 1645 cm⁻¹, and possess near-identical evolution kinetics, implying that the two vibrational features may likely arise from the same β -sheet structure and not necessarily from polymorphs. On increasing the temperature from 298 K to 353 K, two distinct behaviors of the hydrogel are observed; while some batches of the hydrogel form RC from fibers followed by fiber reassembly on cooling the peptide to room temperature, for several other batches the fibers are resistant to temperature and do not undergo LLPS. Interestingly, irrespective of the batch, the hydrogels undergo reversible LLPS with respect to concentration i.e. on serially diluting the hydrogel, solution phase is formed at the expense of the gel phase, implying that fiber formation is reversible. We find that ¹³C isotope labelling of GQ-12 (GGGGLFG*NTQTQ, *: ¹³C labelled residue) leads to a slightly broadened diagonal spectrum of RC, specifically between ~1610-1620 cm⁻¹ than in the unlabeled peptide, due to contribution from down-shifted frequency of uncoupled ¹³CO oscillators. However, the 2D-IR spectra of the labelled and unlabeled hydrogel look identical, implying that either the parallel β -sheets which compose the fiber are out of register or the labelled residue participates in forming a β -turn, leading to negligible coupling amongst the ¹³CO oscillators.

Q13

pH-Dependent Structural Alterations in LHCII Revealed by Two-Dimensional Infrared Spectroscopy

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Light-harvesting complex II (LHCII) is the major light-harvesting antenna, which is responsible for harvesting sunlight energy and photoprotection of Photosystem II. In this study, we assessed the structural changes in LHCII that are induced by lowering the pH. Absorption, circular dichroism and emission spectra of LHCII indicate that the trimeric LHCII samples were well intact and maintained in the light-harvesting state, exhibiting a single 3-ns exponential fluorescence decay. Fourier-transform and two-dimensional infrared spectroscopies reveal a frequency upshifting of Amide I band of some α -helical elements upon increasing acidity, which could be interpreted as a combination of refolding of a 3_{10} helix and a C-terminal coil to form new α -helices. Some sub-stoichiometry formations of β -motifs were also observed upon acidification with the β -sheet's size estimated at around 30-40 amino acids. This β -sheet can be formed with the N-terminal coil or co-folding between the N-terminus and the stromal loop. Our results demonstrate that lowering the pH induces significant structural changes in LHCII, but that they are not sufficient to switch LHCII into a quenched state. In the thylakoid membrane, such pH-dependent structural changes may enable specific protein-protein interactions between LHCII and other membrane proteins, in particular PsbS, that are required for non-photochemical quenching.

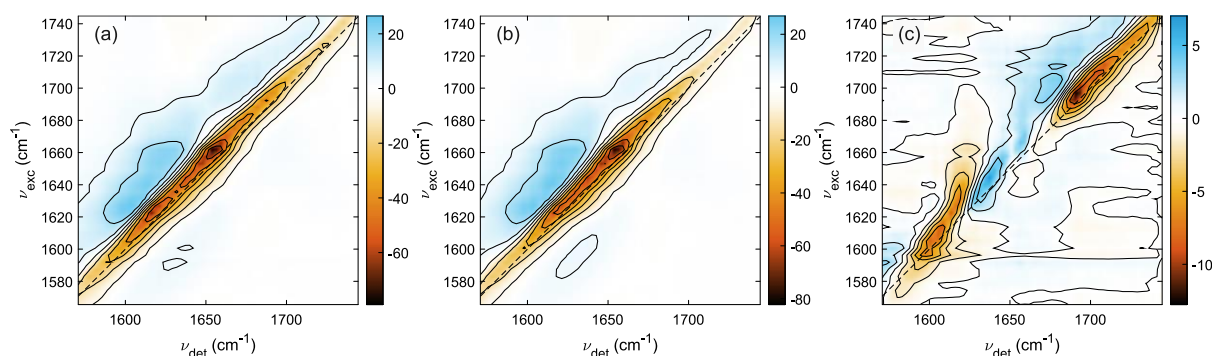


Fig. 1. (a-b) 2DIR spectra of Light-harvesting complex II measured at acidic (a) and neutral (b) pH. (c) The acidic-minus-neutral difference Δ 2DIR spectrum.

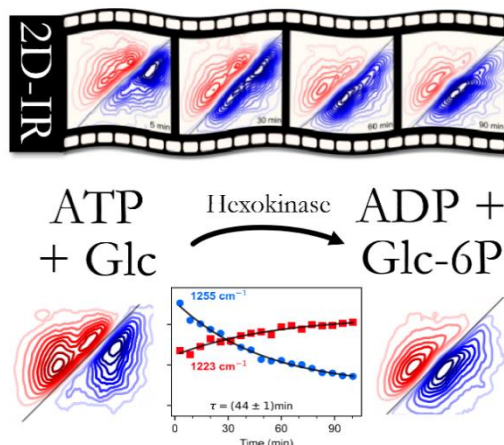
Q14

Tracking enzymatic activity with 2D-IR spectroscopy

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Two-dimensional IR (2D-IR) spectroscopy is now an established technique. However, so far it is mostly used to observe static samples. Here we use 2D-IR spectroscopy to measure the enzymatic activity of Hexokinase. Hexokinase catalyzes the first step of glycolysis, transferring a phosphate group from adenosine triphosphate (ATP) to glucose (Glc) to produce glucose-6-phosphate (Glc-6P) and adenosine diphosphate (ADP). This enzymatic reaction is crucial in biochemistry and medical diagnostics, such as Glc level assessments using the hexokinase/glucose-6-phosphate dehydrogenase method.

To monitor hexokinase activity in real time we observe the asymmetric stretching vibrations of the $(\text{PO}_2)^-$ -groups in ATP and ADP, located in 1100-1200 cm^{-1} region. Compared to Fourier transform infrared (FTIR), 2D-IR offers enhanced structural resolution and reduced solvent interference. By modelling the reaction kinetics, we extract the species associated 2D-IR spectra. Our results reveal distinct cross-peaks in the ATP spectrum absent in ADP, attributed to intramolecular dipolar coupling of $(\text{PO}_2)^-$ -groups.

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Q15

Carrier-donor Scattering Enhanced Multiple Carrier Generation at an Exceptionally Low Energy Threshold

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Multiple carrier generation (MCG), a process that generates two or more carriers from a single high-energy absorbed photon, holds immense promise for quantum technologies and photovoltaics [1]. However, high threshold photon energy of at least twice the bandgap ($2E_g$) and low efficiency hinders its practical applications [2]. Here, we address these challenges by leveraging carrier-donor scattering to excite secondary electrons from donor states positioned below the conduction band [3]. We experimentally demonstrated this idea in a model system of monolayer MoS_2 by exploiting electron-donating sulfur vacancy states. We observed MCG threshold energy at $1.12E_g$, a 44% reduction from its conventional energy limit of $2E_g$, and quantum yield increases up to >3 at $1.65E_g$ excitation energy. Our findings open a new avenue for MCG in the next-generation high-performance photovoltaics, biosensing, and quantum metrology applications.

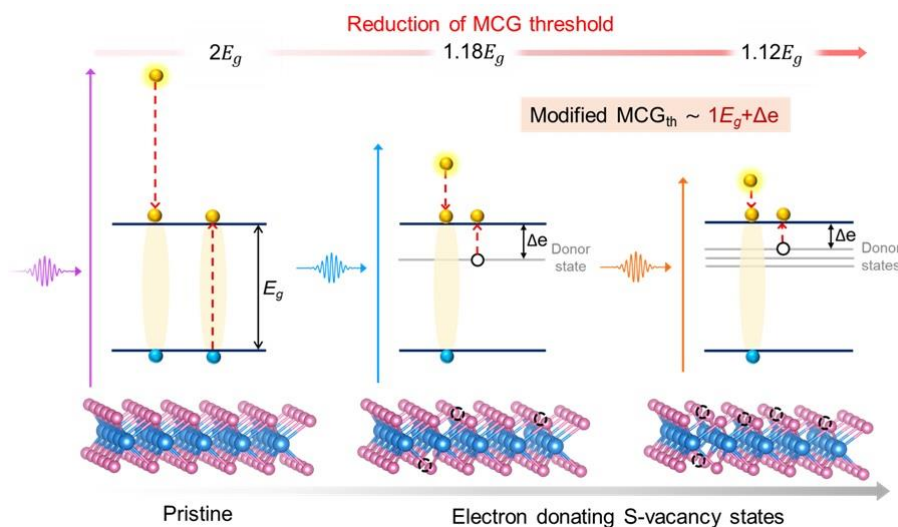


Fig 1. MCG in monolayer MoS_2 . With the increase in the density of electron-donating sulfur vacancy centers, the energy difference between conduction band minima and donor states (Δe) decreases, which subsequently decreases the MCG threshold energy ($1E_g + \Delta e$).

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Q16

Q17

Investigation Of Coherent Phonon Dynamics And ASE Performance In FAPbBr₃ With MACl Additives

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Metal halide perovskites are promising optoelectronic materials due to their high efficiency and simple fabrication [1]. While formamidinium (FA)-based perovskites offer a narrower bandgap and better efficiency than methylammonium (MA)-based counterparts, they suffer from lower structural stability. To address this, we used methylammonium chloride (MACl) as an additive to improve film quality [2].

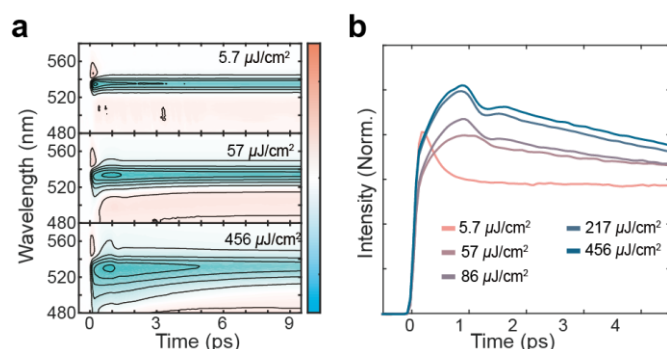


Fig. 1. (a) Fluence-dependent TA spectra of FAPbBr₃. (b) Time trace of PIB at 530 nm.

To investigate the relationship between MACl concentration and ASE efficiency, we performed Transient Absorption (TA) spectroscopy. By varying the excitation fluence, we observed significant changes in the photoinduced bleaching (PIB) band near 530 nm (Fig. 1a). Specifically, while the decay feature remained simple below the ASE threshold, a clear oscillation pattern emerged at fluences above the threshold (Fig. 1b). As our previous work indicates that multiexciton formation induces the population inversion required

for ASE, these oscillations are assigned to coherent phonons generated via exciton-phonon coupling under high carrier density. The simultaneous emergence of ASE and coherent phonons suggests a potential correlation between lattice dynamics and the lasing process.

Furthermore, analysis of PIB time traces revealed that the phonon-mean frequency shifts with MACl concentration, following a non-monotonic trend similar to the ASE threshold. Notably, the phonon frequency exhibited a distinct red-shift in samples with lower ASE thresholds, demonstrating a clear correlation between lattice vibrational modes and lasing efficiency. These results demonstrate how structural changes modulate exciton-phonon interactions to optimize ASE performance.

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Q18

Excitonic Interactions and Dynamics in Mono- and Bi-Layer WSe₂

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Interactions between particles underpin the properties and functionality of solid state materials. In the case of two-dimensional semiconducting transition metal dichalcogenides (TMDCs) the key particle is the exciton (and excitonic complexes) which dominates the optoelectronic properties of these materials [1]. Understanding how the exciton interacts with the environment around it – and how these interactions may be controlled – can enable further development of novel 2D optoelectronic devices, as well as an opportunity to explore fundamental physics questions in 2D systems and realise new quantum phases. Multidimensional Coherent Spectroscopy (MDCS) is well suited to measuring complicated interactions in a wide range of systems [2]. In this work, we perform rephasing measurements on a sample containing tungsten di-selenide (WSe₂) which is grown by chemical vapor deposition as both a monolayer, as well as bilayers with varying interlayer twist angle.

By measuring the homogeneous linewidth as a function of excitation density we can extract the exciton-exciton interaction strength, as well as the exciton coherence time at zero excitation density. The exciton-exciton interaction strength, found by the slope of the fit, is effectively the same for the monolayer and bilayers. However, the zero-density homogeneous linewidth is larger in the bilayers and increases as the twist angle goes from 30° to 3°. We attribute this to the transfer of an electron or hole to the other layer in the bilayer, and the formation of an interlayer exciton. By comparing the monolayer and different bilayer dynamics, we extract charge transfer times of 440fs and 1300fs for the bilayers with 3° and 30° twist angles. We suggest that the faster transfer time for the 3° twist angle is due to stronger coupling between the layers with small twist angle, and better wavevector matching at the band extrema.

We also determine the exciton-phonon interaction strengths by performing temperature dependent measurements of the homogeneous linewidth. The decoherence of the exciton in the monolayer varies slightly with temperature at $T < 35\text{K}$, driven by weak acoustic phonon scattering before the contribution from optical phonon interactions begins to dominate at higher temperatures. By fitting the temperature dynamics to the equation in Fig. 1b we determine an acoustic phonon coupling strength of 4.8×10^{-3} meV/K, and optical phonon coupling strength of 35 meV/K. For the bilayer region, the enhanced dephasing at low temperature due to fast charge transfer masks the impact of acoustic phonon scattering. At higher temperatures the optical phonon scattering rapidly becomes the dominant decoherence mechanism, revealing a coupling strength similar to that observed in the monolayer region.

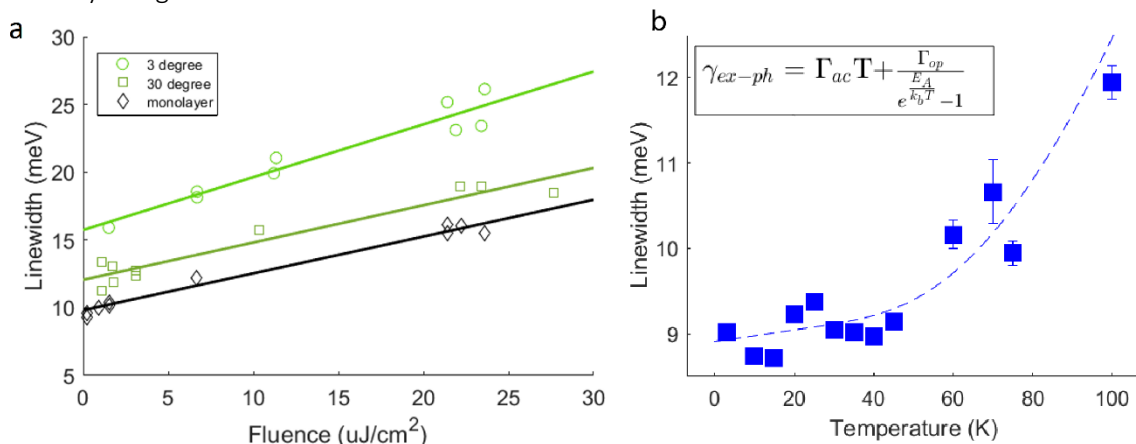


Fig. 1. a) Linear fits to the fluence dependent linewidth. b) Temperature dependent linewidth in the monolayer region.

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Q19

Ultrafast charge generation dynamics through interfacial energetic modulation for high-performance single-component organic solar cells with 14.8% Efficiency

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Bulk heterojunction (BHJ) organic solar cells have achieved high efficiencies but suffer from poor morphological stability due to phase separation after long-term operation. Single-component OSCs based on double-cable polymers, offer improved stability through covalently linked donor and acceptor units.[1] However, their efficiency remains limited by inefficient charge generation arising from extensive intermixed morphologies. Here, we report a fluorinated double-cable polymer, DCPY2-F, which achieves an outstanding efficiency of 14.8% with high short-circuit current density of 26.83 mA cm⁻². Ultrafast pump-probe transient absorption spectroscopy reveals that fluorination of DCPY2 into DCPY2-F accelerates interfacial charge transfer and long-range charge separation dynamics. The pump-push-probe spectroscopy and steady-state electroluminescence show that the faster interfacial charge transfer arises from a reduced reorganization energy and a correspondingly accelerated molecular reorganization process (2.5 ps vs. 0.8 ps). Despite comparable acceptor aggregate sizes with DCPY2, DCPY2-F also shows faster long-range charge separation dynamics, which we attribute to a narrower charge transfer states (CTs) energetic distribution. Molecular dynamics simulations further show that fluorination strengthens non-covalent interactions, promoting well-aligned intermolecular donor–acceptor interfaces. These structurally and energetically ordered interfacial CTs enable ultrafast, efficient charge generation, establishing interfacial energetic modulation via fluorination as a powerful strategy for high-efficiency and stable organic photovoltaics.

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Q20

Tracking the impact of ultrafast measurement on strongly confined perovskite nanoplatelets using asynchronous and interferometric transient absorption

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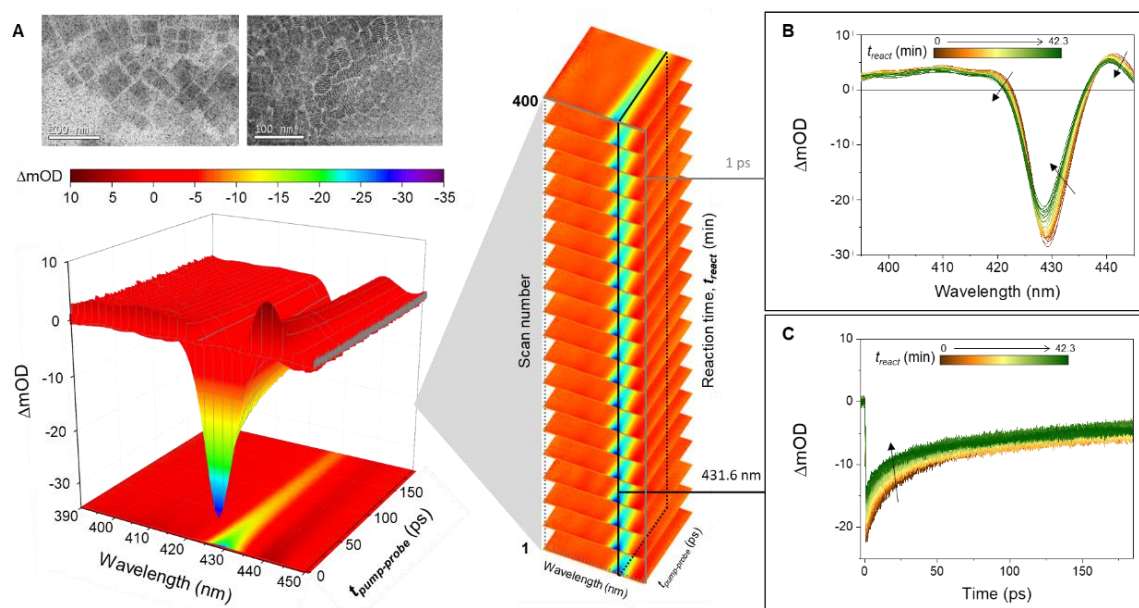


Fig. 1. **A.** TEM images of perovskite nanoplatelets and an averaged map of 400 asynchronous and interferometric transient absorption measurements. **B.** Spectral evolution during the measurement. **C.** Temporal evolution during the measurement

Developed and refined by our group, asynchronous and interferometric transient absorption (AI-TA) spectroscopy provides a rapid and powerful framework for observing excited-state dynamics with high temporal and spectral resolution. This presentation highlights our recent work, where we monitored the photophysics of in situ, light-induced transformation of colloidal CsPbBr₃ nanoplatelets (PeNPLs) using AI-TA. While these 2D nanomaterials show unique photophysical properties originating from strong quantum confinement, their structural instability poses quite a challenge for characterization. Using AI-TA, we captured the simultaneous evolution of excited-state spectra and exciton dynamics, as two-monolayer nanoplatelets coalesced under femtosecond laser irradiation. By analyzing continuously acquired datasets separately, we could follow the evolution of the transient absorption map over the course of the measurement. Spectrally, we observed a blueshift in the transient absorption signal, which we attribute to the preferential degradation of larger platelets. In the temporal domain, we could observe a progressive increase in the hot carrier thermalization lifetime. These findings allow us to establish the relationship between light-induced sample changes to excited-state dynamics. Moreover, it highlights the unique area of spectroscopy accessible by AI-TA, demonstrating this as an ideal tool for studying photochemical dynamics in unstable materials.

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Q21

Two-Dimensional Coherent Spectroscopy Provides Unambiguous Separation of Linewidths in MoSe₂

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Distinguishing homogeneous (dephasing) and inhomogeneous (disorder) broadening is crucial for understanding coherence in quantum materials. Conventional photoluminescence (PL) cannot decouple these effects due to correlated parameters. Two-dimensional coherent spectroscopy (2DCS) overcomes this limitation through simultaneous analysis of orthogonal spectral projections, achieving a 2-3x increase in parameter confidence and less correlated parameters.

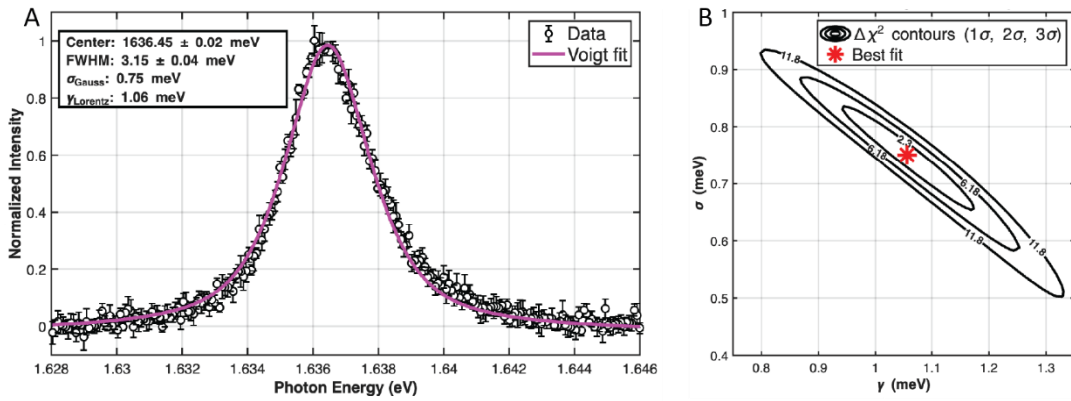


Fig. 1: Conventional PL analysis of MoSe₂ monolayer excitons. (A) PL spectrum with Voigt fit. (B) χ^2 error landscape exhibits diagonal, elongated "cigar-shaped" confidence contours, demonstrating intrinsic correlation and ambiguity in parameter estimation.

PL Voigt fits using calibrated uncertainties (N=5) gave strongly correlated linewidths (σ , γ) and total linewidths (~ 3 meV). χ^2 contours persisted across locations, indicating time-averaged PL's inability to rigorously resolve broadening sources.

x2DCS offers orthogonal slices: diagonal (disorder-dominated, σ) and cross-diagonal (dephasing-dominated, γ). By fitting averaged 2D spectra (N=10) with shared parameters, joint constraints from both projections yield circular confidence landscapes, even within simple models, robust against excitation-induced artifacts.

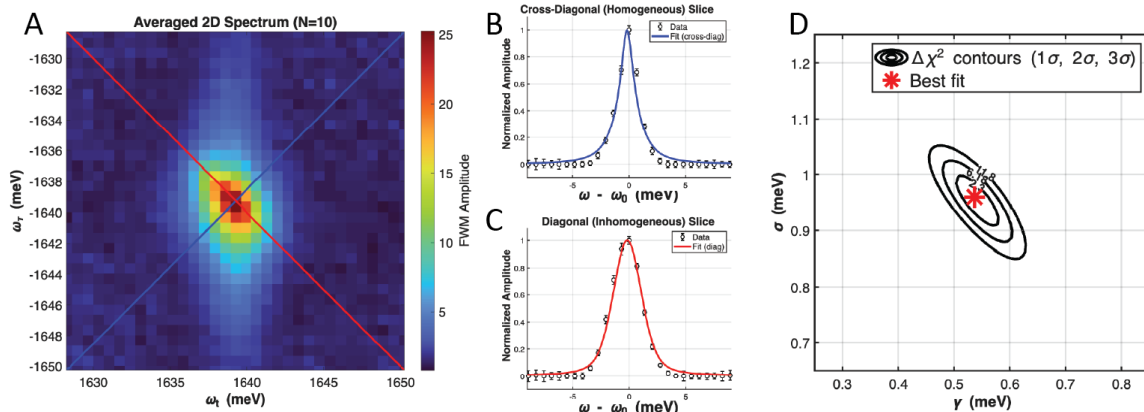


Fig. 2: 2DCS analysis enables clear separation. (A) Averaged 2D FWM spectrum of the MoSe₂ exciton. (B/C) Diagonal/Cross-Diagonal fits as defined by ref. [1]. (D) χ^2 contours from simultaneous fit show compact and more circular confidence contours than the linear case.

Both methods yield similar total linewidths, but only 2DCS ensures unambiguous separation via compact, circular error regions, enabling quantification of coherence dynamics in quantum materials.

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Q22

Investigating The Ultrafast Carrier Dynamics Of Two-Dimensional Semiconductors With Time- And Energy-Resolved Photoemission Electron Microscopy

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Semiconducting transition metal dichalcogenides (TMDs) are layered two-dimensional materials which are promising candidates in optoelectronic device applications. The well-studied group 6 TMDs such as WSe₂, WS₂, MoS₂ and MoSe₂ demonstrates in-plane isotropic optical response because of its symmetric crystal lattice. In contrast, group 7 TMDs like ReS₂ display in-plane anisotropic optical response with respect to the crystal b-axis because of the reduced crystal symmetry from the additional Re-Re bonds along the b-axis. The optical polarization anisotropy in ReS₂ could serve as an additional tuning parameter in using TMDs for optoelectronic device applications.

We investigate the ultrafast carrier dynamics in bulk ReS₂ using time- and energy-resolved photoemission electron microscopy with sub-50-fs pump and probe pulses, and sub-200-meV energy resolution. With a bulk band gap of ~1.4 eV, above-gap excitation is initiated by 2.41 eV pump pulses. The electron population in the conduction band is subsequently probed by 4.82 eV probe pulses. A photoemission spectrum is obtained at each time delay (Fig. 1a). Integrating the signals obtained at higher energies reveal dynamics that are dependent on the pump- and probe- polarization. Using a fixed probe polarization parallel to the b-axis, at an initial excited carrier density at $\sim 3.3 \times 10^{12} \text{ cm}^{-2}$, for example, excitation perpendicular to the b-axis yields a fast decay of $\tau_{fast}^{\perp} = 0.49 \pm 0.05 \text{ ps}$, whereas excitation parallel to the b-axis results in a slower decay of $\tau_{fast}^{\parallel} = 1.3 \pm 0.3 \text{ ps}$ (Fig. 1b). Both cases exhibit a shared slow decay constant. Increasing the photoexcited carrier density results in a decreasing τ_{fast} , suggesting that the fast decay arises from carrier thermalization via carrier-carrier scattering. An additional rise component, $\tau_{rise} = 0.10 \pm 0.03 \text{ ps}$ is also observed when probing only with a polarization perpendicular to the b-axis. Preliminary analysis of the calculated excited polarization-resolved density of states suggests that a larger population of high energy carriers perpendicular to the b-axis may result in larger probability of electron-electron interactions and therefore a shorter τ_{fast} . The rise component is likely attributed to the probing of p_y -like states as compared to the p_x - and p_z -like states with different probe polarizations. These results highlight that bulk ReS₂ exhibits anisotropic ultrafast dynamics to linearly polarized light and could serve as a promising platform for optoelectronic applications with an additional tuning parameter controlled by light polarization.

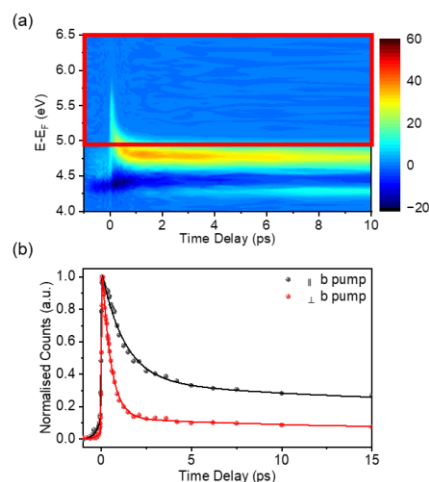


Fig. 1. (a) Contour plot of photoemission spectra as a function of time delay. Red box represents the energy integration for time-trace analysis. (b) Time traces for different pump polarization at a given excitation density.

Femtosecond Hyperspectral Source - HYPUS

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We developed a high-power, tunable femtosecond light source offering multiple spectral channels in 200 -12000 nm range without the complexities of OPAs, OPCPAs or NOPAs. A LightConversion CARBIDE (330 fs, 2 mJ, 80 W) laser was compressed to sub-10fs [1] with a spectrum ranging from 750-1250 nm, channel 3 (yellow) in Fig.1 serves as the “front-end”. We subsequently exploit its spectral content and short pulse duration, respectively, to derive new wavelengths in an OPA-free manner.

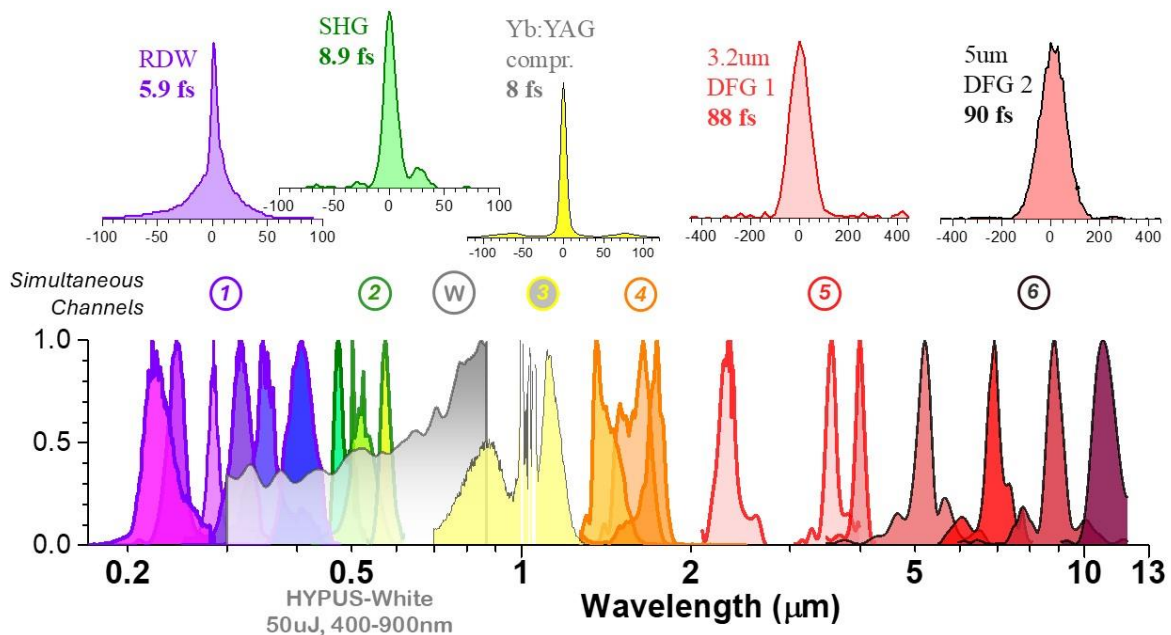


Fig. 1. Six tunable output channels (#1-6) of HYPUS that can be available simultaneously plus an extra white light channel. The independent tuning within each channel is depicted by exemplary spectra of same line color. Top row shows experimentally measured pulse durations.

Our source offers up to five other channels (UV, VIS, NIR, IR, mid IR) that are optically synchronized to the broadband HCF output (channel 3). The deep UV (channel 1) is achieved via resonant dispersive wave (RDW) emission to span the range 200-450 nm at repetition rates up to 100 kHz. Two separately tunable VIS pulses (channel 2) with up to 3W and 4W of sub 50fs pulses can be obtained between 450 nm and 560 nm. IR generation (channels 4-6) is achieved by DFG between parts of the broadband spectrum from channel 3. It was possible to cover the range of 2.5-12 μm with pulse durations between 80-150 fs. The 3-5 μm (channel 5) and 5-10 μm ranges (channel 6) can be obtained simultaneously and tuned independently if the energy is split between the two ranges. The IR and mid IR ranges are passively CEP-stabilized, while other channels follow the CEP stability of the driving laser. Alternatively, a continuous white light source spanning 300 nm – 1200nm can be available. Notably, the shot-to-shot timing jitter between the UV and mid IR has been measured to be on the 1 fs level. These results highlight the potential of this platform for broadband, ultrafast applications across diverse spectral regions.

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Q24

From Pulse Shaping To Batteries: New Approaches And Applications In 2D Raman Spectroscopy

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2D Raman is a fifth-order technique that can provide unique access to low frequency mode-mode couplings in chemical systems. Despite its outstanding potential, third-order cascading signals that can contaminate the 2D Raman spectra have made the technique challenging to apply. While a variety of approaches have been used to eliminate such cascades, including the use of pulse trains, polarization control and resonant excitation, each has had drawbacks limiting widespread uptake of the technique.

Here, using a high-repetition rate laser source, we develop a scheme based on pulse pairs and detection of spectral shifts in a compressed high repetition probe pulse to improve the resolution, accuracy and applicability of 2D Raman.

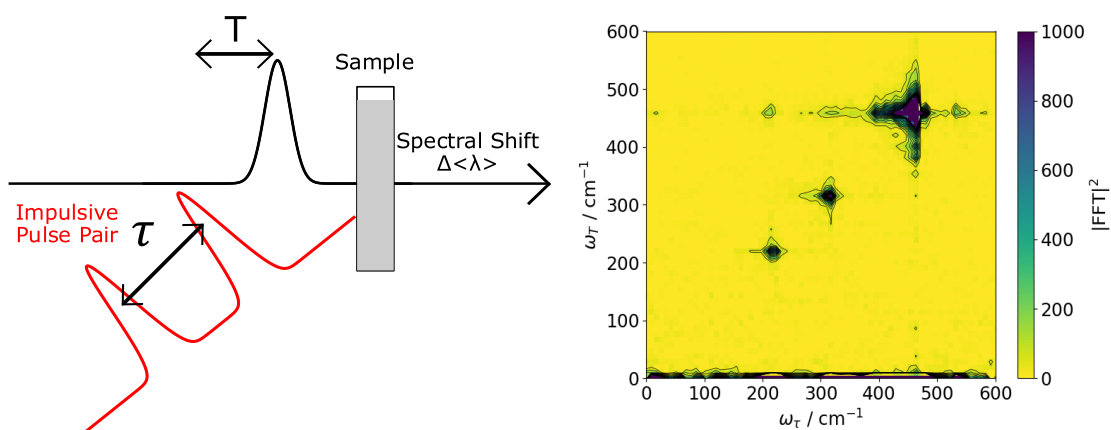


Fig. 9. Experimental schematic and 2D Raman spectrum of CCl_4 obtained using the cascade-free spectral shift.

We demonstrate our method by exploring mode-mode couplings and anharmonicity in a topical Li-ion battery solid (lithium lanthanum zirconate) and liquid (LiTFSI in water) electrolyte. In the liquid system we find evidence for coupling between the librational solvent modes and anions, and in the solid system coupling between framework modes and mobile ion containing modes.

Our results take us a step further to a more widespread use of 2D Raman methods and for performing time resolved 2D Raman spectroscopy where one can follow the evolution of mode-mode couplings following photoexcitation.

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Q25

New Polarization Tricks For Background Removal In Action Detection Spectroscopies

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Action detection spectroscopy is a method based on detecting a population state with a measurable observable rather than traditional optical absorption. The observable (or action) might be fluorescence, thermal expansion, or electrical current to name a few.[1-3] These types of detection methods offer several advantages over optical detection including increased sensitivity, pathway selectivity, and no phase matching conditions, but they are not without their downsides. Large background known as incoherent mixing often plagues action-detected spectra, obscuring lineshapes and system dynamics.[4] Unfortunately,

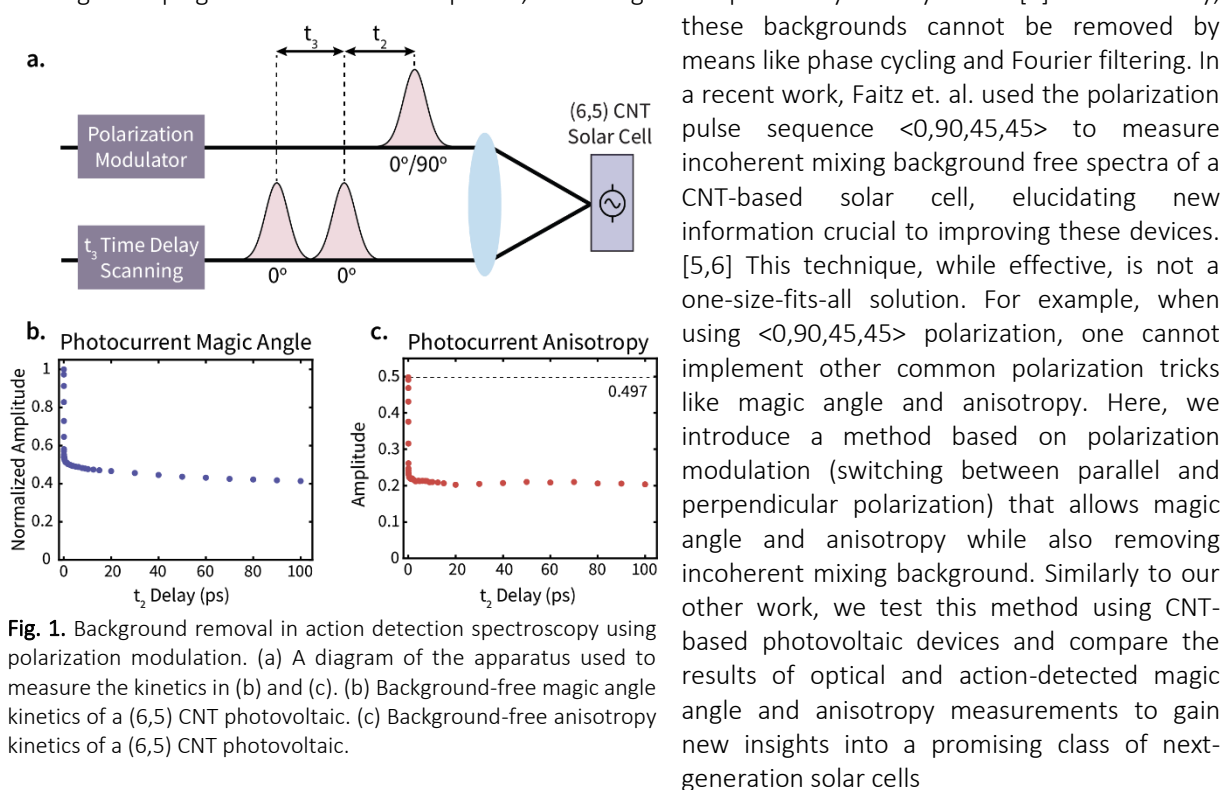


Fig. 1. Background removal in action detection spectroscopy using polarization modulation. (a) A diagram of the apparatus used to measure the kinetics in (b) and (c). (b) Background-free magic angle kinetics of a (6,5) CNT photovoltaic. (c) Background-free anisotropy kinetics of a (6,5) CNT photovoltaic.

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SpecSuite: A toolbox for modelling ultrafast experiments

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SpecSuite is an HPC software package comprising approximately 300 subroutines, written predominantly in Python. It has been in continuous development since 2016 and is now available for download via GitHub.

The code allows modelling of a variety of 2DES experiments [1], femtosecond coherence spectroscopy [2], photon correlation experiments [3], and quantum information protocols [4]. New modules such as quantum tomography are now under development. The code is specifically designed for modelling molecules and includes prebuilt codes for vibrational molecules, vibronic dimers and trimers. It has subroutines suitable for modelling molecular cavity quantum electrodynamics. The open system dynamics can be run under the Lindblad equation (Markovian dynamics) or the hierarchical equations of motion (HEOM, non-Markovian dynamics) with both overdamped and underdamped baths. The codes allow for multiple bath coupling to the same system.

In this poster we overview some of the features of SpecSuite and provide details on where to download it.

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Q27

The new HiLUX-ULTRA instruments at the Central Laser Facility

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The UK's Central Laser Facility has hosted user facilities for time-resolved Raman and IR spectroscopy over three decades and SSFG and 2D-IR spectroscopy facilities for one and two decades. The laser technology used has spanned several generations of dye lasers and multiple synchronised Ti:Sapphire amplifiers, with Yb:KGW amplifiers entering around a decade ago with the LIFETIME facility. Many hundreds of user experiments have been delivered over the years. In this poster we present our progress and efforts to build a new generation of Yb:KGW-based facility spectrometers for 2D-IR, 2D-IR-Raman, 2D-ES, transient IR and visible, SSFG, 2D-IR SSFG and Kerr-gated Raman. This collection of instruments (called 'HiLUX') draws on a platform of five synchronisable Yb:KGW amplifiers and a dozen or-so optical parametric amplifiers (OPAs). We are incorporating the engineering ethos, motion controls and software approaches of our department's large-scale petawatt laser facilities. We are automating and simplifying experimental reconfigurations and through the exceptional stability / ease-of-use of the Yb:KGW platform expect HiLUX to significantly speed up routine measurements and lower the barriers to difficult experiments for years to come.

Q28

Asynchronous Interferometric Two-Dimensional Electronic Spectroscopy of Surface-Accumulation-Layer Dynamics in InAs

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Surface band bending in InAs creates an electron accumulation layer whose role in ultrafast carrier relaxation has remained difficult to isolate under conventional above-gap excitation, where bulk hot-carrier signatures dominate the transient response. Here, we employ two-dimensional (2D) electronic spectroscopy based on asynchronous interferometric (AI) method [1] to disentangle excitation-energy-specific relaxation pathways in submicrometer InAs particles. The 2D approach correlates excitation frequency with probe response across the band-edge and surface-subband spectral region, providing

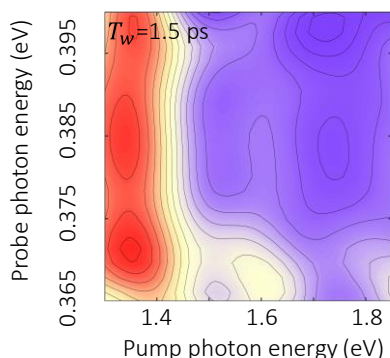


Fig. 1. 2D spectra at the waiting time (T_w) of 1.5 ps.

the excitation-axis resolution that one-dimensional pump–probe lacks and enabling direct separation of resonant, intermediate, and far-above-gap dynamics within a single coherent framework. Near-resonant 0.38 eV excitation reveals a long-lived 400–600 ps recovery assigned to remote recombination between bulk-like conduction-band electrons and the surface-subband manifold, while strongly above-gap excitation produces a distinct response dominated by ~ 1.5 ps hot-carrier cooling, lattice heating, and 20–30 ps heat-assisted release and recovery of accumulation-layer electrons. Excitation at 0.8 eV bridges these limits. Crucially, two-color 2D electronic spectroscopy resolves a heterogeneous ~ 2.4 ps non-radiative channel associated with surface-mediated recombination — a feature that is averaged out in conventional pump–probe and only emerges along the excitation-frequency axis. These results establish 2D electronic spectroscopy as a discriminating probe of surface-versus-bulk relaxation in narrow-gap semiconductors and show that the InAs accumulation layer acts as an active kinetic reservoir that redirects carrier relaxation depending on initial excitation energy, with implications for engineering ultrafast processes in infrared optoelectronic devices.

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Q29

Probing Coupling To and Through Metal-Metal Multiple Bonds With DOVE Spectroscopy

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Sparking from the first description of a Rhenium-Rhenium quadruple bond 1964, the field of metal-metal multiply bonded systems remains vibrant with applications in catalytic ammonia oxidation, medicine, nitrogen insertion, and molecular electronics. While metal-metal multiple bonds have been frequently characterized via resonance Raman and X-ray crystallography, identification and analysis of vibrational modes coupled to the metal-metal stretching mode is absent. Here, Doubly Vibrational Enhanced Four Wave Mixing spectroscopy (DOVE-FWM) is used to probe the vibrational coupling between the metal-metal multiply bonded stretches and pseudohalide asymmetric stretches in $\text{Ru}_2(\text{dpf})_4\text{N}_3$ and $\text{Mo}_2\text{Ni}(\text{dedpa})_4(\text{NCS})_2$, where dpf = N,N'-diphenylformamidine and dedpa = 4,4'-diethyl-2,2'-dipyridylamine. The coupling of the pseudohalides to the metal-metal stretches in these complexes are implicated in concerted motion theoretically predicted for the activation of $\text{Ru}_2(\text{dpf})_4\text{N}_3$ to the reactive species $\text{Ru}_2(\text{dpf})_4\text{N}$, and in the electron transport through $\text{Mo}_2\text{Ni}(\text{dedpa})_4(\text{NCS})_2$. Of note, the coupling of the metal-metal stretches to the strong oscillators of the pseudohalides and resonant electronic transitions allows identification of combination bands with intensities below the detection threshold in Fourier-transform infrared spectroscopy. Additionally, DOVE-FWM is used to probe for coupling between the axial isothiocyanates in $\text{Mo}_2\text{Ni}(\text{dedpa})_4(\text{NCS})_2$ and in conjunction with the coupling to the Molybdenum-Molybdenum stretch describes communication across the core of the complex. The identification of vibrational coupling between the metal-metal multiple bonds and the axial ligands serves as the initial steps in experimentally validating computational descriptions of the mechanism of $\text{Ru}_2(\text{dpf})_4\text{N}_3$ conversion to $\text{Ru}_2(\text{dpf})_4\text{N}$, and electron transport through $\text{Mo}_2\text{Ni}(\text{dedpa})_4(\text{NCS})_2$.

Q30

Disentangling electronic and vibrational signatures via simulated femtosecond coherence spectroscopy

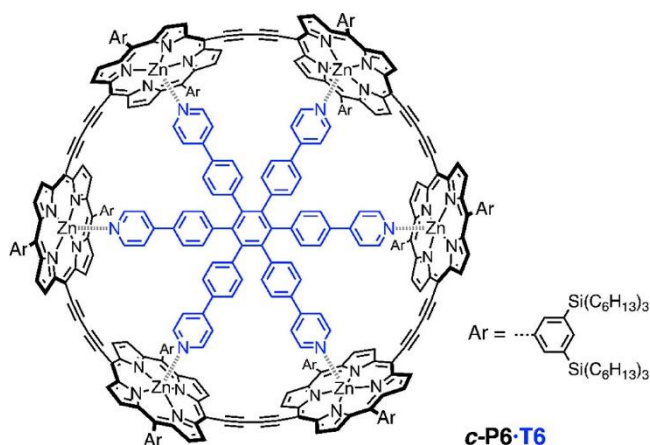
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Coherent dynamics in photosynthetic complexes, such as some extended π -conjugated systems, have been broadly investigated, yet the relative contributions of electronic and vibrational coherences remain difficult to disentangle[1,2]. In complex systems, oscillatory features observed in ultrafast spectroscopy can arise from both excitonic superpositions and vibrational wavepackets, making their interpretation non-trivial. Here, we investigate coherence dynamics in a 6-membered Zn-porphyrin nanoring system simulating femtosecond coherence spectroscopy (FCS) to separate electronic from vibrational contributions[3,4]. A finite-field excitation scheme is employed to enhance the ground-state vibrational wavepackets. By analysing probe-frequency-resolved coherence signals following ultrafast excitation, we identify distinct signatures associated with excitonic coupling around the ring and localised vibrational motion, corresponding to 380 cm^{-1} Zn-N stretching modes localised on each Zn-porphyrin core[2]. Environmental effects are incorporated through a combined open quantum system framework, where electronic dephasing and local disorder are modelled using Lindblad operators, while vibrational relaxation is captured using the hierarchical equations of motion (HEOM). This enables realistic treatment of decoherence processes across electronic and nuclear degrees of freedom. Our results reveal how inter-nanoring interactions promote the emergence and persistence of coherent dynamics, and clarify the extent to which vibrations contribute to experimentally observed coherence signals.



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Fig. 1. Porphyrin structures consist of nanorings composed by six meso-substituted Zn-porphyrin cores (bearing aryl groups (3,5-bis(trihexylsilyl)benzene) at meso-positions) forming a circular structure through butadiyne links. c-P6·T6 is the templated ring.

Q31

Kramers–Kronig–Constrained Reconstruction of Optical Absorption from Interference-Dominated Transmission Spectra

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Layered materials, such as transition metal dichalcogenides, van der Waals heterostructures, and organic–inorganic hybrid crystals, exhibit rich optical properties and strong visible-light responses, making them ideal platforms for investigating ultrafast carrier and excitonic dynamics via transient absorption spectroscopy. However, for thin-layer samples with thicknesses ranging from several hundred nanometers to a few micrometers, multiple internal reflections and Fabry–Pérot interference significantly modulate the transmitted spectra, introducing pronounced oscillatory artifacts that complicate quantitative extraction of absorption coefficients.

Here, we develop a theoretical inversion algorithm to reconstruct the intrinsic extinction (imaginary refractive index) spectrum of thin layered samples from interference-dominated transmission spectrum. By incorporating an approximate refractive-index range and employing an iterative Kramers–Kronig constrained reconstruction scheme, we retrieve the extinction coefficient as a function of wavelength. Numerical simulations demonstrate that, although direct interpretation of the transmission spectra fails to yield reliable absorption information, the reconstructed extinction spectrum closely reproduce the predefined ground-truth extinction spectrum.

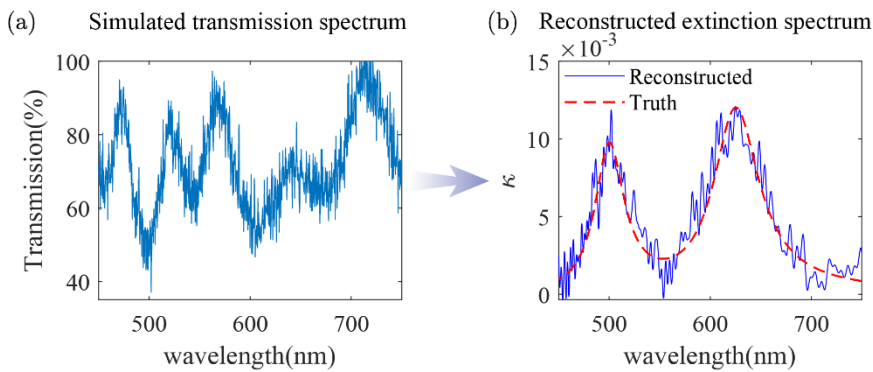


Fig. 1. Demonstration of the reconstruction algorithm on simulated thin-film spectra. (a) Simulated transmission spectrum of a 1.5- μm -thick layered sample with refractive index $n \approx 1.9$, embedded in argon on both sides. Two absorption bands are introduced at 500 and 625 nm, and 5% Gaussian noise is added. (b) Comparison between the predefined extinction coefficient κ (red dashed line) and the reconstructed extinction coefficient κ obtained from the transmission data (blue solid line), showing excellent agreement.

Our method provides a general framework for correcting interference artifacts in optical spectroscopy of thin layered materials. It can be directly applied to steady-state and ultrafast spectroscopic measurements, enabling accurate determination of intrinsic absorption properties and facilitating quantitative studies of light–matter interactions in low-dimensional systems.

Q32

Ultrafast Vibrational Wave Packet Dynamics of the Aqueous Adenine Radical Induced by Photodetachment

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An understanding of the elementary reactions that follow the ionization of biomolecules in an aqueous medium forms the basis of radiation chemistry and radiation biology [1]. As one of the primary targets of radiation damage, the ionization of DNA and its constituents are of particular significance in these fields, with nucleobase radicals playing major roles in DNA damage. While the ionization of nucleobases has been studied extensively [2,3], the sub-picosecond structural rearrangement dynamics of the generated radical species remain vastly unexplored.

Extending on a previous study on guanine [4], here, we utilize femtosecond optical pump-probe spectroscopy to elucidate the ultrafast vibrational wave packet dynamics of the aqueous adenine radical induced by photodetachment. Strong-field detachment by intense, few-cycle, broadband visible-NIR pump pulses drives the formation of the hydrated electron and the aqueous adenine radical. Absorption signatures of the hydrated electron and the aqueous adenine radical are observed in the transient absorption spectra, probed in the visible-NIR and UV range. In addition to the increase in ΔA signal below 350 nm, attributed to the absorption of the aqueous adenine radical [5], the transient absorption spectra probed in the UV also feature pronounced modulations of the ΔA signal as a function of pump-probe time delay (Fig. 1a). These modulations originate from the vibrational wave packet motion of the aqueous adenine radical.

The vibrational wave packet motion is resolved into ten vibrational modes, which encode the structural reorganization that occurs upon photodetachment. The observed vibrational modes are 623 cm^{-1} , 721 cm^{-1} , 965 cm^{-1} , 1051 cm^{-1} , 1136 cm^{-1} , 1246 cm^{-1} , 1320 cm^{-1} , 1368 cm^{-1} , 1405 cm^{-1} and 1454 cm^{-1} (Fig. 1b).

These modes are assigned with the aid of ab initio density functional theory (DFT) calculations. Consistent with the observed vibrational modes exhibiting significant in-plane ring deformation, DFT calculations reveal picometer changes in the bond lengths as the equilibrium geometry of the aqueous adenine anion shifts to that of the adenine radical, with these changes concentrated on the purine ring. This work provides insights into the ultrafast structural rearrangement dynamics following the ionization of nucleobases in an aqueous environment.

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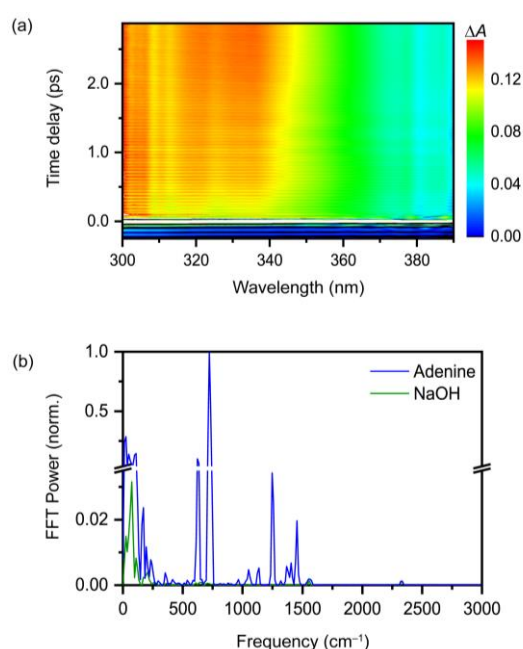


Fig. 1. (a) Differential absorption (ΔA) spectra of the photodetached aqueous adenine anion as a function of pump-probe time delay. (b) Normalized fast Fourier transform (FFT) power spectra of the first-moment time trace showing vibrational modes of the adenine radical.

Q33

Ultrafast Vibrational Wave Packet Dynamics of the Aqueous Ionized Dipeptide, L-Phenylalanyl-L-alanine (Phe-Ala)

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Photoionization of proteins and other biomolecules (e.g., DNA, amino acids, lipids, metabolites) in aqueous environments occurs when they are subjected to high-energy radiation, which consequently generates the radical species of that molecule and leads to radiation damage to amino acids in proteins [1]. Previous work done in our group uncovered the ultrafast vibrational wave packet dynamics of the aqueous phenylalanine radical (Phe[•]) following strong-field photodetachment of the phenylalaninate anion [2]. This study elucidates the vibrational modes that participate in ultrafast structural rearrangement upon the ejection of an electron from the aqueous dipeptide, L-Phenylalanyl-L-alanine (Phe-Ala). Here, we extend our previous work by exploring the ionization-induced ultrafast vibrational wave packet dynamics of the aqueous Phe-Ala.

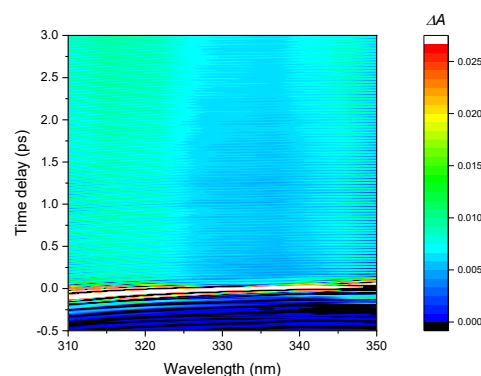


Fig. 1. Differential absorption (ΔA) spectra of ionized aqueous Phe-Ala (at pH 7.7) as a function of pump-probe time delay.

We use transient absorption spectroscopy, ionizing aqueous Phe-Ala with an intense, visible pump pulse via a multiphoton process, and probing the resultant ultrafast dynamics with a broadband UV pulse. The differential absorption (ΔA) spectra reveals modulations, which arise from vibrational wave packet motion, as a function of pump-probe time delay (Figure 1). Reverse-phase high performance liquid chromatography (RP-HPLC) shows that sample degradation over the course of the experiment is negligible. The chromatogram obtained from the HPLC run on the stock Phe-Ala solution subjected to the laser pulses indicates the absence of Phe, which strongly confirms that the dynamics observed in the differential absorption spectra can be attributed to Phe-Ala. The spectral first moment time trace, $\langle E(t) \rangle$, which encodes the vibrational wave packet dynamics, was calculated. A biexponential line was fitted to the first moment time trace and a Fast Fourier Transform (FFT) was performed on the residual of the fitted line. Ultrafast structural rearrangement drives coherent vibrational motion involving 12 fundamental frequencies. Geometry optimization using Molecular mechanics followed by an optimization + frequency DFT calculation shows the dominant conformer of Phe-Ala based on the Boltzmann population at room temperature.

The vibrational frequencies observed in the residual first moment time trace are assigned to the vibrational modes found from the DFT calculations. A comparison of the vibrational modes of Phe-Ala and Phe shows 5 vibrational frequencies that take part in ultrafast structural rearrangement in both Phe-Ala and Phe, and 7 vibrational frequencies unique to Phe-Ala. As this research is currently in progress, further work is required to determine the dephasing time of Phe-Ala compared to Phe, $\tau_1 = 1540$ fs, based on the previous work done [2].

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Q34

Excited-State-Selective 2D Electronic Spectroscopy: A Role-Reversed Time-Dependent Perturbation Approach with Quantum Photonic States

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Photosynthetic light harvesting complexes transfer photoexcitation energy efficiently through networks of coupled chromophores. Small energy gaps in the exciton manifold make it difficult to resolve exciton dynamics with clear spectral separation. To disentangle these complex spectral features, two-dimensional electronic spectroscopy (2DES) is widely used for these systems by correlating excitation and detection frequencies [1]. However, it is limited by coherent artifacts near time zero and spectral congestion arising from the overlap of ground state bleach (GSB), stimulated emission (SE), and excited state absorption (ESA). These effects obscure early-time signals and complicate interpretation.

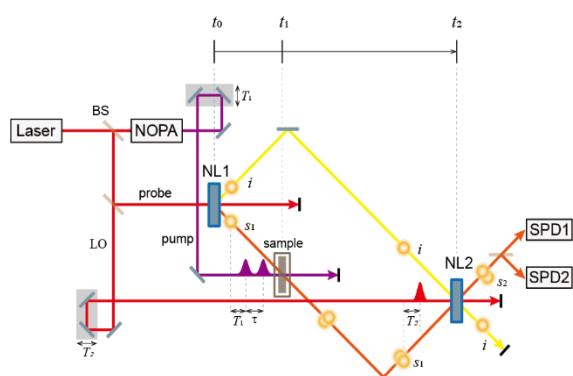


Fig. 1. Optical layout of 2DTSE. The coincidence counting of SE photons is achieved via stimulated parametric down-conversion.

To mitigate these limitations, the Mukamel group introduced a hybrid concept using a classical pump with a single-photon probe [2]. The key advantage of this approach lies in the use of a classical pump to create a substantial excited state population. Consequently, the low interaction probability of single photons no longer limits the measurement.

Motivated by this idea, we propose two-dimensional transient stimulated emission (2DTSE). This method is specifically designed for pathway selectivity, which isolates the stimulated emission contribution from other nonlinear responses. By effectively filtering out competing signals, 2DTSE provides a significantly cleaner readout of exciton dynamics.

In our formulation, the probe is treated as a quantum photonic wavefunction, and the sample response is represented as an operator acting on the photon state in the photonic Hilbert space. Since a single-photon probe cannot interfere with the pump to form transient gratings, the formalism naturally excludes the coherent-artifact contributions associated with pump–probe interference. As a result, we expect that 2DTSE can, in principle, reduce spectral contamination from time-zero artifacts and from spectral overlap among GSB, SE, and ESA, enabling clearer access to exciton dynamics in complex molecular systems.

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Q35

Anharmonic Phonon-Phonon Coupling in LaAlO_3 Revealed by Two-Dimensional Raman-Terahertz Spectroscopy

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Anharmonicities govern many of the key thermal, electronic, and optical properties of solids. They can be broadly classified into mechanical and electrical anharmonicities [1–3]. Hybrid two-dimensional Raman-THz-THz (2D-RTT) spectroscopy is a novel spectroscopic technique extensively used to study anharmonic coupling in molecular systems [4–6]. In this work, we report the observation of anharmonic phonon-phonon coupling in LaAlO_3 (LAO) using 2D-RTT spectroscopy. Here, the Raman-active A_{1g} phonon mode is coherently excited through an impulsive interaction with the Raman pulse. This coherent A_{1g} phonon anharmonically drives the low-energy E_g phonon mode, which exhibits weak but measurable IR activity due to inversion-symmetry breaking. Anharmonic coupling between the A_{1g} and E_g phonon modes is revealed by the appearance of a localized cross-peak feature at ($f_1 \sim 3.7$ THz, $f_2 \sim 1$ THz) in the 2D Raman-THz spectrum, consistent with theoretical modeling. Furthermore, the good agreement between the experiment and simulations confirms that both mechanical and electrical anharmonicities contribute to the observed 2D signal.

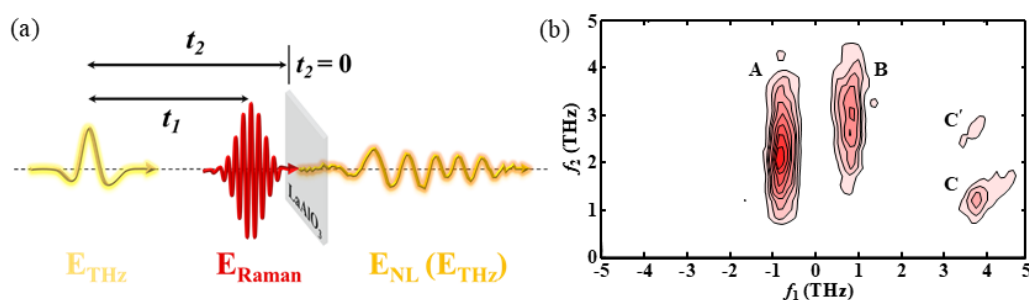


Fig. 1. (a) Schematic of the experimental geometry for hybrid 2D Raman-THz spectroscopy with the Raman-THz-THz (RTT) pulse sequence. (b) Experimental 2D Raman-THz spectrum of LAO.

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Q36

Decoding the THz–IR–Vis Spectrum of Water

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The interaction between low-frequency (THz) and high-frequency (IR, mid-IR) modes plays a fundamental role in physical and chemical processes in the condensed phase. Two-dimensional Terahertz-Infrared-Visible (2D TIRV) spectroscopy is a promising technique for elucidating the couplings between low- and high-frequency molecular vibrations. It is a four-wave mixing technique in which a femtosecond broadband terahertz (THz) pulse coherently excites low-frequency modes of the sample. The system is then probed by a femtosecond mid-infrared (IR) pulse followed by a picosecond narrowband visible (VIS) pulse. The system produces a second order response signal where the crosspeak appears at $\omega_{\text{VIS}} + \omega_{\text{IR}} \pm \omega_{\text{THz}}$ being the signature of the mode-mode coupling [1].

Recently, 2D TIRV spectroscopy has been employed to measure intramolecular vibrational coupling in liquid water [2]. Their work revealed a coupling between low- and high-frequency modes (Figure 1, right). We used simulations to analyze the experimental 2D TIRV spectrum of water. The response functions were simulated quantum mechanically using the Redfield theory. By comparing simulated and experimental spectra, shown in Figure 1, we were able to extract the essential parameters of the low-frequency hydrogen-bond stretching mode and the high-frequency OH stretching modes and the coupling between them. Then, we varied the parameters to investigate the spectral signatures of intermode and system–bath couplings, as well as the mechanical and electrical anharmonicity of the low-frequency collective hydrogen-bond stretching mode. By analyzing the negative- ω_1 frequency region, which is experimentally challenging to access, we elucidate the key mechanisms underlying the 2D TIRV spectrum of water.

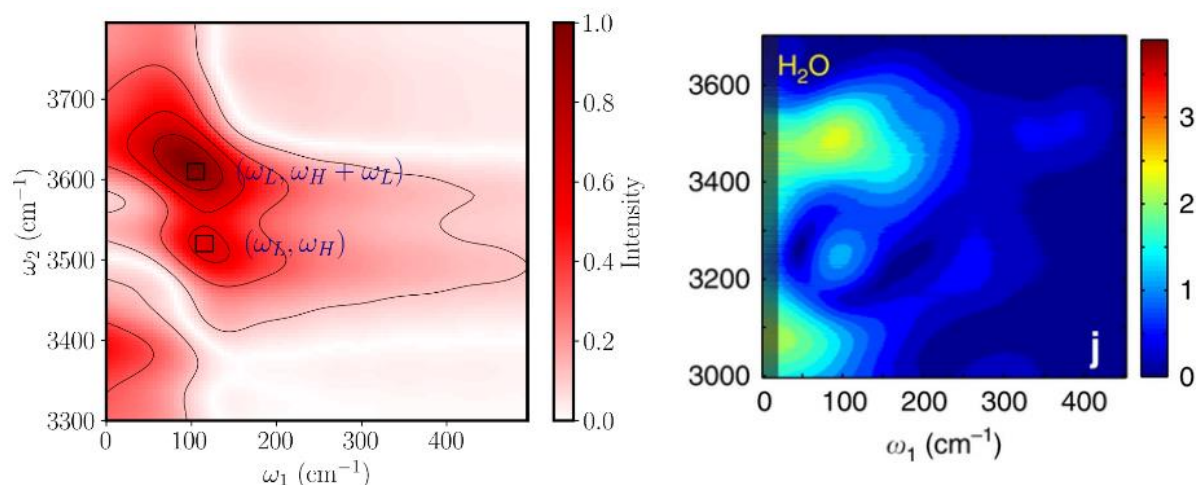


Fig. 1. Left: Absolute-value 2D TIRV simulated spectrum. Low-frequency mode $\omega_L = 115$ cm⁻¹ and high-frequency mode $\omega_H = 3520$ cm⁻¹. Squares mark signals at $(\omega_1, \omega_2) = (\omega_L, \omega_L + \omega_H)$ and (ω_L, ω_H) . Right: Absolute-value 2D TIRV experimental spectrum from Ref [2].

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Q37

Two-dimensional Raman-infrared Spectroscopy for Reaction Dynamics in the Electronic Ground state

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Chemical reactions predominantly occur in the electronic ground state, where they are driven by thermal fluctuations that activate nuclear motions, leading the system along the reaction coordinate. Because these nuclear rearrangements are mapped onto vibrational modes, tracking the temporal evolution of vibrational modes provides a window into the reaction dynamics. Femtosecond time-resolved spectroscopy reveals the reaction coordinate by analyzing vibrational wavepackets, probing how the vibrational signatures of molecular species evolve during the chemical transformation. This fundamental principle can be applied to two-dimensional infrared (2D-IR) spectroscopy, which theoretically allows for a similar dynamical analysis. However, generating broadband mid-infrared (mid-IR) excitation pulses is technologically more challenging than generating mid-IR probe pulses, thereby restricting the accessible excitation frequency window and dynamic information.

To overcome these limitations, two-dimensional Raman-infrared (2D Raman-IR) spectroscopy has been proposed as an alternative method [1-2]. This multidimensional spectroscopic method utilizes non-resonant impulsive stimulated Raman scattering (ISRS) to generate vibrational coherence in the electronic ground state, which is measured by a broadband mid-IR probe. By capturing motion of vibrational wavepackets, 2D Raman-IR spectroscopy maps the correlation between various vibrational modes onto a two-dimensional frequency space, analogous to 2D-IR spectroscopy but with an expanded excitation bandwidth. Although 2D Raman-IR spectroscopy provides a lower dimensionality compared to 2D-IR spectroscopy, it holds significant potential for investigating reversible chemical reactions at thermal equilibrium. However, the experimental realization of this technique has yet to be demonstrated.

In this work, we present the experimental implementation of 2D Raman-IR spectroscopy. The experimental setup of 2D Raman-IR spectroscopy employs a 10-fs near-infrared (NIR) pump pulse

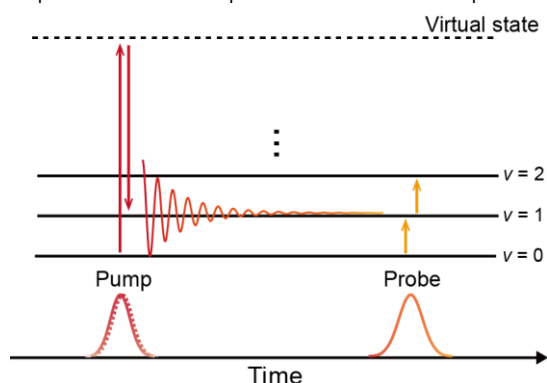


Fig. 6. Schematic of 2D Raman-IR spectroscopy.

paired with a broadband mid-IR probe. The ultrashort NIR pump simultaneously excites a broad manifold of vibrational modes, spanning a wide spectral range of up to 1600 cm^{-1} via non-resonant ISRS. This process generates vibrational coherence in the electronic ground state, which can be then interrogated by the mid-IR probe to monitor the structural evolution associated with the chemical process.

This technique facilitates a comprehensive analysis of vibrational dynamics over an extensive spectral range, elucidating the motions that mediate chemical reactions. By tracking these active vibrational modes, the technique resolves the underlying reaction pathways at chemical equilibrium. We anticipate that

this method will become an effective tool for investigating the detailed mechanisms of chemical processes.

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